

Variational Closure of Finite Einstein–Standard-Model Actions

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Abstract

We prove a variational closure theorem for finite Lorentzian gravity–matter systems with Standard-Model field content and a common action. Under compactness, first-variation consistency, and physically scaled stationarity, subsequences of reconstructed fields converge to distributional solutions of the classical Einstein–Standard-Model equations. The argument requires neither uniform coframe convergence nor strong spinor-derivative convergence, while compactness of the covariant Higgs gradient suffices to close the quartic sector. When bosonic compactness fails, the limit retains explicit stress defects whose vanishing is characterized by positive energy measures. For an explicit finite action, a gauge-invariant Wilson-transport criterion yields zero-defect closure at critical gauge regularity. On regular branches we also obtain quantitative convergence and an exact solution shadowing a single finite record from corrected initial data, without prescribing its future trajectory. Renewal Geometry provides one realization of the finite interface, but the closure theorem is independent of that construction.

Keywords. Einstein–Standard-Model system; finite gravity–matter actions; variational closure; continuum limits; nonlinear compactness; stress defects; hyperbolic stability; Renewal Geometry.

1 Introduction

The Standard Model and general relativity organize two different kinds of geometric structure. General relativity identifies gravitation with the Lorentzian geometry of spacetime, whereas the Standard Model introduces an internal gauge structure, chiral fermion representations, a Higgs sector, Yukawa couplings, and a three-generation matter multiplicity over an already given spacetime. In the usual continuum formulation these ingredients are specified before the coupled field equations are written.

A natural finite-to-continuum question is whether a gravity–matter theory specified at finite resolution can recover the classical Einstein–Standard-Model system without inserting the limiting field equations by construction. This question is logically distinct from the microscopic origin of the finite field content itself. A finite model may carry the correct Lorentzian signature, gauge group, matter representations, and common gravity–matter action while nevertheless failing to possess a controlled continuum limit. Conversely, a convergent discretization of a prescribed continuum theory need not explain why those particular structural data were present in the first place.

The present paper isolates this analytic closure problem. We formulate a finite Lorentzian–Standard-Model interface containing the structural bundle data, finite Dirac/Yukawa incidence, positive source geometry, and one common gravity–matter action. Reconstruction maps, physical variation maps, compactness, and stationarity are imposed separately. The principal question is then whether these finite data have the classical Einstein–Yang–Mills–Higgs–Dirac–Yukawa equations as a variational continuum limit.

Geometric and variational context. There is a long history of describing Standard-Model structure in geometric terms. In noncommutative geometry, the Connes–Lott construction realizes electroweak and Standard-Model data through a finite noncommutative geometry [1], finite spectral triples admit a diagrammatic classification [2], and the spectral action produces gravitational, Yang–Mills, Higgs, and fermionic terms from one spectral datum [3]. The almost-commutative construction was later extended to include neutrino mixing and Majorana masses [4]; Lorentzian formulations of the noncommutative Standard-Model geometry have also been investigated, including the fermionic and doubling issues specific to indefinite signature [5]. Causal fermion systems provide a different finite or microscopic entrance in which effective gauge and gravitational dynamics arise from a causal action principle [6].

A complementary literature concerns the preservation of geometric and variational structure under discretization. Regge calculus initiated a simplicial treatment of gravitational curvature [7]; piecewise-flat curvature measures were subsequently shown to converge under refinement [8], and more recent work has placed linearized Regge calculus, densitized scalar curvature, and the Einstein tensor in rigorous finite-element and distributional frameworks [9, 10, 11, 12]. Wilson lattice gauge theory retains exact finite gauge invariance [13], while discrete variational field theories preserve action-derived multisymplectic and Noether structures [14]. Finite element exterior calculus provides stable discrete differential complexes [15, 16]. Gauge-invariant Maxwell and Maxwell–Klein–Gordon discretizations give complementary scheme-specific consistency and convergence results [17, 18, 19], and simplicial gauge actions can combine finite gauge invariance, consistency, and discrete Noether identities [20]. In these approaches the continuum geometric or gauge structure is specified before the discretization. The formulation used here instead isolates structural, compactness, first-variation consistency and stationarity hypotheses that can be checked for different finite gravity–matter realizations. It is therefore complementary to, rather than a replacement for, scheme-specific stability and convergence analyses.

One upstream realization: Renewal Geometry. Renewal Geometry provides one realization of the finite interface considered below. Predictive-state constructions characterize state by the information required for future prediction rather than by a separately postulated hidden coordinate [21], while multitime operational formalisms provide a related language for processes with memory [22, 23]. In the Renewal Geometry construction developed in [24, 25], operational histories are quotiented by equality of conditional future statistics, and the resulting minimum persistent carrier $Z_X^{\min}$, represented history algebra $\mathcal{A}_X^{\text{hist}}$, and positive word-Gram hierarchy are organized into the *Renewal Grand Tensor*,

$$\mathbb{T}_{\text{Grand},X} = \left(\mathcal{A}_X^{\text{hist}}, \{ \mathbb{K}_{X,r} \}_{r \geq 0}, Z_X^{\min} \right). \quad (1.1)$$

The terminology “tensor” refers to this finite predictive data structure, not to a tensor field on a pre-existing manifold.

On certified branches, the corresponding Lorentzian construction provides a relational carrier, Lorentzian signature, ADM-type data, and gravitational normalization [24]; the spectral construction provides compatible differential and spin data [25]; and the structural gravity–matter construction provides a Standard-Model matter packet and common finite gravity–matter action [26]. These constructions establish one nonempty realization of the finite interface. None of their microscopic reconstruction arguments is used in place of the compactness or first-variation estimates proved here.

The continuum obstruction. The main analytical difficulty is that neither convergence of fields nor convergence of action values is sufficient to pass to the coupled classical equations. The

Yang–Mills stress is quadratic in curvature, the Higgs stress contains quadratic covariant derivatives and a quartic potential, and the metric variation of the fermionic action also varies the spin geometry. Weak convergence of the primitive fields can therefore fail to determine the limit of the quantities that gravitate.

This issue is familiar from several directions. Gauge compactness requires careful control of both gauge choice and curvature [27], and at the critical four-dimensional Yang–Mills energy scale minimizing sequences can develop curvature concentration and bubbling [28]. Critical Sobolev embeddings likewise permit concentration phenomena [29]. More generally, weak limits of nonlinear PDEs can retain oscillation or concentration information through measure-valued and microlocal descriptions, including DiPerna–Majda measures, generalized Young measures, H-measures and microlocal defect measures [30, 31, 32, 33]. The covariance measures used below are deliberately coarser: they remain in physical space and retain only the quadratic information needed by the Yang–Mills and Higgs metric variations. In general relativity, high-frequency gravitational waves already produce an effective averaged stress tensor in the Isaacson expansion [34]; Burnett’s weak limit formulation and its modern developments give a rigorous continuation of this effective-stress perspective [35, 36]. Spinors introduce an additional metric-dependence through the spin bundle and Dirac operator, whose variation can be organized using the metric-spinor constructions of Bourguignon–Gauduchon [37] and the universal spinor-bundle framework of Müller–Nowaczyk [38]. We work throughout on a torsion-free metric branch; independently varying the Cartan connection would instead lead generically to Einstein–Cartan dynamics in the presence of spin current [39, 40].

For this reason the relevant convergence object is the *complete physical first variation*. At cutoff h , let z_h^d denote the finite configuration and $z_h = \mathcal{R}_h z_h^d$ its reconstructed field tuple. The finite first variation evaluated on a lifted physical test must approach the continuum first variation evaluated on z_h , while the finite common-action residual must vanish in the same physical normalization. If the native action covector has squared residual R_h , the lift of a unit physical test has source norm at most L_h , and the identification error is e_h , then

$$\|\mathcal{F}_h\|_{(\mathcal{V}_K^r)^*} \leq L_h \sqrt{R_h} + e_h. \quad (1.2)$$

Thus the natural stationarity condition is $L_h \sqrt{R_h} + e_h \rightarrow 0$, not merely $R_h \rightarrow 0$.

Main results. We first prove a regulator-independent subsequential variational closure theorem in a reduced action topology. For any finite-interface sequence satisfying the stated reduced compactness, first-variation consistency, and physically scaled stationarity hypotheses, every cutoff sequence has a subsequence whose reconstructed fields solve

$$\mathbf{G}_{\mu\nu}(g) + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{SM}}, \quad \mathcal{E}_A = \mathcal{E}_H = \mathcal{E}_\Psi = \mathcal{E}_{\bar{\Psi}} = 0 \quad (1.3)$$

in distributions. No candidate continuum solution, limiting stress tensor, or summable adjacent-cutoff transport is assumed. It appears as one additional condition that upgrades subsequential closure to a unique route-independent limit; uniqueness in a prescribed Cauchy class gives an alternative without summable transport.

The compactness assumptions can be obtained in several ways. On a regular branch, Sobolev bounds imply strong bosonic compactness, while Lorentzian symmetric-hyperbolic stability gives strong spinor H^1 compactness. At lower regularity, compact screens control the nonlinear bosonic packets. The reduced closure criterion further shows that strong compactness of the covariant Higgs gradient, together with $A_h \rightarrow A$ in L^4 , already closes the critical Higgs endpoint, whereas bounded weak H^1 spinors suffice for the complete bilinear fermionic first variation. On the geometric side,

strong H^1 convergence of the coframe on a common compact nondegenerate chart is sufficient for continuity of the complete first variation; no strong L^∞ convergence of the coframe is needed for zero-defect closure.

We also prove stronger quantitative results for an explicit finite local gravity–Yang–Mills–Higgs–Dirac action. A small finite Euler row, controlled initial and harmonic-gauge mismatch, and measured high-frequency leakage imply quantitative convergence of the actual reconstructed fields, curvature, and complete Standard-Model stress. On a regular data-centred branch, positive full-record reader control upgrades the finite Euler row to the differentiated physical forcing topology; a small composite initial constraint residual is then corrected only at the initial slice, and the physical Cauchy problem produces a unique exact Einstein–Standard-Model solution shadowing the observed record on the certified slab. Thus no exact future comparison trajectory is supplied in advance. The same-source refinement shows how the required finite budgets can be obtained simultaneously on one actually observed record without replacing that record by an expectation or a filtered configuration.

Finally, when the bosonic compactness required by the classical theorem fails, the weak limit retains the unresolved nonlinear information as explicit Yang–Mills and Higgs stress defects. The classical Einstein–Standard-Model system is therefore identified as the zero-defect branch of a broader weak-limit theory rather than being obtained by formally discarding oscillation or concentration. At the critical four-dimensional Yang–Mills energy endpoint we obtain this defect limit modulo cutoff-dependent local internal gauges, using an equivariant physical-test topology. For the explicit local action, a gauge-invariant Wilson translation screen of the literal curvature and Higgs-gradient packets yields strong local L^2 packet compactness and hence the zero-defect branch without a separate packet-energy uniform-integrability hypothesis. For minimal coupling, the timelike energy contraction independently characterizes vanishing of the complete bosonic stress defect in terms of strong curvature and covariant-gradient compactness. For the explicit finite Higgs packet we also isolate a strictly stronger higher-integrability certificate: the positive companion kinetic density is uniformly comparable to $|K_h^H|^2$, so its $L^{4/3}$ bound is equivalent to an $L^{8/3}$ bound for the covariant Higgs difference. Strong L^2 compactness alone yields only L^1 convergence of this positive density and does not imply the stronger endpoint.

The results are classical and finite-to-continuum in scope. They do not address the construction of a renormalized quantum Standard Model coupled to gravity, nor do they assert that every microscopic dynamics realizes the finite interface or selects the controlled branch. The analysis instead provides a portable closure theorem: any finite gravity–matter realization satisfying the stated structural and analytic hypotheses has the corresponding classical Einstein–Standard-Model system as its variational continuum limit.

Organization. [Section 2](#) specifies the finite interface, the continuum action and its physical variations. [Section 3](#) states the reduced closure theorem before developing its two endpoint mechanisms and proof. [Section 4](#) passes from the joint weak-limit identity to positive defects, critical gauge-quotient closure and the finite Wilson-screen criterion. [Section 5](#) gives the quantitative finite-action realization and culminates in the data-centred exact-shadow theorem. [Section 6](#) discusses interpretation and scope. The appendices retain the supporting estimates and realization-specific refinements; the guide preceding them identifies their dependencies.

2 Finite Einstein–Standard-Model interface

The analytic problem begins with finite data carrying one common gravity–matter action. We first separate these structural data from the reconstruction and convergence assumptions, then specify the continuum variation that the finite action must approximate. Renewal Geometry supplies one structural realization; its representation and source identities are recorded in [Section D.2](#) and are not compactness hypotheses.

Definition 2.1 (Finite classical Einstein–Standard-Model interface). At cutoff h , a finite classical Einstein–Standard-Model interface $\mathcal{J}_h$ consists of the following data.

- (I1) A finite configuration space $\mathcal{Q}_h$ with a reconstructed oriented, time-oriented Lorentzian carrier and local coframe variables, together with a fixed structural realization on a smooth comparison cylinder M whenever the reconstruction map below is applied.
- (I2) A fixed finite-rank Standard-Model bundle type with

$$G_{\text{SM}} = S(\text{U}(3) \times \text{U}(2)) \cong [\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)]/\mathbb{Z}_6, \quad (2.1)$$

the chiral fermion representations of [Table 2](#), one rank-three generation factor, the Higgs doublet, and any optional neutral/Majorana blocks declared as part of the branch.

- (I3) A gauge-covariant finite Dirac/Yukawa incidence operator $D_{F,h}$ on that fixed structural carrier and a coefficient bank $\theta_h = (\kappa_h, \Lambda_h, g_{1,h}, g_{2,h}, g_{3,h}, \lambda_{H,h}, v_{H,h}, \mathbf{Y}_h)$.
- (I4) A differentiable common finite action

$$\mathcal{S}_h = \mathcal{S}_{g,h} + \mathcal{S}_{\text{SM},h}$$

with one physical relative normalization between gravity and matter and with the corresponding finite gauge/relabeling covariance on the declared branch.

- (I5) A positive source geometry for physical variations, including a faithful source quotient or equivalent positive Gram on the retained variation directions. This is used only through the physical test-lift estimates stated explicitly below; no algebraic commutant gap is treated as an analytic compactness estimate.

No reconstruction-to-continuum map, compactness, stationarity, stress convergence, or field equation is included in this structural definition. Reconstruction maps and physical test lifts are introduced separately in [Definition 2.5](#).

2.1 Fields, bundles and physical tests

The fixed representation packet and its covariant Yukawa blocks are given in [Table 2](#) and [Lemma D.6](#). The analytic arguments below use their finite rank, unitarity of the internal representations and the affine Higgs dependence of the Yukawa map, rather than the mechanism by which that packet was obtained.

We work on an oriented, time-oriented spin cylinder

$$M = (0, T) \times \Sigma, \quad \partial\Sigma = \emptyset, \quad (2.2)$$

where Σ is a compact smooth three-manifold. All conclusions are local on compact subcylinders. There is no physical timelike boundary in the main theorem, and test variations vanish near the

initial and final time slices. A boundary extension requires its own flux and trace estimates; it is not implicit in (2.2).

Choose a smooth auxiliary Riemannian metric r and density dV_0 on M . All L^p , Sobolev, and Hilbert-space norms refer to these positive comparison data and fixed smooth bundle metrics. Lorentzian contractions occur only in the signed action. On a compact nondegenerate coframe chart the different positive comparison norms are uniformly equivalent. No positivity of the Lorentzian action is assumed.

Let $P \rightarrow M$ be a fixed principal G_{SM} bundle of the type specified by Definition 2.1. The chiral fermion bundle, generation factor, and Higgs bundle E_H are the corresponding associated bundles. We write

$$z = (e, A, H, \Psi, \bar{\Psi}), \quad g_{\mu\nu} = \eta_{ab} e_\mu^a e_\nu^b, \quad \eta = \text{diag}(-1, 1, 1, 1). \quad (2.3)$$

We use the curvature convention $[\nabla_\mu, \nabla_\nu]v^\rho = R^\rho_{\sigma\mu\nu}v^\sigma$ and $\text{Ric}_{\sigma\nu} = R^\rho_{\sigma\rho\nu}$. The coframe and spinor expressions are local in a fixed finite atlas, with compatible frame changes on overlaps. A global coframe is not required. The fields $\Psi, \bar{\Psi}$ are classical realified spinor variables, varied independently before imposing the physical duality relation. They are not a Grassmann integration measure.

Put

$$F_A = dA + A \wedge A, \quad K = D_A H = dH + \rho_H(A)H. \quad (2.4)$$

The spin derivative $\nabla^{e,A}$ uses the Levi-Civita spin connection of e and the retained gauge representation. The map $\mathcal{M}_Y(H)$ is the finite Dirac/Yukawa coupling on the fixed fermion carrier. In real Higgs coordinates it is affine, with a possible constant Majorana block, and satisfies

$$\|\mathcal{M}_Y(H)\| \leq C_Y(1 + |H|), \quad \|D_H \mathcal{M}_Y(H)[\eta_H]\| \leq C_Y|\eta_H|. \quad (2.5)$$

Its covariance is part of the finite incidence interface. If the optional Majorana sector is retained, the fermion carrier includes its charge-conjugate blocks and the realified bilinear is understood on that doubled carrier with its fixed normalization.

Definition 2.2 (Physical variation space). For a compact region $K \Subset M$, let $\mathcal{V}_K$ consist of smooth compactly supported tuples

$$v = (k, a, \eta_H, \eta_\Psi, \eta_{\bar{\Psi}}), \quad k \in \mathcal{D}(M; \text{Sym}^2 TM), \quad a \in \mathcal{D}(M; T^*M \otimes \text{ad } P),$$

with the remaining entries in the indicated matter bundles. Thus $k^{\mu\nu} = \delta g^{\mu\nu}$. Fix an integer $r_0 \geq 4$ and give this space the sum of the C^{r_0} norms in the chosen atlas. Its completion is denoted $\mathcal{V}_K^{r_0}$. More generally, $\mathcal{V}_K^r$ denotes the completion in the C^r test norm for an integer $r \geq 1$; r_0 remains the fixed regulator convention. A determining core means a subspace dense in this test space, with the stated uniform bounds extending to its completion.

The symmetric coframe lift of k is

$$\dot{e}_\mu^a(k) = -\frac{1}{2}e_\alpha^a g_{\mu\beta} k^{\alpha\beta}. \quad (2.6)$$

It gives $\dot{g}_{\mu\nu} = -g_{\mu\alpha} g_{\nu\beta} k^{\alpha\beta}$. Spinor components are transported in the same local spin-frame identification. Antisymmetric coframe variations are local Lorentz gauge directions, not additional metric tests.

2.2 The action and its complete metric variation

Let $\theta = (\kappa, \Lambda, g_1, g_2, g_3, \lambda_H, v_H, \mathbf{Y})$ be a fixed finite coefficient bank, with $\kappa, g_j, \lambda_H > 0$. The matter action is

$$\mathcal{S}_{\text{SM},\theta}(z) = \int_M \mathcal{L}_{\text{SM},\theta}(z) dV_g, \quad (2.7)$$

$$\begin{aligned} \mathcal{L}_{\text{SM},\theta} &= -\frac{1}{4} \langle \mathbf{F}_{\mu\nu}, \mathbf{F}^{\mu\nu} \rangle_g - g^{\mu\nu} \text{Re} \langle D_\mu H, D_\nu H \rangle - V(H) + \mathcal{L}_D, \\ V(H) &= \lambda_H (|H|^2 - v_H^2)^2, \\ \mathcal{L}_D &= \text{Re} \left\{ \frac{i}{2} [\bar{\Psi} \gamma^\mu \nabla_\mu^{e,A} \Psi - (\nabla_\mu^{e,A} \bar{\Psi}) \gamma^\mu \Psi] - \bar{\Psi} \mathcal{M}_Y(H) \Psi \right\}. \end{aligned} \quad (2.8)$$

The positive invariant Lie-algebra metric in the first term has the three coefficients g_j^{-2} . Pairings in the fermionic expression include all retained chiral and generation indices. The gravitational action is

$$\mathcal{S}_g(g) = \frac{1}{2\kappa} \int_M (\text{Scal}(g) - 2\Lambda) dV_g, \quad \mathcal{S}_\theta = \mathcal{S}_g + \mathcal{S}_{\text{SM},\theta}. \quad (2.9)$$

Only compactly supported variations are used. At limited regularity the gravitational first variation is defined by the first-derivative bulk expression obtained by integrating the torsion-free Palatini action by parts; [Section A.5](#) fixes this definition. It agrees with (2.9) on smooth fields up to the appropriate boundary term.

Remark 2.3 (Metric gravity versus independent Cartan gravity). In (2.9) the spin connection is the torsion-free function $\omega(e)$, including in the matter action. This is a declared second-order branch. If instead ω is varied independently in the total Palatini–fermion action, its Euler equation contains the spin current. Eliminating that connection generally produces Einstein–Cartan interactions, not (2.9). The spinless Palatini elimination in the Lorentzian construction [24] does not establish torsionlessness for a newly coupled spin-current sector. [Section A.5](#) retains the two alternatives explicitly.

The Hilbert stress convention is

$$D_g \mathcal{S}_{\text{SM},\theta}(z)[k] = -\frac{1}{2} \int_M T_{\mu\nu}^{\text{SM}}(z) k^{\mu\nu} dV_g. \quad (2.10)$$

For smooth fields the gauge and Higgs contributions are

$$T_{\mu\nu}^{\text{YM}} = \langle \mathbf{F}_{\mu\alpha}, \mathbf{F}_\nu{}^\alpha \rangle_g - \frac{1}{4} g_{\mu\nu} \langle \mathbf{F}_{\alpha\beta}, \mathbf{F}^{\alpha\beta} \rangle_g, \quad (2.11)$$

$$T_{\mu\nu}^H = 2 \text{Re} \langle D_\mu H, D_\nu H \rangle - g_{\mu\nu} (|D_A H|_g^2 + V(H)). \quad (2.12)$$

The Dirac–Yukawa contribution is always taken from the complete metric variation of (2.8). In particular, the spin-connection variation is not dropped by holding a metric-dependent Dirac operator fixed.

At the reduced regularity, the gravitational distribution is fixed by the local first-derivative representative

$$\frac{1}{2\kappa} [-dB(e) \wedge \omega(e) + B(e) \wedge \omega(e) \wedge \omega(e)] - \frac{\Lambda}{\kappa} dV_g,$$

where $B(e)$ is normalized by $B_{ab}(e) \wedge R^{ab}(\omega(e)) = \text{Scal}(g) dV_g$. Its complete variation contains only bounded coframe coefficients, first coframe derivatives and first derivatives of the metric test. The cutoff construction and agreement with the smooth Einstein variation are established in [Section A.5](#). This specifies the distribution used in the main theorem without assigning pointwise meaning to a rough curvature tensor.

Remark 2.4 (Chirality, finite covariance, and ultraviolet modes). The chiral labels specify the fixed continuum representation packet, not a claim that the finite difference operator is a doubler-free quantum chiral regulator. The lattice-chirality obstruction is a spectral issue [41, 42]; describing the variational spinors as classical does not remove it. For example, a centered free derivative has symbol $i \sin(h\xi_\mu)/h$, which is not uniformly coercive on the full lattice spectrum. The present argument instead imposes compactness, or the explicit measured-tail bounds, on the *complete reconstructed record*. It does not discard ultraviolet modes or infer spinor compactness from a small lattice Dirac row. Nor does it construct a chiral determinant, a fermionic integration measure, or quantum anomaly cancellation. Exact internal gauge covariance of the finite common action and consistency on the controlled classical branch are the finite statements used below.

2.3 Finite reconstruction and physically scaled stationarity

Definition 2.5 (A finite-interface regulator sequence). A finite-interface regulator sequence is a cofinal family $(\mathfrak{I}_h)_{h \downarrow 0}$ of structural interfaces from Definition 2.1, with finite configurations $z_h^d \in \mathcal{Q}_h$, together with analytic reconstruction maps giving smooth local fields

$$z_h = \mathcal{R}_h(z_h^d) = (e_h, A_h, H_h, \Psi_h, \bar{\Psi}_h)$$

on the same bundles over M , with the coefficient bank θ_h carried by $\mathfrak{I}_h$. For every $K \Subset M$ there are linear physical test lifts $\mathcal{I}_h : \mathcal{V}_K \rightarrow T_{z_h^d} \mathcal{Q}_h$, supported in a fixed slightly larger region. Define

$$c_h(K) = \sum_{b \in \{\text{g, SM}\}} \sup_{\|v\|_{C^{r_0}} \leq 1} |D\mathcal{S}_{b,h}(z_h^d)[\mathcal{I}_h v] - D\mathcal{S}_{b,\theta_h}(z_h)[v]|, \quad (2.13)$$

$$\varepsilon_h(K) = \sup_{\|v\|_{C^{r_0}} \leq 1} |D\mathcal{S}_h(z_h^d)[\mathcal{I}_h v]|. \quad (2.14)$$

The supremum is over $\mathcal{V}_K$. First-variation consistency means $c_h(K) \rightarrow 0$. Physical common-action stationarity means $\varepsilon_h(K) \rightarrow 0$.

The comparison in (2.13) is with the action evaluated on the *finite reconstructed fields*, not on their prospective limit. It can be checked by local interpolation, quadrature, and differentiated stencil estimates. It therefore does not assume the continuum Euler equation. Section D.3 supplies a concrete smooth comparison realization. The existence of such estimates on a prescribed mesh must not be confused with their validity on every microscopic renewal complex.

Convergence of action values alone cannot replace (2.13): $s_n(x) = n^{-1} \sin(nx)$ converges uniformly to zero, whereas $s'_n(0) = 1$. Likewise, matching only normalized probabilities loses the affine action normalization needed for (2.14).

Proposition 2.6 (Scaled source control of physical stationarity). *Let a_h be a finite action covector on a positive source space with Gram G_h . Suppose*

$$\|a_h\|_{G_h^{-1}}^2 \leq R_h, \quad \|v_h(v)\|_{G_h} \leq L_h \|v\|_{C^{r_0}},$$

and

$$|D\mathcal{S}_h[\mathcal{I}_h v] - a_h[v_h(v)]| \leq e_h \|v\|_{C^{r_0}}.$$

Then $\varepsilon_h(K) \leq L_h \sqrt{R_h} + e_h$. In particular, the sufficient vanishing condition is $L_h \sqrt{R_h} + e_h \rightarrow 0$, not merely $R_h \rightarrow 0$.

Proof. Write the source pairing in the G_h metric and apply Cauchy–Schwarz:

$$|a_h[v_h(v)]| \leq \|a_h\|_{G_h^{-1}} \|v_h(v)\|_{G_h} \leq L_h \sqrt{R_h} \|v\|_{C^{r_0}}.$$

Adding the identification error and taking the supremum proves the claim. $\square$

The same estimate holds for an averaged source after Cauchy–Schwarz in the probability variable. An averaged Euler identity, however, is not a fieldwise equation for one limiting metric. Similarly, stationarity restricted to a constraint surface does not control a missing normal action row; [Section M.4](#) gives the precise decomposition. The reduced closure theorem is deterministic; [Corollary B.15](#) states the additional requirements for a realized probabilistic version.

2.4 The local finite realization

The regulator-independent theorem does not prescribe a stencil. The finite Wilson criterion and the quantitative results use the particular link/plaquette action S_h^{loc} specified in [Section D.1](#). Its Cartan and spin links share the same coframe-derived torsion-free connection, and all sectors retain the relative normalization of [\(2.9\)](#).

For that realization fix a periodic comparison box $\mathcal{Q}$ of period 2π in each coordinate, an odd integer n , and $h = 2\pi/n$. The physical cylinder lies strictly inside a buffered coordinate region; the auxiliary temporal seam imposes no physical periodic-time condition. The notation R_h^0 denotes the piecewise-constant reconstruction, $\mathcal{I}_h^{\text{trig}}$ the full trigonometric reconstruction, and $D_{\mu,h}^+$ the ordinary forward difference. Their comparison is proved in [Lemma E.3](#). For internal links U_μ , let $U_{\mu\nu}$ be the ordered plaquette product in [\(D.2\)](#). The literal packets are

$$F_{\mu\nu}^h = h^{-2} \text{Log } U_{\mu\nu}, \quad K_\mu^h = \frac{\rho_H(U_\mu(x))H(x + he_\mu) - H(x)}{h}.$$

Their logarithms are always taken on the declared admissible chart. The complete physical covector uses the positive mass metric with link tangent $h^{-1}\dot{U}_\mu U_\mu^{-1}$; its norm, the action and the packet norms are preserved by actual site gauges. These conventions do not filter or replace the finite record.

3 Variational closure

We now state the principal closure theorem. Its hypotheses are expressed in the topology of the complete first variation, rather than the topology of the primitive fields alone. The proof has three steps: extract the nonlinear packets, establish continuity of the complete first variation, and pass physically scaled finite stationarity to the limit. The two sector-specific steps are proved in [Section 3.2](#); the elementary product and screen estimates are collected in [Appendix A](#).

3.1 Reduced topology and the main theorem

The following convergence class records the different roles of first-order bilinear and second-order quadratic sectors. It is a sufficient action topology; no claim of optimal regularity for the Cauchy problem is intended.

Definition 3.1 (Reduced action convergence). On each compact smooth chart K , reduced action convergence means

$$\begin{aligned} e_h &\rightarrow e \quad \text{strongly in } H^1, & e_h(x), e(x) &\in \mathcal{K}_e \quad \text{a.e.}, \\ A_h &\rightarrow A \quad \text{in } L^4, & F_{A_h} &\rightarrow F_A \quad \text{in } L^2, \\ H_h &\rightarrow H \quad \text{in } L^2, & D_{A_h} H_h &\rightarrow D_A H \quad \text{in } L^2, \\ \Psi_h &\rightharpoonup \Psi \quad \text{in } H^1, & \bar{\Psi}_h &\rightharpoonup \bar{\Psi} \quad \text{in } H^1, \end{aligned} \tag{3.1}$$

where $\mathcal{K}_e$ is a fixed compact subset of one oriented, time-oriented, nondegenerate coframe chart, together with convergence of the coefficient bank in a compact physical parameter set. All bundle identifications are fixed. Strong $H^1 \cap L^4$ convergence of H_h follows from [Proposition 3.4](#); it is not an independent condition. The geometric condition is strong first-jet convergence on a bounded chart and does not assert $\|e_h - e\|_\infty \rightarrow 0$.

A common compact screen means the approximation property in [Lemma A.3](#); [Proposition A.4](#) gives a sufficient spatial-screen and time-regularity criterion. The next definition turns those estimates into a compactness certificate for the reduced topology.

Definition 3.2 (Reduced compactness certificate). A regulator sequence satisfies the reduced compactness certificate on $K \subseteq M$ when, after fixed local gauge and spin-frame identifications,

- (R1) the coframe values lie almost everywhere in one fixed compact nondegenerate coframe chart, the coframes are bounded in H^1 , and their first-derivative packet has a common compact screen in L^2 ;
- (R2) the connections are precompact in L^4 , and the curvature packet is bounded and has a common compact screen in L^2 ;
- (R3) the Higgs fields are bounded in L^2 , and the covariant-gradient packet is bounded and has a common compact screen in L^2 ;
- (R4) Ψ_h and $\bar{\Psi}_h$ are bounded in H^1 ; no spinor first-jet screen is required;
- (R5) the coefficient banks remain in a compact subset of the physical parameter region.

The first item deliberately requires only boundedness of the coframe values, not their precompactness in L^∞ . Rellich compactness supplies the zeroth-order L^2 compactness, while the screen supplies strong compactness of the first derivatives.

Theorem 3.3 (Reduced Einstein–Standard-Model variational closure). *Let $(\mathfrak{J}_h)$ be a finite-interface regulator sequence satisfying [Definition 3.2](#) on every compact region, first-variation consistency $c_h(K) \rightarrow 0$, and scaled common-action stationarity $L_h \sqrt{R_h} + e_h \rightarrow 0$. Then every cutoff sequence has a subsequence for which*

$$\begin{aligned} e_h &\rightarrow e & \text{strongly in } H^1, \\ A_h &\rightarrow A & \text{in } L^4, \quad F_{A_h} \rightarrow F_A & \text{in } L^2, \\ H_h &\rightarrow H & \text{in } H^1 \cap L^4, \quad D_{A_h} H_h \rightarrow D_A H & \text{in } L^2, \\ \Psi_h &\rightharpoonup \Psi & \text{in } H^1, \quad \bar{\Psi}_h \rightharpoonup \bar{\Psi} & \text{in } H^1. \end{aligned} \tag{3.2}$$

The coframe values of the subsequence and its limit remain in the same compact nondegenerate chart almost everywhere; uniform convergence of the coframes is not asserted. Along this subsequence the

complete reconstructed first variation converges in $(\mathcal{V}_K^{\tau_0})^*$ and the limit is a distributional solution of all classical Einstein–Standard-Model Euler equations. In particular, with the gravitational distribution defined by [Section A.5](#),

$$(\mathbf{G}(g) + \Lambda g)dV_g = \kappa T^{\text{SM}}(z)dV_g. \quad (3.3)$$

Thus neither strong L^∞ coframe convergence, an independent Higgs quartic compactness condition, nor strong spinor first-jet compactness is part of this closure criterion.

The conclusion is distributional stationarity of the classical, non-quantized field theory. It does not by itself assert smoothness or well-posedness of a Cauchy problem. In particular, the theorem needs strong compactness of the bosonic quantities entering quadratic stresses, but only weak first-jet control of the bilinear fermionic sector.

3.2 The two endpoint mechanisms

The apparent critical Higgs obstruction is removed by using the covariant gradient itself. Once the connections are compact in L^4 , that gradient controls the ordinary first derivative and closes the quartic potential.

Proposition 3.4 (Covariant-gradient compactness closes the Higgs endpoint). *Let $K \Subset M$ be a bounded smooth chart and let H_h be the smooth reconstructed Higgs fields. Suppose*

$$A_h \rightarrow A \quad \text{in } L^4(K), \quad \sup_h \|H_h\|_{L^2(K)} < \infty, \quad D_{A_h} H_h \rightarrow Y \quad \text{in } L^2(K).$$

Then (H_h) is precompact in $H^1(K)$. Every subsequential limit H satisfies $Y = D_A H$, and the convergence along that subsequence is strong in $H^1(K)$ and hence in $L^4(K)$. If $H_h \rightarrow H$ in $L^2(K)$ is already known, the whole sequence converges strongly in these spaces. In particular the quartic potential and the complete Higgs stress converge in L^1 without an independent Orlicz or $L^{4+\delta}$ hypothesis.

Proof. The closure of $\{A_h\} \cup \{A\}$ is norm compact in $L^4(K)$. Multiplication by a fixed L^4 coefficient is compact $H^1 \rightarrow L^2$: approximate the coefficient in L^4 by bounded functions and compose multiplication by the bounded approximation with the compact embedding $H^1 \Subset L^2$. Uniformity on the compact coefficient set therefore gives, for every $\epsilon > 0$,

$$\|BH\|_2 \leq \epsilon \|H\|_{H^1} + C_\epsilon \|H\|_2 \quad (B \in \overline{\{A_h\} \cup \{A\}}). \quad (3.4)$$

Since $dH_h = D_{A_h} H_h - \rho_H(A_h)H_h$, the ordinary H^1 norm and (3.4) give a uniform H^1 bound after the first term on the right is absorbed. Extract $H_h \rightharpoonup H$ in H^1 and $H_h \rightarrow H$ in L^2 .

The product identity

$$A_h H_h - A H = (A_h - A) H_h + A (H_h - H)$$

converges to zero in L^2 : the first term uses boundedness of H_h in L^4 , and the second uses compact multiplication by the fixed $A \in L^4$. Passing to distributions in the formula for dH_h identifies $Y = D_A H$. Applying the same graph estimate to $H_h - H$ gives strong H^1 convergence. The continuous embedding $H^1(K) \hookrightarrow L^4(K)$ in four dimensions gives strong L^4 convergence, and the quadratic and quartic stress terms then converge by the product estimates of [Lemma A.2](#). $\square$

The fermionic mechanism is different. Each differentiated term in the Dirac action has an undifferentiated spinor factor, so weak–strong pairing can pass the first variation. The metric dependence of the spin connection must be included in that argument, as in the following proposition.

Proposition 3.5 (Weak- H^1 continuity of the complete fermionic first variation). *On $K \Subset M$, suppose the coframes e_h, e take values almost everywhere in one compact nondegenerate coframe chart and*

$$\begin{aligned} e_h &\rightarrow e \quad \text{strongly in } H^1, & A_h &\rightarrow A \quad \text{in } L^4, \\ H_h &\rightarrow H \quad \text{in } L^4, & \Psi_h &\rightarrow \Psi \quad \text{in } H^1, \\ & & \bar{\Psi}_h &\rightarrow \bar{\Psi} \quad \text{in } H^1, \end{aligned}$$

and let the finite Yukawa/Majorana coefficients converge. Then the complete first variation of the Dirac–Yukawa action (2.8), including its metric and spin-connection variation, converges against every physical test tuple. For every integer $r \geq 2$,

$$\|D\mathcal{S}_{D,h} - D\mathcal{S}_D\|_{(\mathcal{V}_K^r)^*} \longrightarrow 0. \quad (3.5)$$

The same conclusion holds if either or both of the gauge and Higgs hypotheses are weakened to strong L^2 convergence with uniform L^4 bounds. In particular $A_h \rightarrow A$ strongly in L^2 with $\sup_h \|A_h\|_4 < \infty$ suffices for the fermionic sector. No strong convergence of the spinor first derivatives, and no uniform convergence of the coframes, is required. In particular the fermionic Hilbert stress has no independent defect in this class.

Proof. Weak H^1 convergence and Rellich compactness give strong L^2 convergence of both undifferentiated spinors and uniform L^4 bounds. Their bilinears converge strongly in L^1 and remain bounded in L^2 , hence converge weakly in L^2 . A differentiated Dirac term pairs one weakly convergent L^2 derivative with one strongly convergent L^2 undifferentiated spinor. By Lemma A.2, the coframe Clifford coefficients are bounded, converge in measure, and preserve this strong L^2 factor. The spin connection and its metric variation converge strongly in L^2 by (A.7). Gauge coefficients need only converge strongly in L^2 , as does the affine Yukawa coefficient. These coefficients pair with weakly convergent L^2 spinor bilinears. In a spinor variation a bilinear is replaced by one spinor and one bounded test; this factor is strongly convergent in L^2 . The L^4 bounds retain uniform control of the covariant products. This proves both relaxed hypotheses. Thus every term in the complete first variation converges for a fixed smooth test, including the complete metric–spin-connection chain.

The same expansions give a cutoff-independent bound by $C\|v\|_{C^1}$ on bounded H^1 spinor sets and on the fixed compact coframe chart. The C^r unit ball supported in a fixed compact set is totally bounded in C^1 for $r \geq 2$. A finite-net argument upgrades pointwise convergence to the dual norm (3.5). $\square$

3.3 Continuity and proof of closure

The two endpoint propositions supply precisely the sectorwise convergence summarized in Table 1. The remaining gravity and Yang–Mills terms are bounded algebraic coefficients multiplying strongly convergent quadratic packets. No limiting stress tensor is supplied as independent data.

Proposition 3.6 (Continuity in the reduced action topology). *Under Definition 3.1, for every integer $r \geq 2$,*

$$\|D\mathcal{S}_{\theta_h}(z_h) - D\mathcal{S}_\theta(z)\|_{(\mathcal{V}_K^r)^*} \longrightarrow 0. \quad (3.6)$$

Table 1: Sectorwise hypotheses for reduced variational closure. All norms are local positive comparison norms. The strong curvature and gradient requirements concern nonlinear packets, not independent tensor variables.

Sector	Convergence used	Consequence
Gravity	$e_h \rightarrow e$ strongly in H^1 on one compact nondegenerate coframe chart	Continuity of the distributional Einstein first variation.
Yang–Mills	$A_h \rightarrow A$ in L^4 , $F_{A_h} \rightarrow F_A$ in L^2	Identification of curvature and strong L^1 convergence of gauge stress.
Higgs	$H_h \rightarrow H$ in L^2 , $D_{A_h} H_h \rightarrow D_A H$ in L^2	Strong $H^1 \cap L^4$ convergence and closure of the quartic potential.
Fermions	Weak H^1 convergence of both spinor variables	Continuity of the complete bilinear first variation; no independent fermionic stress defect.

The conclusion holds separately for gravity, Yang–Mills, Higgs and the complete Dirac–Yukawa contribution, including the metric–spin connection chain. In particular the limiting matter metric variation is the Hilbert stress of the limiting fields, not an independently prescribed tensor.

Proof. [Proposition 3.4](#) gives strong L^4 convergence of H_h . The curvature, covariant-gradient and quartic products then converge in L^1 by [Lemma A.2](#) and [\(A.11\)](#). The first-order gravitational representative [\(A.12\)](#) is a finite sum of bounded smooth coframe coefficients multiplying products of first coframe derivatives, plus terms linear in a first derivative of the metric test. The derivative packets converge strongly in L^2 , while the coefficients remain uniformly bounded and converge in measure; [Lemma A.2](#) therefore passes every term uniformly on bounded C^1 test sets. The Yang–Mills and Higgs metric rows have the same bounded-coefficient/strong-product structure. The complete fermionic contribution follows from [Proposition 3.5](#). A finite-net argument on the C^r unit ball yields [\(3.6\)](#). $\square$

It remains to combine this continuity with compactness and finite stationarity. The abstract triangle inequality is recorded in [Lemma A.1](#); the following proof applies it to the complete coupled action.

Proof of Theorem 3.3. On a compact chart, the bounded H^1 coframes are precompact in L^2 . The common screen lemma gives, after extraction, strong convergence of their first derivatives. Hence $e_h \rightarrow e$ strongly in H^1 . After a further subsequence the coframes converge almost everywhere, so the closed compact coframe range is preserved by the limit and all algebraic coefficient and inverse bounds remain uniform. The same screen argument gives strong L^2 convergence of the curvature and covariant Higgs-gradient packets, while the assumed zeroth-order compactness gives $A_h \rightarrow A$ in L^4 . The identity $dA_h = F_{A_h} - A_h \wedge A_h$ identifies the curvature limit with F_A in distributions.

By [Proposition 3.4](#), the bounded L^2 Higgs family with strong covariant gradients has a subsequence converging strongly in $H^1 \cap L^4$, and its limiting gradient is $D_A H$. The two spinor families have weakly convergent H^1 subsequences, with strong L^2 convergence by Rellich compactness. Extract the finite parameter bank as well and use a diagonal argument over the fixed atlas and a countable compact exhaustion. This gives [\(3.2\)](#).

Let

$$\omega_h(K) = \|D\mathcal{S}_{\theta_h}(z_h) - D\mathcal{S}_\theta(z)\|_{(\mathcal{V}_K^{r_0})^*}.$$

It tends to zero by [Proposition 3.6](#). The abstract closure estimate and [Proposition 2.6](#) give

$$\|D\mathcal{S}_\theta(z)\|_{(\mathcal{V}_K^{r_0})^*} \leq L_h \sqrt{R_h} + e_h + c_h(K) + \omega_h(K) \longrightarrow 0. \quad (3.7)$$

Independent gauge, Higgs, spinor and dual-spinor tests give their Euler identities. An inverse-metric test gives the gravitational variation $(G(g) + \Lambda g)dV_g/(2\kappa)$ and the matter variation $-T^{\text{SM}}dV_g/2$, proving (3.3). The first-order gravitational representative fixes this identity at the stated metric regularity; no multiplication of an arbitrary curvature distribution by an uncontrolled coefficient is used. $\square$

Remark 3.7 (Scope of the weaker geometric topology). The preceding weakening concerns the zero-defect first-variation passage. It is not used to strengthen a theorem with a nonzero concentrating stress measure. A uniformly bounded coefficient that converges only in measure need not commute with a singular weak-star measure. Accordingly the defect results below retain the stronger geometric hypotheses stated there. This separates the regularity needed for the classical zero-defect branch from that needed to identify an unresolved measure-valued source.

3.4 Stress and uniqueness consequences

Corollary 3.8 (Reconstructed and operational stress in the reduced class). *Along a subsequence of [Theorem 3.3](#), the Yang–Mills and Higgs stress densities converge strongly in L^1 . The complete matter metric first variation converges in the dual C^r test norm for $r \geq 2$, hence in H^{-m} for every integer $m > 4$ on a fixed compact test region. For the operational stress*

$$\langle T_h, k \rangle = -2D\mathcal{S}_{\text{SM},h}[Z_h(k, 0, 0, 0, 0)],$$

sectorwise consistency in (2.13) gives convergence to $T^{\text{SM}}dV_g$ in $(\mathcal{V}_K^{r_0})^$ and in H^{-m} for $m > r_0 + 2$. Strong L^1 convergence of the fermionic stress is not asserted under weak spinor H^1 convergence.*

Proof. The first assertion follows from the quadratic and quartic product estimates. Apply [Proposition 3.6](#) to metric tests and use $H_0^m(K) \hookrightarrow C^r(K)$ for $m > r + 2$ in four dimensions. Taking $r = 2$ proves the reconstructed statement. The operational statement adds the separately controlled matter consistency error, which is measured in the C^{r_0} test norm. $\square$

Subsequential closure and selection of a unique continuum trajectory are separate assertions. A unique Cauchy class removes subsequence ambiguity provided its initial traces are identified independently.

Proposition 3.9 (Uniqueness without summable adjacent-cutoff transport). *Suppose the hypotheses of [Theorem 3.3](#) hold for the full regulator family with one limiting coefficient bank. Suppose further that all accumulation points, in fixed gauge and frame identifications, belong to a solution class in which the limiting Cauchy data determine a unique solution z_* . Assume convergence of the relevant Cauchy traces separately in that class. Then the entire family converges to z_* in reduced action convergence. No summability of adjacent-cutoff errors is needed.*

Proof. On bounded H^1 sets the weak spinor topology is metrizable because H^1 is a separable Hilbert space; the other factors in (3.1) are norm topologies. A countable exhaustion therefore metrizes the relevant local product topology on the bounded family. If convergence failed, a sequence outside a fixed neighborhood of z_* would have, by reduced compactness, a convergent subsequence. Its limit is a solution with the prescribed traces and hence equals z_* , a contradiction. Trace convergence is an additional hypothesis: spacetime compactness alone does not identify data on an initial slice. $\square$

The conditional Sobolev upgrade is [Corollary B.16](#). Strong spinor first-jet convergence, regular Sobolev criteria, summable cofinal transport and realized-random variants are retained in [Appendix B](#). They strengthen the conclusions on their stated branches without becoming hypotheses of the reduced theorem.

4 Weak limits and zero-defect closure

The main theorem identifies the classical limit when the nonlinear bosonic packets are compact. We next retain, rather than discard, the information lost when those packets converge only weakly. The order of the argument is: identify the joint defect source, characterize its vanishing by positive energies, pass to cutoff-dependent gauge representatives, and finally give a finite criterion that restores strong compactness.

4.1 The joint weak-limit alternative

On the weak-limit branch, assume compatible coframes converge strongly in H^1 and uniformly to continuous nondegenerate coframes, with uniform inverse bounds. Let $A_h \rightarrow A$ in L^4 , let the Higgs fields converge weakly in H^1 , and identify the weak L^2 limits of F_{A_h} and $D_{A_h}H_h$ with F_A and D_AH . Both spinor variables remain weakly bounded in H^1 , and the parameters converge in the physical region. Uniform geometric convergence is retained here because a singular stress measure must be multiplied by a continuous limiting coefficient.

[Theorem C.2](#) and [Proposition C.4](#) construct the measure limits

$$T_h^{\text{YM}} dV_{g_h} \rightharpoonup^* T^{\text{YM}} dV_g + \mathfrak{S}_{\text{YM}}, \quad T_h^H dV_{g_h} \rightharpoonup^* T^H dV_g + \mathfrak{S}_H,$$

where $\mathfrak{S}_H = \mathfrak{K}_H - \lambda_H g \nu_H$: $\mathfrak{K}_H$ is the kinetic stress defect and $\nu_H \geq 0$ the quartic concentration measure. Their construction and the matter-equation passage are proved in [Appendix C](#). They assemble into the following limit identity.

Theorem 4.1 (Joint matter-defect Einstein limit). *Assume the critical Higgs hypotheses of [Theorem C.2](#), the Yang–Mills hypotheses of [Proposition C.4](#), $A_h \rightarrow A$ strongly in L^4 , the compatible coframe and weak- H^1 spinor hypotheses of [Proposition C.5](#), and convergence of the gravitational first variation. The latter follows in particular from the strong first-coframe-jet hypothesis of the reduced theorem. If the operational metric residual and its reconstruction-consistency error vanish, then*

$$(\mathbf{G}(g) + \Lambda g) dV_g = \kappa (T^{\text{SM}} dV_g + \mathfrak{S}_{\text{YM}} + \mathfrak{S}_H). \quad (4.1)$$

If additionally $g_h \rightarrow g$ in C^1 , the total reconstructed stress has the uniform measure bound and vanishing conservation residual of [Lemma C.3](#), and the regular limiting matter stress satisfies its Noether identity, then $\mathfrak{S}_{\text{YM}} + \mathfrak{S}_H$ is distributionally conserved. These additional conditions concern conservation only, not (4.1). In particular, the classical Einstein–Standard-Model equation is the zero-defect branch of the weak-limit identity.

Proof. Test the finite metric equation against a compactly supported smooth symmetric tensor and pass the gravitational, regular fermionic, Yang–Mills, and Higgs terms separately using [Proposition C.4](#) and [Theorem C.2](#) and the complete fermionic continuity in [Proposition C.5](#). This gives (4.1). Under the additional C^1 hypothesis, [Lemma C.3](#) passes the finite conservation identity; subtraction of the separately valid regular Noether identity gives conservation of the sum of the defects. Neither conservation nor the Ward identity forces that sum to vanish. $\square$

The matter equations and the metric equation have different weak-limit requirements. The former pass by weak–strong pairing, whereas metric variation retains two weak bosonic factors. The critical bubble and null-wave examples in [Examples C.11](#) and [C.12](#) illustrate respectively concentration and oscillatory kinetic stress. Neither matter stationarity nor conservation alone removes either phenomenon.

4.2 Positive defects and exact zero-defect characterization

The signed tensor defect is controlled by positive covariance measures. For a weak L^2 packet $Y_h \rightharpoonup Y$, write

$$Y_h \otimes Y_h dV_0 \rightharpoonup^* Y \otimes Y dV_0 + Q_Y, \quad \mu_Y = \text{tr } Q_Y \geq 0.$$

The covariance measure is positive semidefinite, and on a compact domain including its boundary, $\mu_Y = 0$ is equivalent to strong L^2 convergence. [Lemma C.6](#) proves this assertion and its bounded-contraction estimate. Applied to curvature and covariant Higgs gradients, it yields the measures μ_{YM} and $\mu_{\nabla H}$.

Apply this lemma to the realified packets $Y_h^{\text{YM}} = F_{A_h}$ and $Y_h^H = D_{A_h} H_h$, using the fixed positive comparison metrics. Write their trace defects as μ_{YM} and $\mu_{\nabla H}$. The volume factors and Lorentzian contractions in the Hilbert stress are included in B_h in [\(C.11\)](#). On each compact nondegenerate chart,

$$|\mathfrak{S}_{\text{YM}}| \leq C\mu_{\text{YM}}, \quad |\mathfrak{K}_H| \leq C\mu_{\nabla H}, \quad |\mathfrak{S}_H| \leq C(\mu_{\nabla H} + \nu_H). \quad (4.2)$$

Thus positive packet defects control every component of the stress defects. The local formulation uses compactly supported tests on a larger chart; vanishing there yields strong convergence on smaller compact sets. Including the boundary in the compact-domain formulation prevents undetected loss of mass at the edge of that domain.

Theorem 4.2 (Timelike positivity and exact zero-defect equivalence). *Assume the weak Yang–Mills and Higgs hypotheses of [Theorem 4.1](#), with positive limiting gauge coefficients and $\lambda_H > 0$, and let n be a continuous future-directed unit timelike vector field for g on the compact chart. There are $c, C > 0$ such that, as scalar Radon measures,*

$$c(\mu_{\text{YM}} + \mu_{\nabla H} + \nu_H) \leq (\mathfrak{S}_{\text{YM}} + \mathfrak{S}_H)(n, n) \leq C(\mu_{\text{YM}} + \mu_{\nabla H} + \nu_H). \quad (4.3)$$

Consequently the following conditions are equivalent along the extracted sequence on the compact domain:

$$\begin{aligned} \mathfrak{S}_{\text{YM}} + \mathfrak{S}_H &= 0; \\ (\mathfrak{S}_{\text{YM}} + \mathfrak{S}_H)(n, n) &= 0; \\ \mu_{\text{YM}} = \mu_{\nabla H} = \nu_H &= 0; \\ F_{A_h} \rightarrow F_A \text{ in } L^2, \quad D_{A_h} H_h \rightarrow D_A H \text{ in } L^2, \quad H_h \rightarrow H \text{ in } L^4. \end{aligned} \quad (4.4)$$

In particular, independent Yang–Mills and Higgs stress defects cannot cancel to produce a zero total bosonic defect in this minimally coupled class.

Proof. Choose a local orthonormal frame (n, e_1, e_2, e_3) and put $E_i = F(n, e_i)$ and $B_i = \frac{1}{2}\varepsilon_{ijk}F(e_j, e_k)$. Direct substitution into [\(2.11\)](#) gives

$$T^{\text{YM}}(n, n) = \frac{1}{2} \sum_i (|E_i|_g^2 + |B_i|_g^2). \quad (4.5)$$

This is positive definite on the complete six-component two-form packet. Likewise, the kinetic part of (2.12) obeys

$$T^{H,\text{kin}}(n, n) = |D_n H|^2 + \sum_i |D_{e_i} H|^2. \quad (4.6)$$

Both forms, multiplied by the positive volume density, are uniformly comparable on the compact chart to the respective reference squared norms. Contracting their covariance measures therefore gives nonnegative measures bounded above and below by the corresponding trace defects. The potential defect is $-\lambda_H g \nu_H$, whose contraction with $n \otimes n$ is $\lambda_H \nu_H$. The lower-degree potential terms converge strongly and contribute no defect. Uniform convergence of g_h , its inverse and the parameters justifies all coefficient replacements by Lemma C.6. A finite frame cover gives (4.3).

Vanishing of the full tensor implies vanishing of its timelike contraction. The lower bound then forces all three positive measures to vanish. Lemmas C.1 and C.6 identify this with the three strong convergences in (4.4). Those convergences, by the quadratic and quartic estimates, make the tensor defects vanish, closing the cycle. $\square$

Corollary 4.3 (Independent compactness certificates). *Under the same hypotheses, $\mu_{\nabla H} = 0$ already implies $H_h \rightarrow H$ in $H^1 \cap L^4$ and hence $\nu_H = 0$. Thus $\mu_{\text{YM}} = \mu_{\nabla H} = 0$ is sufficient and necessary for the zero-defect branch in Theorem 4.2; a separate quartic certificate is redundant once the covariant gradient is compact.*

Proof. Rellich compactness provides $H_h \rightarrow H$ in L^2 along the weak H^1 subsequence. Trace-defect zero gives strong convergence of the covariant gradient. Apply Proposition 3.4, followed by Lemma C.1. $\square$

Remark 4.4 (Scope of the converse). The converse uses the coercive *timelike Hilbert-energy contraction*, not the Lorentzian trace of the stress or the signed action density. For example the four-dimensional Yang–Mills stress is trace free even when $\mu_{\text{YM}} \neq 0$. Conservation alone also does not imply vanishing. The conclusion does not include nonminimal improvement terms, a weakly converging gravitational first jet, or an additional fermionic defect outside the weak- H^1 class. Those cases require their own defect analysis.

Thus the minimal theory has a genuine two-sided criterion: zero total bosonic stress defect means that the unresolved positive packet energies vanish. The nonminimal improvement and the stronger Higgs $L^{8/3}$ reserve are separate refinements, treated in Sections C.6 and D.1.1.

4.3 Critical gauge-quotient closure

The preceding defect identities were formulated after choosing local bundle representatives. At the four-dimensional Yang–Mills energy endpoint it is preferable to state the compactness and first-variation hypotheses modulo the internal gauge group itself. Two additional ingredients are needed. First, cutoff-dependent gauge normalizations must be paired with a test topology that is exactly equivariant. Second, at critical curvature regularity one must separate concentration control from the stronger compactness needed to remove quadratic stress oscillations.

Fix on each local spinor–generation bundle a smooth positive-metric reference connection ∇^{ref} acting trivially on the internal gauge factor and write

$$\nabla^{\text{ref},A} \Psi := \nabla^{\text{ref}} \Psi + \rho(A) \Psi. \quad (4.7)$$

For a compactly supported physical test $v = (k, a, \eta_H, \eta_\Psi, \eta_{\bar{\Psi}})$ define

$$\begin{aligned} \mathbf{n}_z(v) := & \|k\|_{W^{1,\infty}} + \|a\|_{L^\infty} + \|D_A a\|_{L^2} \\ & + \|\eta_H\|_{L^\infty} + \|D_A \eta_H\|_{L^2} \\ & + \|\eta_\Psi\|_{L^\infty} + \|\nabla^{\text{ref}, A} \eta_\Psi\|_{L^2} + \|\eta_{\bar{\Psi}}\|_{L^\infty} + \|\nabla^{\text{ref}, A} \eta_{\bar{\Psi}}\|_{L^2}. \end{aligned} \quad (4.8)$$

All norms use the fixed positive comparison geometry.

The test size is exactly invariant under internal gauge transformations: $\mathbf{n}_{u \cdot z}(u \cdot v) = \mathbf{n}_z(v)$, and the complete first variation is bounded by $C_K \mathbf{n}_z(v)$ on bounded weak packets. The verification is [Lemma C.7](#). These facts allow tests in a cutoff-dependent normalized gauge to be pulled back without introducing derivatives of the normalizing gauge into the finite budget.

We say that a finite regulator has *equivariant consistency and stationarity budgets* $c_h^{\text{eq}}(K)$ and $\varepsilon_h^{\text{eq}}(K)$ if

$$|D\mathcal{S}_h(z_h^{\text{d}})[\mathcal{I}_h v] - D\mathcal{S}_{\theta_h}(z_h)[v]| \leq c_h^{\text{eq}}(K) \mathbf{n}_{z_h}(v), \quad (4.9)$$

$$|D\mathcal{S}_h(z_h^{\text{d}})[\mathcal{I}_h v]| \leq \varepsilon_h^{\text{eq}}(K) \mathbf{n}_{z_h}(v) \quad (4.10)$$

for every smooth compactly supported physical test. Because of [\(C.12\)](#), these inequalities can be applied to a fixed test in a cutoff-dependent normalized gauge by pulling that test back to the original finite record; no derivative of the normalizing gauge enters the budget.

Definition 4.5 (Uniformly integrable critical curvature energy). A family of reconstructed connections has uniformly integrable critical curvature energy on a compact chart K if

$$\limsup_{\delta \downarrow 0} \sup_h \sup_{\substack{E \subset K \text{ measurable} \\ \text{vol}_0(E) \leq \delta}} \int_E |F_{A_h}|_r^2 dV_0 = 0. \quad (4.11)$$

This property is internal-gauge invariant.

Uniform integrability of critical curvature energy supplies a common small-energy cover and local Coulomb representatives by [Proposition C.8](#). This supplements the classical critical four-dimensional Yang–Mills compactness picture, where curvature concentration and bubbling can occur even for bounded-energy sequences [\[27, 28\]](#): uniform integrability rules out such concentration on the chosen compact chart, while a separate compactness mechanism is still needed to remove quadratic stress oscillation. Positive covariant graph bounds then give ordinary H^1 bounds for the matter variables in those representatives ([Lemma C.9](#)). The resulting theorem concerns the gauge quotient; it does not select a canonical representative.

Theorem 4.6 (Critical-energy quotient defect closure). *Let $(\mathfrak{I}_h)$ be a finite-interface regulator sequence. On every compact chart assume:*

- (Q1) *the coframes are precompact in $L^\infty \cap H^1$, uniformly nondegenerate, and the coefficient banks lie in a compact physical set;*
- (Q2) *$\sup_h \|F_{A_h}\|_2 < \infty$ and the curvature energies satisfy [\(4.11\)](#);*
- (Q3) *with $K_h = D_{A_h} H_h$, $\sup_h (\|H_h\|_2 + \|K_h\|_2) < \infty$;*
- (Q4) *the spinors satisfy the gauge-invariant comparison bounds*

$$\sup_h (\|\Psi_h\|_2 + \|\nabla^{\text{ref}, A_h} \Psi_h\|_2 + \|\bar{\Psi}_h\|_2 + \|\nabla^{\text{ref}, A_h} \bar{\Psi}_h\|_2) < \infty; \quad (4.12)$$

$$(Q5) \quad c_h^{\text{eq}}(K) + \varepsilon_h^{\text{eq}}(K) \rightarrow 0.$$

Then every cutoff sequence has a subsequence and cutoff-dependent local internal gauges such that, on smaller compact charts,

$$\begin{aligned} e_h &\rightarrow e && \text{in } L^\infty \cap H^1, \\ A_h &\rightharpoonup A && \text{in } W^{1,2}, \quad A_h \rightarrow A \quad \text{in every } L^q, \quad q < 4, \\ F_{A_h} &\rightharpoonup F_A && \text{in } L^2, \\ H_h &\rightharpoonup H && \text{in } H^1, \quad H_h \rightarrow H \quad \text{in every } L^q, \quad q < 4, \\ D_{A_h} H_h &\rightharpoonup D_A H && \text{in } L^2, \\ \Psi_h &\rightharpoonup \Psi, \quad \bar{\Psi}_h &\rightharpoonup \bar{\Psi} && \text{in } H^1. \end{aligned} \tag{4.13}$$

The Yang–Mills, Higgs, Dirac and dual-Dirac equations hold distributionally, and the metric equation is

$$(\mathbb{G}(g) + \Lambda g) dV_g = \kappa \left(T^{\text{SM}}(z) dV_g + \mathfrak{S}_{\text{YM}} + \mathfrak{S}_H \right). \tag{4.14}$$

If, in addition, $|D_{A_h} H_h|^2$ and $|H_h|^4$ are uniformly integrable, then the complete bosonic defect is an L^1 tensor density and the quartic concentration measure vanishes. The conclusion is a quotient statement: no fixed gauge representative is selected canonically.

Proof. Extract the geometric and parameter subsequence and apply [Proposition C.8](#) on a finite cover of each smaller compact chart. The covariant graph estimates give weak H^1 matter compactness, and strong subcritical convergence identifies the curvature and covariant derivatives. Pulling back fixed normalized tests preserves their $\mathfrak{n}$ -size, so the equivariant budgets pass the matter Euler rows. The relaxed fermionic continuity theorem supplies the complete fermionic metric limit. Only the quadratic bosonic metric terms and the quartic potential retain defects, giving (4.14). Uniform integrability makes the corresponding covariance measures absolutely continuous, and Vitali removes the quartic measure under the additional hypothesis. The complete sectorwise passage is given in [Section C.4](#). $\square$

4.4 The finite Wilson-screen criterion

The quotient theorem permits a defect. To exclude it directly from the explicit local finite action, we now screen the literal packets rather than a continuum surrogate. The crucial distinction is between scalar magnitude compactness, which follows from unitarity alone, and vector packet compactness, which also uses a controlled gauge representative.

For the explicit local action one can recognize the missing strong packet compactness directly on the same finite links that enter the action. Let $U_\mu(x)$ be the internal link of [Section D.1](#). For an integer $m > 0$ let

$$\mathcal{U}_{\mu,m}(x) := U_\mu(x) U_\mu(x + h e_\mu) \cdots U_\mu(x + (m-1) h e_\mu), \tag{4.15}$$

and use the inverse ordered product when $m < 0$. For a finite packet Y_h^{d} in a unitary representation define the Wilson shift

$$\mathbb{W}_{\mu,m}^U Y_h^{\text{d}}(x) := \rho(\mathcal{U}_{\mu,m}(x)) Y_h^{\text{d}}(x + m h e_\mu), \tag{4.16}$$

with $\rho = \text{Ad}$ for curvature. Put

$$\Omega_h(Y^{\text{d}}; \varrho) := \max_{\mu} \max_{0 < |m| h \leq \varrho} \left(h^4 \sum_{x \in Q'_{h,\mu,m}} |\mathbb{W}_{\mu,m}^U Y_h^{\text{d}}(x) - Y_h^{\text{d}}(x)|^2 \right)^{1/2}. \tag{4.17}$$

The endpoint gauges telescope in (4.15), so (4.17) is exactly site-gauge invariant. Throughout the native finite-packet statements below we use

$$\|Y_h\|_{p,h}^p = h^4 \sum_x |Y_h(x)|^p \quad (4.18)$$

for finite p , with the usual maximum norm for $p = \infty$.

Fix, independently of the record, a rooted spanning tree $\mathfrak{T}_h$ of the periodic comparison grid. Spanning- or maximal-tree gauges are standard tools for eliminating redundant lattice link variables [43]. The role of the tree here is more specific: it produces a gauge-invariant same-record logarithmic seed whose smallness can be fed into the exact finite Coulomb normalization below. For a vertex x , let $C_x(U)$ be the ordered product of the actual internal links along the tree path from the root to x , with inverses on reversed edges, and define the root-based closed words

$$W_\mu^\mathfrak{T}(x) = C_x(U) U_\mu(x) C_{x+h\epsilon_\mu}(U)^{-1}, \quad B_{\mu,h}^\mathfrak{T}(x) = h^{-1} \text{Log } W_\mu^\mathfrak{T}(x). \quad (4.19)$$

The seed is declared inadmissible when one of the required logarithms leaves the fixed conjugation-invariant logarithm chart. On the admissible set put

$$\mathfrak{c}_h^\mathfrak{T}(U) := \|B_h^\mathfrak{T}\|_{4,h} + \|\mathbb{F}_h\|_{2,h}. \quad (4.20)$$

Endpoint cancellation shows that this quantity is invariant under arbitrary finite site gauges, up to the irrelevant constant conjugation at the root. The construction and the associated exact finite Coulomb normalization are proved in [Appendix F](#).

Theorem 4.7 (Finite Wilson-screen zero-defect criterion). *Consider the explicit local action of [Section D.1](#) on the periodic buffered comparison grid, with conclusions restricted to an interior physical cylinder. Let*

$$\mathbb{F}_h = \{F_{\alpha\beta}^h\}_{\alpha<\beta}, \quad \mathbb{K}_h = \{K_\alpha^h\}_{\alpha=0}^3 \quad (4.21)$$

be its literal curvature and Higgs-gradient packets. In this theorem all packet norms and Wilson translation sums are taken over the entire periodic comparison box, including periodic shifts; only the final Euler tests are restricted to the interior physical cylinder. Assume:

- (F1) *the nodal coframes E_h take values in one fixed compact nondegenerate coframe chart, the families $R_h^0 E_h$ and $R_h^0 D_h^+ E_h$ are precompact in L^2 , the scaled connection margin [\(E.13\)](#) holds, and the complete unfiltered reconstructed coframes remain in the same compact chart;*
- (F2) *for one predeclared rooted tree the same-record certificate [\(4.20\)](#) obeys $\mathfrak{c}_h^\mathfrak{T}(U_h) \leq \epsilon_c$, where ϵ_c is the cutoff-independent threshold in [Theorem F.6](#);*
- (F3) *$\|\mathbb{F}_h\|_{2,h} + \|\mathbb{K}_h\|_{2,h} + \|H_h\|_{2,h}$ is bounded and*

$$\lim_{\varrho \downarrow 0} \limsup_{h \downarrow 0} [\Omega_h(\mathbb{F}_h; \varrho) + \Omega_h(\mathbb{K}_h; \varrho)] = 0; \quad (4.22)$$

- (F4) *the spinor and dual-spinor blocks have uniformly bounded positive internal-link graph norm [\(E.12\)](#), and the coefficient banks remain in a compact physical parameter set;*
- (F5) *the norm σ_h^{phys} of the complete finite action covector in the positive physical link/matter mass metric tends to zero.*

Then there are site gauges acting on the same finite records for which the normalized logarithmic connections are in exact discrete Coulomb gauge and, after extraction,

$$\begin{array}{ll}
e_h \rightarrow e & \text{strongly in } H_{\text{loc}}^1, \\
A_h \rightarrow A & \text{strongly in } L_{\text{loc}}^4, \\
F_{A_h} \rightarrow F_A & \text{strongly in } L_{\text{loc}}^2, \\
H_h \rightarrow H & \text{strongly in } H_{\text{loc}}^1 \cap L_{\text{loc}}^4, \\
D_{A_h} H_h \rightarrow D_A H & \text{strongly in } L_{\text{loc}}^2, \\
\Psi_h \rightharpoonup \Psi, \quad \bar{\Psi}_h \rightharpoonup \bar{\Psi} & \text{weakly in } H_{\text{loc}}^1.
\end{array} \tag{4.23}$$

The complete sectorwise first-variation consistency errors relative to the unfiltered reconstructed fields tend to zero. Every subsequential limit therefore satisfies the ordinary distributional Einstein–Standard-Model equations, the bosonic stress densities converge strongly in L_{loc}^1 , and the complete fermionic metric variation converges in the dual C^r test norm for every $r \geq 2$.

No growing-band or high-derivative reserve, independently assumed matter first-variation consistency, separate quartic compactness condition, strong spinor first-jet compactness, or independent packet-energy uniform-integrability hypothesis is used. The small rooted certificate is a substantive finite non-Abelian gauge-normalization condition; it is not inferred from the curvature bound alone.

Proof. By [Proposition F.7](#), the rooted words in (4.19) are an exact tree representative of the original links, and the small certificate constructs an exact finite site gauge in which the logarithmic connection obeys $\delta_h A_h = 0$ and $\|A_h\|_{4,h} \leq \epsilon_*$. The same-record gauge transformation leaves the complete action, the physical action-covector norm, the literal curvature and Higgs-gradient norms, their Wilson moduli, and the positive internal-link spinor graph norms unchanged.

The normalized L_h^4 bound gives a uniform L_h^2 connection bound on the fixed-volume box. Apply [Theorem F.1](#) separately to $\mathbb{F}_h$ and $\mathbb{K}_h$. Their raw reconstructions are therefore strongly precompact in L^2 , and the associated positive packet energies are uniformly integrable. The critical discrete Coulomb absorption theorem [Theorem F.3](#) converts strong literal curvature compactness into strong L^4 connection convergence and strong convergence of the full ordinary connection first differences. The native Higgs graph argument [Proposition E.4](#) then gives the Higgs statements in (4.23); the positive spinor graph gives the weak H^1 spinor subsequence.

The coframe precompactness gives, after extraction, strong L^2 convergence of both values and first differences; discrete summation by parts identifies the latter limit with the derivative of the former. Hence [Proposition E.7](#) gives strong H^1 geometric convergence without uniform coframe convergence. The all-sector native theorem [Theorem E.8](#) now shows that the unchanged local finite action has the complete classical first variation as its limit, uniformly on bounded C^r physical tests. Because the gauge normalization is an isometry for the positive physical tangent metric, assumption (F5) makes this limiting first variation vanish. Independent metric, gauge, Higgs, spinor and dual-spinor tests give the full Euler system. Strong L^2 convergence of the curvature and covariant Higgs gradient, together with strong L^4 Higgs convergence, gives strongly convergent quadratic and quartic densities. Their bounded metric coefficients converge in measure, so [Lemma A.2](#) gives strong L^1 convergence of the bosonic stress densities directly. This step does not apply a merely measure-convergent coefficient to a nonzero defect measure. The complete fermionic metric variation is covered by [Proposition 3.5](#). $\square$

The proof therefore has a fixed direction: actual site-gauge normalization, compactness of the literal packets, identification of the reconstructed fields, and only then use of the finite action

covector. The small rooted certificate is not inferred from a curvature bound. The *determinant-resolved* extension in [Corollary G.8](#) restricts the smallness requirement to the semisimple factor on the stated zero-flux branch; its complete construction is in [Section G.1](#).

The Wilson criterion is qualitative and requires no high-order Cauchy reserve. Quantitative control of the fields, curvature and complete stress belongs to a different, regular branch, developed next.

5 Quantitative finite-action realization

We now strengthen qualitative closure to quantitative comparison on a regular slab. The first result compares the explicit finite action with a specified regular solution. The second constructs the comparison future from a corrected initial datum of one finite record. The high-order forcing estimates and the constraint solver are kept separate: a small finite Euler row alone supplies neither a regular Cauchy solution nor compatible initial data.

Throughout this section the action is the unchanged S_h^{loc} of [Section D.1](#); Fourier decompositions are estimates on its complete record. [Appendix H](#) proves the consistency and forcing estimates, and [Section J.1](#) proves the coupled actual-jet stability theorem used in both comparisons.

5.1 Finite residual and leakage budgets

Fix buffered slabs $Q \subseteq Q'$ inside a periodic auxiliary coordinate box $\mathcal{Q}$ of period 2π in each coordinate, with the auxiliary temporal seam outside Q' . Let n be odd, $h = 2\pi/n$, and

$$\Lambda_h = \{\ell \in \mathbb{Z}^4 : |\ell_\mu| \leq (n-1)/2\}.$$

Let u_h be the calibrated nodal physical-coordinate record $(e, A, H, \Psi, \bar{\Psi})$ in the open nondegenerate chart of [Section D.1](#), and let $z_h = \mathcal{I}_h^{\text{trig}} u_h$ be its trigonometric reconstruction. For $1 \leq K < (n-1)/2$ define

$$\tau_{h,j}(K) = \sum_{\substack{\ell \in \Lambda_h \\ |\ell|_\infty > K}} \left(1 + \frac{|\ell|_1}{K}\right)^j |\hat{u}_h(\ell)|, \quad j \geq 0. \quad (5.1)$$

These are measured tails of the existing record; no Fourier coefficient is deleted.

For the finite action S_h^{loc} of [\(D.6\)](#), define its mass-normalized interior physical Euler row E_h^{raw} by

$$DS_h^{\text{loc}}(u_h)[v_h] = h^4 \sum_{x \in Q'_h} \langle E_h^{\text{raw}}(u_h)(x), v_h(x) \rangle \quad (5.2)$$

for every interior nodal physical variation, with the fixed finite coefficient and representation metrics. Write

$$\|E_h^{\text{raw}}(u_h)\|_{0,h;Q'} \leq \sigma_h. \quad (5.3)$$

The quantity σ_h is an actual finite-action covector norm, not a continuum residual.

The actual-jet state $\mathcal{U}(z)$ contains the reconstructed metric and its first derivatives, the connection and curvature, the Higgs field and covariant derivative, and both spinors with their tangential covariant first derivatives. Its exact components and normal-spinor elimination are given in [\(J.6\)](#). No independent jet variables are supplied to the comparison theorem.

The finite consistency estimate is $O(hK^3)$ on the declared growing smoothness scale ([Proposition H.1](#)). The leakage transfer retains the full tail and gives the following differentiated forcing budget.

Set $m = k + 2$ for an integer $k \geq 4$ and define

$$\begin{aligned} \varepsilon_{0,h} &= \sigma_h + hK^3, & \varepsilon_{c,h} &= \varepsilon_{0,h} + K^2\tau_{h,2}(K), \\ L_{h,k} &= 1 + \log \left(2 + \frac{C_k K^{k+5}}{\varepsilon_{c,h}} \right), & F_{h,k} &= \varepsilon_{c,h} K^{k+1} L_{h,k}^{k+1} + K^{k+2} \tau_{h,m}(K). \end{aligned} \quad (5.4)$$

Under the calibrated field-value and inverse-metric margins, $hK \leq c_{\text{res}}$, $\tau_{h,m} \leq \tau_*$ and $\varepsilon_{c,h} \leq 1$, [Theorem H.2](#) proves

$$\mathcal{Y}_k(z_h) := \|\mathcal{R}_B(z_h)\|_{L_t^2 H_x^k} + \|\mathcal{R}_D(z_h)\|_{L_t^2 H_x^{k+1}} \leq CF_{h,k}.$$

Here $\mathcal{R}_B$ collects the physical Einstein, Yang–Mills and Higgs rows, and $\mathcal{R}_D$ the Dirac and dual-Dirac rows. The additional spatial derivative in the Dirac source is part of the coupled prolongation estimate, not a freely interchangeable norm.

5.2 Reference-centred quantitative closure

Let z_* be a regular torsion-free Einstein–Standard-Model solution on $Q = [0, T] \times \mathbb{T}^3$, written in harmonic coordinates and temporal internal gauge, and let $\mathcal{U}_*$ denote its actual-jet state from [Section J.1](#). For an actual reconstructed record put

$$i_{h,k} = \|\mathcal{U}_h(0) - \mathcal{U}_*(0)\|_{H^k}, \quad \gamma_{h,k} = \|C(g_h)\|_{L_t^2 H_x^{k+1}} + \|\partial_t C(g_h)\|_{L_t^2 H_x^k}, \quad (5.5)$$

where $C^\mu(g) = g^{\alpha\beta} \Gamma_{\alpha\beta}^\mu(g)$. The state $\mathcal{U}_h$ is built from the actual derivatives and curvatures of z_h ; no independent jet variables are supplied.

Theorem 5.1 (Quantitative finite-action stability and Einstein–Standard-Model closure). *Let $k \geq 4$. Suppose the complete finite records exist on Q , are in temporal internal gauge, and satisfy the calibrated field-value, inverse-metric and buffered comparison hypotheses of [Theorem H.2](#). No all-time Sobolev bound for their actual-jet states is imposed. Suppose*

$$K_h \rightarrow \infty, \quad hK_h \rightarrow 0, \quad \tau_{h,k+2}(K_h) \leq \tau_*,$$

together with the finite-action row bound [\(5.3\)](#). Define

$$D_{h,k}^{\text{full}} = i_{h,k} + \gamma_{h,k} + F_{h,k}. \quad (5.6)$$

If $D_{h,k}^{\text{full}} \rightarrow 0$, then the all-time $C_t H_x^k$ bound for the actual state is a conclusion, and

$$\begin{aligned} &\|\mathcal{U}_h - \mathcal{U}_*\|_{C_t H_x^k} + \|\text{Riem}(g_h) - \text{Riem}(g_*)\|_{L_t^2 H_x^{k-1}} \\ &\quad + \|T^{\text{SM}}(z_h) - T^{\text{SM}}(z_*)\|_{L_t^2 H_x^k} \leq CD_{h,k}^{\text{full}}. \end{aligned} \quad (5.7)$$

In particular the compactness, stress convergence and zero-defect conclusions needed for the classical limit are derived on this branch from the finite physical row, initial/gauge mismatch and measured leakage. Existence of the finite record on the slab is a premise; no existence or selection theorem for an unspecified microscopic rule is inferred.

Proof. [Theorem H.2](#) provides the physical forcing of the complete record before any comparison with z_* . The gauge-defect-aware equation [\(J.10\)](#) applies to these actual records even when $C(g_h) \neq 0$; its added source is controlled by the measured $\gamma_{h,k}$. Apply the one-sided coupled actual-jet stability

estimate [Theorem J.6](#) with $d = i_{h,k} + \gamma_{h,k} + \mathcal{Y}_k(z_h)$. A first-exit argument makes the temporary bounded-state hypothesis self-improving and yields (5.7). The stress estimate follows because the complete torsion-free Standard-Model stress is a smooth algebraic function of the controlled actual jets, with the normal spinor derivative recovered from the Dirac equation. Thus no separately convergent stress tensor or curvature sequence is assumed. $\square$

This theorem derives the all-time state bound by a first-exit argument. The explicit rate and calibration consequences are [Corollaries H.9](#) and [K.4](#). They remain conditional on the stated finite-record budgets; no microscopic existence or selection law is inferred.

5.3 A posteriori exact physical shadows

The preceding theorem compares a finite record with a regular exact solution that is supplied as comparison data. On a regular physical Cauchy chart that external future trajectory can be removed. The finite record itself supplies the approximate forcing; only its initial datum is retracted to the exact constraint manifold, after which the exact future is generated by the Einstein–Standard-Model Cauchy problem.

Let $\mathcal{X}_k$ be the declared initial actual-jet chart and let $\mathfrak{C}_k : \mathcal{X}_k \rightarrow \mathcal{Y}_k^{\text{con}}$ be its complete continuum constraint map, with the Sobolev orders of the target chosen for the differential orders of the constraints. It includes the Einstein normal constraints, internal Gauss constraints, defining-jet identities and the chosen initial gauge completion. Put

$$\mathcal{C}_k = \{d \in \mathcal{X}_k : \mathfrak{C}_k(d) = 0\}. \quad (5.8)$$

The word “exact” in this subsection refers to this zero set, not to the zero set of a sampled or projected constraint operator.

Assumption 5.2 (Verified physical Cauchy tube). Fix $k \geq 4$ and a harmonic-coordinate/temporal-internal-gauge actual-jet chart. There is a declared compatible regular family $\mathcal{C}_k^{\text{reg}} \subset \mathcal{C}_k \cap H^{k+1}$, carrying a fixed higher-order bound, and an open neighborhood $\mathcal{O}_k$ in the initial H^k topology such that every datum in $\mathcal{C}_k^{\text{reg}} \cap \mathcal{O}_k$ generates a unique physical Einstein–Standard-Model solution on the common slab $Q = [0, T] \times \mathbb{T}^3$. These solutions remain in one compact nondegenerate chart, with a common spacelike foliation and a uniform $C_t H_x^{k+1}$ bound for their actual-jet states. The constants in [Theorem J.6](#) are uniform throughout this family.

The regular family is part of the hypothesis: a bounded open H^k ball is not asserted to acquire an extra derivative under hyperbolic evolution. Only the exact initial datum, not its future trajectory, will be supplied to this Cauchy theorem.

Definition 5.3 (Exact finite-dimensional constraint reduction). Let $\widehat{z}_h$ denote the record after the regular normalization used in [Theorem 5.4](#); put $\widehat{z}_h = z_h$ when no normalization is needed. An exact reduction at its normalized initial record x_h consists of a C^1 correction family $\iota_h(x_h, \lambda) \in \mathcal{X}_k$, $\lambda \in \mathbb{R}^{m_h}$, $|\lambda| \leq r_*$, a fixed injective linear map $j_h : \mathbb{R}^{m_h} \rightarrow \mathcal{Y}_k^{\text{con}}$, and a C^1 map $\Phi_h(x_h, \lambda) \in \mathbb{R}^{m_h}$ such that

$$\iota_h(x_h, 0) = \mathcal{U}(\widehat{z}_h)(0), \quad (5.9)$$

$$\mathfrak{C}_k(\iota_h(x_h, \lambda)) = j_h \Phi_h(x_h, \lambda) \quad (|\lambda| \leq r_*). \quad (5.10)$$

Thus all complementary field-valued constraints are solved exactly throughout the correction family, not just at its base point. Equivalently, a splitting of the constraint target has identically zero

complementary component on this family and retained component Φ_h . We require

$$\|\iota_h(x_h, \lambda) - \iota_h(x_h, \mu)\|_{H^k} \leq B_* |\lambda - \mu|. \quad (5.11)$$

The reduced coordinate norms and the constants in the ensuing estimates are fixed in the physical normalization.

The subscript h labels the record and its correction family; it does not make Φ_h a discretization of the remaining continuum constraints. Constructing ι_h may itself use an independently verified elliptic complement solver. A finite list of nodal constraint equations alone does not establish (5.10).

The exact factorization is the substantive constraint hypothesis. [Proposition I.1](#) shows that a retained residual bounded by β_h , a uniform inverse margin γ_* and correction Lipschitz constant B_* produce an exactly compatible datum at distance at most $2B_*\gamma_*^{-1}\beta_h$. Validated evaluation errors and complementary corrections are charged separately in [Corollary I.2](#) and [Remark I.3](#).

The positive reader criterion of [Corollary L.3](#) has a particularly useful regular specialization. Fix integers

$$q > s + 4, \quad s > k + 1, \quad k \geq 4. \quad (5.12)$$

If one selected record has $V_h \leq R^2$ and the reader LMI of (L.15), then

$$\|z_h\|_{H^q(Q)} \leq C_R, \quad \max_{|\alpha| \leq s+2} \|\partial^\alpha z_h\|_{L^\infty(Q)} \leq C_R. \quad (5.13)$$

Thus no growing Fourier scale is needed on this branch.

The fixed reserve converts a small complete finite Euler row $\sigma_h \leq s_h$ into a strong physical forcing estimate. [Proposition I.4](#) proves

$$\varepsilon_h := \|\mathcal{R}_B(z_h)\|_{L^2} + \|(r_D, \bar{r}_D)(z_h)\|_{L^2} \leq C_R(s_h + h), \quad \eta_{h,k} := C_R(\varepsilon_h^{1-k/s} + \varepsilon_h^{1-(k+1)/s}),$$

with bosonic forcing bounded in H^k and both spinor rows in H^{k+1} by $\eta_{h,k}$. The comparison future can now be generated from the corrected datum rather than supplied in advance.

Theorem 5.4 (A posteriori exact Einstein–Standard-Model shadow). *Assume [Assumption 5.2](#). Let z_h be one complete finite record satisfying the reader hypothesis leading to (5.13), the finite Euler bound $\sigma_h \leq s_h$, and the exact initial reduction of [Definition 5.3](#) and the retraction hypotheses of [Proposition I.1](#) with $c_h \leq \beta_h$. Suppose the record is already in the harmonic/temporal chart of [Assumption 5.2](#), or admits a certified regular normalization $z_h \mapsto \hat{z}_h$ that fixes its initial embedding and first differential and changes the residual norms in (I.9) by at most a cutoff-independent factor. Define*

$$\mathfrak{d}_h := 2B_*\gamma_*^{-1}\beta_h + C_R\sqrt{T}\eta_{h,k}. \quad (5.14)$$

Assume also that the initial retraction stays in the verified initial tube, so $d_h \in \mathcal{C}_k^{\text{reg}} \cap \mathcal{O}_k$ with its stated regularity bounds. If $\mathfrak{d}_h$ lies below the uniform stability radius of that tube, then there exists a unique exact physical Einstein–Standard-Model solution $z_{,h}$ on Q with initial datum d_h from [Proposition I.1](#), and*

$$\begin{aligned} & \|\mathcal{U}(\hat{z}_h) - \mathcal{U}(z_{*,h})\|_{C_t H_x^k} + \|\text{Riem}(\hat{g}_h) - \text{Riem}(g_{*,h})\|_{L_t^2 H_x^{k-1}} \\ & + \|T^{\text{SM}}(\hat{z}_h) - T^{\text{SM}}(z_{*,h})\|_{L_t^2 H_x^k} \leq C_R \mathfrak{d}_h. \end{aligned} \quad (5.15)$$

Uniqueness here is for the corrected datum d_h in the declared gauge class, not among all exact solutions lying near the record. In particular, the exact comparison future is a conclusion of the

finite record certificate and the verified Cauchy tube; it is not supplied in advance. If the corrected initial data d_h converge in a Cauchy class for which the physical solution is unique, the shadows $z_{,h}$ converge to that unique exact solution and the certified finite records have the same continuum limit.*

Proof. By [Proposition I.4](#), the complete normalized record has bosonic forcing in $L_t^2 H_x^k$ and Dirac forcing in $L_t^2 H_x^{k+1}$ with total size at most $C_R \eta_{h,k}$. [Proposition I.1](#) supplies a compatible exact initial datum whose H^k distance from the normalized actual initial state is at most $2B_* \gamma_*^{-1} \beta_h$. [Assumption 5.2](#) therefore constructs the exact physical solution $z_{*,h}$ from that datum on the same slab.

Apply [Theorem J.6](#) with reference solution $z_{*,h}$. The stated harmonic normalization makes $C(\hat{g}_h) = 0$ on this branch; otherwise the additional budget of [Corollary I.5](#) is used. Cauchy–Schwarz in time bounds the forcing contribution to its integrated energy estimate by $C_R \sqrt{T} \eta_{h,k}$. The sum of this contribution and the initial mismatch is [\(5.14\)](#), and the assumed tube smallness closes the first-exit argument. The state, curvature and complete stress estimates are exactly [\(5.15\)](#). The final assertion follows from continuous dependence and uniqueness in the stated Cauchy class. $\square$

The harmonic-defect version and a transparent $O_R(h^{1/2})$ rate are [Corollaries I.5](#) and [I.6](#). The correction acts only on the initial comparison datum; the observed spacetime record is unchanged. Uniqueness is attached to that corrected datum, not to an unspecified collection of nearby exact trajectories.

Obtaining the budgets on one record. [Appendix L](#) gives a source-level entrance to the deterministic results. Signed action drift, positive reader energy and initial alignment or constraint control are tested simultaneously on one actually observed record. The corresponding selection theorems retain the original failure probabilities and do not replace the field by an expectation. The reader certificates, alternative prefix-balance entrance and generated comparison families are recorded in [Sections K.2](#), [M.1](#) and [M.3](#). These are sufficient realization mechanisms, not additional premises of reduced or Wilson-screen closure.

6 Interpretation and discussion

The central result of this paper is a finite-to-continuum statement rather than a reconstruction theorem for the microscopic origin of the Standard Model. We begin from a finite Lorentzian–Standard-Model interface carrying one gravity–matter action and ask under which analytic conditions its refinement has the classical coupled field equations as a continuum limit. The structural origin of the finite carrier, the analytic control required for the limit, and the eventual uniqueness of that limit are logically distinct questions.

This separation is important because the presence of the correct finite field content and a common variational principle does not by itself imply classical dynamics. Nonlinear products must converge in the topologies in which they enter the complete first variation, finite stationarity must remain small after being tested in physical directions, and gauge, spin, and Lorentzian geometric structures must remain compatible across the refining family. The Einstein–Standard-Model equations arise here from the simultaneous control of these ingredients rather than being built into the definition of the finite interface.

Several neighboring convergence theories should therefore be distinguished from the present statement. Rigorous Regge and finite-element results control distributional scalar or Einstein

curvature for specified approximating metrics [8, 10, 11, 12], while gauge-invariant Maxwell–Klein–Gordon schemes establish convergence for specific discretizations by combining discrete symmetry, energy estimates and compactness [18, 19]. Those results provide stronger scheme-specific stability information than is assumed abstractly here. The present closure theorem addresses a different question: conditional on compactness, first-variation consistency and physically scaled stationarity, it passes the complete coupled Lorentzian gravity–gauge–Higgs–spinor variation to the limit and records any unresolved bosonic quadratic products as explicit stress defects. Thus regulator independence is obtained by stating the analytic hypotheses explicitly, not by asserting that every finite regulator satisfies them.

6.1 Closure, quantitative comparison and uniqueness

The natural convergence object is the complete physical first variation. Convergence of action values does not identify critical configurations, and a small source-coordinate residual must still be tested against the physical lift. This is why the condition is

$$L_h \sqrt{R_h} + e_h \longrightarrow 0. \quad (6.1)$$

rather than $R_h \rightarrow 0$ alone. The lift cost is part of the finite-to-continuum dictionary. Under [Theorem 3.3](#), controlled sequences have accumulation points satisfying

$$\mathbf{G}_{\mu\nu}(g) + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{SM}}, \quad \mathcal{E}_A = \mathcal{E}_H = \mathcal{E}_\Psi = \mathcal{E}_{\bar{\Psi}} = 0 \quad (6.2)$$

in distributions. The compactness step produces the candidate fields; continuity and stationarity identify their equations. Cofinal uniqueness requires an additional mechanism, either summable transport or uniqueness in a fixed Cauchy class with independently convergent traces.

The quantitative theorems answer a stronger question on a regular branch. The reference-centred result bounds the actual fields, curvature and stress in terms of a finite action row, initial/gauge mismatch and measured leakage. The data-centred theorem instead uses a positive reader reserve, an exact initial-constraint reduction and a verified physical Cauchy tube to construct the comparison future. It does not assume that a bounded H^k neighborhood acquires an extra derivative or that finitely many sampled constraints solve the full continuum constraint system.

The source-level theorems supply a further, conditional entrance to these budgets on one observed record. Their probabilistic estimates retain failure events and do not define a mean-field geometry. Whether a particular microscopic dynamics satisfies their drift, reader and legal-domain hypotheses remains a separate selection question.

6.2 Nonlinear compactness and the defect alternative

The distinction between primitive fields and nonlinear packets is visible in the weak-limit equation

$$(\mathbf{G}(g) + \Lambda g) dV_g = \kappa \left(T^{\text{SM}} dV_g + \mathfrak{S}_{\text{YM}} + \mathfrak{S}_H \right). \quad (6.3)$$

The defects retain information invisible in the weak field limits but visible in the metric variation. In the minimally coupled theory, [Theorem 4.2](#) identifies their vanishing with the vanishing of positive curvature and covariant-gradient energy defects; Higgs quartic compactness then follows. Conservation alone is not such a criterion. Nonminimal improvement terms and weaker geometric first-jet limits require their own defect analysis.

The geometry needed for zero-defect closure is weaker than the geometry used to multiply a nonzero singular stress measure. Strong H^1 coframe convergence on a common compact nondegenerate chart suffices for the former, while the defect identities retain uniform coefficient convergence to continuous representatives. The two hypotheses are not interchangeable.

At critical gauge regularity, equivariant tests make quotient closure compatible with cutoff-dependent local gauges. For the explicit action, Wilson translation control first makes scalar packet magnitudes compact; a same-record normalization then gives vector compactness and identifies the limiting fields. The finite rooted certificate remains a substantive non-Abelian condition, and zero determinant flux is a separate topological restriction of the global periodic normalization.

The stronger Higgs diagnostic is distinct from this qualitative closure: $e_h^{H,+} \asymp |K_h^H|^2$ gives equivalence between an $L^{4/3}$ bound for the positive kinetic density and an $L^{8/3}$ covariant-gradient bound. Strong L^2 gradient compactness yields only L^1 convergence of that density. The additional integrability is retained as an optional reserve, not silently inserted into the zero-defect hypotheses.

The fermionic first variation behaves differently because the action is first order and bilinear. Weak H^1 control suffices for its complete weak-strong passage, including the spin-connection variation. On a regular branch, stronger spinor convergence follows from Lorentzian symmetric-hyperbolic stability. Variational stationarity and a controlled Cauchy evolution therefore remain distinct conclusions.

6.3 Structural provenance and classical scope

Renewal Geometry supplies one upstream realization through its predictive carrier, compatible differential/spin structure and structural matter packet [24, 25, 26]. The present analysis begins after that interface has been obtained and is portable to another realization with the same stated hypotheses. The logical separation is

$$\begin{aligned} \text{structural realization:} & \quad \text{a finite Lorentzian-Standard-Model interface is obtained,} \\ \text{analytic closure:} & \quad \text{a controlled refinement satisfies the classical equations,} \\ \text{microscopic selection:} & \quad \text{a microscopic dynamics naturally enters that controlled class.} \end{aligned} \tag{6.4}$$

The second is the principal subject of this paper; the source-level criteria give conditional routes toward the third, not a universality claim for microscopic dynamics.

The endpoint established here is classical. The spinor variables are classical variational fields, and the limiting equations are the classical Einstein–Yang–Mills–Higgs–Dirac–Yukawa equations. The analysis does not construct a fermionic path-integral measure, a renormalized quantum Standard Model coupled to gravity, or a quantum gravitational continuum measure. Operational probabilities in a microscopic predictive theory and a quantum field-theoretic measure play different mathematical roles and should not be identified.

Likewise, the closure theorem does not determine the numerical Standard-Model parameters. Gauge couplings, the Higgs potential parameters, Yukawa eigenvalues and mixing data, and their renormalization-group evolution remain independent inputs or dynamical data on the finite structural carrier.

Outlook. The main remaining questions are consequently more sharply separated. On the analytic side, one may seek weaker compactness hypotheses, a more complete classification of admissible stress defects, boundary versions of the closure theorem, and broader uniqueness criteria for cofinal refinement. On the microscopic side, the central problem is to identify natural classes of finite dynamics for which the interface, stationarity, reader, and compactness hypotheses are

stable or typical rather than externally imposed. Constraint compatibility of a principal operator alone does not control the complete state-dependent source or supply a spatially uniform stationary family.

Within the scope proved here, the central conclusion is that classical Einstein–Standard-Model dynamics can be characterized as the variational closure of a finite gravity–matter theory once the physically relevant nonlinear packets and variations are controlled. When that control is strong enough, the ordinary coupled field equations emerge; when it is not, the unresolved finite structure remains visible through explicit effective stress defects rather than being discarded by definition.

Data and formalization availability

Selected finite algebraic components of the construction have been formalized in `Lean`. Source code is available at <https://github.com/AI-MathPhys/renewal-geometry>.

Guide to the appendices

The appendices are part of the same manuscript and contain the complete supporting proofs. They are organized by their use in the main argument: [Appendices A and B](#) support reduced and strong-packet closure; [Appendix C](#) supports the weak-limit and quotient results; [Appendices D to G](#) specify the finite action and prove its critical packet and gauge estimates; [Appendices H to J](#) supply quantitative forcing, constraint correction and coupled stability. [Appendices K to M](#) retain the comparison constructions and source-level entrances. No external supplement is needed to follow the proof dependencies.

A Continuity, compactness and metric-variation estimates

The estimates in this appendix support the compactness and continuity steps of [Theorem 3.3](#). The abstract closure argument is independent of the field theory; the product and coframe estimates identify the complete first variation in the reduced topology.

A.1 An abstract first-variation closure principle

The passage from finite stationarity to a limiting Euler equation is a first-variation statement. Its abstract form does not require positivity, convexity, minimization, or a prescribed evolution of the finite records. The sector-dependent continuity needed to apply it is proved below.

Lemma A.1 (First-variation closure). *Let $\mathcal{X}$ be a class of reconstructed configurations, with a specified sequential convergence, and let $\mathcal{V}_K$ be a Banach space of physical tests. Let S have first variations $DS(z) \in \mathcal{V}_K^*$ on this class. Suppose $z_h \rightarrow z$ and*

$$\omega_h(K) := \|DS(z_h) - DS(z)\|_{\mathcal{V}_K^*} \longrightarrow 0. \quad (\text{A.1})$$

Let S_h be differentiable finite actions, $z_h = \mathcal{R}_h z_h^d$, and I_h linear test lifts. Assume

$$\begin{aligned} \sup_{\|v\| \leq 1} |DS_h(z_h^d)[I_h v] - DS(z_h)[v]| &\leq c_h(K), \\ \sup_{\|v\| \leq 1} |DS_h(z_h^d)[I_h v]| &\leq \varepsilon_h(K). \end{aligned} \quad (\text{A.2})$$

Then

$$\|DS(z)\|_{\mathcal{V}_K^*} \leq \varepsilon_h(K) + c_h(K) + \omega_h(K). \quad (\text{A.3})$$

In particular, if $c_h(K) + \varepsilon_h(K) \rightarrow 0$, then $DS(z) = 0$. If the reconstructed family is sequentially precompact and these hypotheses hold along every convergent subsequence, all its accumulation points are stationary configurations of S .

Proof. For a unit test, insert and subtract $DS(z_h)[v]$ and $DS_h(z_h^d)[I_h v]$. The three differences are bounded by the three terms in [\(A.3\)](#). Taking the supremum and then the limit proves stationarity. Apply the same argument to each convergent subsequence for the last assertion. $\square$

This principle concerns convergence of critical configurations, rather than only convergence of minimum problems. Plain Γ -convergence of the action functionals [\[44, 45\]](#) does not by itself supply [\(A.1\)](#) or [\(A.2\)](#). There are stronger variational frameworks for critical points: for example, Jerrard–Sternberg obtain critical points of Γ -converging functionals near suitable nondegenerate critical points [\[46\]](#), and Mosco/Attouch-type theories relate variational convergence to convergence

of subdifferentials in convex settings [47]. Those results use additional structure not assumed for the nonconvex Lorentzian gauge system here. The elementary oscillatory example following Definition 2.5 already separates uniform action convergence from first-variation convergence. Variational convergence of gradient flows also requires additional control of slopes and dissipation [48]; no gradient-flow law for the Lorentzian common action is assumed here.

A.2 Products and bounded geometric coefficients

Lemma A.2 (Local product and bounded-coefficient continuity). *If $u_h \rightarrow u$ in L^p , $v_h \rightarrow v$ in L^q , and $1/p + 1/q = 1/r \leq 1$, then $u_h v_h \rightarrow uv$ in L^r . In particular,*

$$\|B(Y_h, Y_h) - B(Y, Y)\|_1 \leq C\|Y_h - Y\|_2(\|Y_h\|_2 + \|Y\|_2) \quad (\text{A.4})$$

for a bounded bilinear contraction B .

Suppose in addition that B_h are finite matrix-valued functions with $\sup_h \|B_h\|_\infty < \infty$ and $B_h \rightarrow B$ in measure. If $u_h \rightarrow u$ in L^p , $1 \leq p < \infty$, then

$$B_h u_h \rightarrow Bu \quad \text{in } L^p. \quad (\text{A.5})$$

If $u_h \rightarrow u$ and $v_h \rightarrow v$ in L^2 , then $B_h(u_h \otimes v_h) \rightarrow B(u \otimes v)$ in L^1 .

Finally, let $\mathcal{K}_e$ be a compact subset of one oriented, time-oriented, nondegenerate coframe chart and suppose $e_h, e \in \mathcal{K}_e$ almost everywhere and $e_h \rightarrow e$ strongly in H^1 . Then every smooth algebraic coefficient $F(e, e^{-1})$ satisfies

$$F(e_h, e_h^{-1}) \rightarrow F(e, e^{-1}) \quad \text{in } H^1 \text{ and in every finite } L^p, \quad \sup_h \|F(e_h, e_h^{-1})\|_\infty < \infty. \quad (\text{A.6})$$

In particular,

$$\omega(e_h) \rightarrow \omega(e) \quad \text{in } L^2, \quad \dot{\omega}(e_h)[\dot{e}_h(k)] \rightarrow \dot{\omega}(e)[\dot{e}(k)] \quad \text{in } L^2, \quad (\text{A.7})$$

uniformly for k in bounded C^1 test sets. No strong L^∞ convergence of e_h is required.

Proof. Subtract products one factor at a time and apply Hölder. For (A.5), split $B_h u_h - Bu = B_h(u_h - u) + (B_h - B)u$. The first term tends to zero in L^p . Every subsequence of the second has a further subsequence on which $B_h \rightarrow B$ almost everywhere; dominated convergence applies because the coefficients are uniformly bounded. The subsequence criterion proves the claim for the full sequence. The tensor-product assertion follows from $u_h \otimes v_h \rightarrow u \otimes v$ in L^1 and the same argument.

Strong H^1 convergence of e_h implies convergence in measure. Smooth coefficients on the compact chart are uniformly bounded with uniformly bounded derivatives, hence converge in measure and in every finite L^p . The chain rule gives

$$\partial F(e_h) - \partial F(e) = DF(e_h)(\partial e_h - \partial e) + (DF(e_h) - DF(e))\partial e,$$

and the second term is controlled by (A.5). This proves (A.6). In local frames the Levi-Civita spin connection is a finite sum $P(e)\partial e$, while its physical metric variation has the form $P_1(e)k\partial e + P_2(e)\partial k$. Applying the same bounded-coefficient argument proves (A.7) and its uniformity on bounded C^1 test sets. $\square$

A.3 Compact screens for the quantities that enter the action

Lemma A.3 (Common compact screen). *Let $Y_h \rightharpoonup Y$ in a fixed Hilbert space $\mathcal{Y}$. Suppose that, for every R , there are bounded $S_{h,R}$ and compact S_R with*

$$\|S_{h,R} - S_R\|_{\text{op}} \longrightarrow 0, \quad \lim_{R \rightarrow \infty} \sup_h \|(I - S_{h,R})Y_h\| = 0, \quad S_R Y \longrightarrow Y. \quad (\text{A.8})$$

Then $Y_h \rightarrow Y$ strongly. The same conclusion holds for a whitened packet $Z_h = \mathbb{A}_h^{1/2} Y_h$ provided $Z_h \rightharpoonup \mathbb{A}^{1/2} Y$, the screen conditions hold for Z_h , and

$$mI \preceq \mathbb{A}_h \preceq MI, \quad \mathbb{A}_h^{-1/2} \rightarrow \mathbb{A}^{-1/2} \text{ strongly,} \quad 0 < m \leq M < \infty.$$

Proof. For fixed R , compactness gives $S_R Y_h \rightarrow S_R Y$, and norm convergence of $S_{h,R}$ gives $(S_{h,R} - S_R)Y_h \rightarrow 0$. Decompose

$$Y_h - Y = (I - S_{h,R})Y_h + (S_{h,R}Y_h - S_R Y) + (S_R - I)Y.$$

Take the upper limit in h and then let $R \rightarrow \infty$. This proves the first assertion. For the second, apply the first assertion to Z_h and use

$$\|Y_h - Y\| \leq m^{-1/2} \|Z_h - Z\| + \|(\mathbb{A}_h^{-1/2} - \mathbb{A}^{-1/2})Z\|.$$

□

A screen for the action must include the full quadratic packet. For example, F_{A_h} , $D_{A_h} H_h$, and the first spinor jets are different components, even when all are represented on the same finite history. A gap in the multiplicity commutant does not control their spatial or temporal frequencies.

Proposition A.4 (From spatial screens to spacetime compactness). *Let $\mathcal{H} = L^2(\Sigma; E)$ and let P_R be finite-rank orthogonal projections with smooth ranges and $P_R \rightarrow I$ strongly. Suppose Y_h is bounded in $L^2(0, T; \mathcal{H})$,*

$$\lim_{R \rightarrow \infty} \sup_h \|(I - P_R)Y_h\|_{L_t^2 \mathcal{H}} = 0,$$

and, for each fixed R , $P_R Y_h$ is bounded in $H^1(0, T; P_R \mathcal{H})$. Then (Y_h) is precompact in $L^2(0, T; \mathcal{H})$. The last condition follows, for example, from a bound on $\partial_t Y_h$ in $L^2(0, T; H^{-m}(\Sigma; E))$ for a fixed m .

Proof. For fixed R , the range is finite dimensional, so one-dimensional compact Sobolev embedding makes $(P_R Y_h)$ precompact in $L_t^2 \mathcal{H}$. The tail estimate gives a finite ϵ -net for (Y_h) by first choosing R and then a finite net for the projected sequence. For the final assertion, pair $\partial_t Y_h$ with a smooth orthonormal basis of $P_R \mathcal{H}$; each coefficient has a bounded L^2 time derivative. □

This elementary special case of spacetime compactness is consistent with the general theory of Simon [49]. The time condition is indispensable: $Y_h(t, x) = \sin(t/h)v(x)$ remains in a single spatial mode but has no strongly convergent subsequence in spacetime L^2 . Thus compact spatial Hodge screens alone do not establish the required spacetime result.

A.4 Sectorwise first-variation estimates

Let $d_K(z_h, z)$ denote the sum of the difference norms in (B.7)–(B.10), together with $|\theta_h - \theta|$. The same formula defines $d_K(z_h, z_j)$ for two reconstructed cutoffs on the common local bundle realization. When necessary, take these norms on a slightly larger compact region containing the supports of the test lifts.

Proposition A.5 (First-variation continuity from primitive fields). *On a uniformly bounded classical strong packet,*

$$\|D\mathcal{S}_{\theta_h}(z_h) - D\mathcal{S}_{\theta}(z)\|_{(\mathcal{V}^{r_0})^*} \leq C_K d_K(z_h, z). \quad (\text{A.9})$$

The same estimate holds separately for the gravitational and matter terms. The matter metric variation has the limit prescribed by (2.10); it is not an independent convergence assumption.

Proof. All arguments take place in one compact region and a finite bundle atlas. A partition of unity introduces only fixed smooth coefficients. We give the separate sector estimates, including the terms specific to metric variation.

Yang–Mills sector. Write its local density as a bounded coframe coefficient contracted with two curvatures. A gauge variation gives

$$\dot{F}_{A_h} = da + [A_h, a],$$

which converges in L^2 uniformly for bounded smooth a , by $A_h \rightarrow A$ in L^4 . The pairing of $\dot{F}_{A_h}$ and F_{A_h} therefore converges in L^1 . A metric variation changes the coefficient and has the form

$$B'_h[k](F_{A_h}, F_{A_h}). \quad (\text{A.10})$$

The coefficient converges in L^∞ and both curvature factors converge in L^2 . Equation (A.4) proves convergence with the stated norm bound. If curvature were only weakly convergent, (A.10) would contain two weak factors; a strongly convergent Hodge coefficient would not repair that loss.

Higgs kinetic sector. With $K_h = D_{A_h} H_h$,

$$\dot{K}_h = D_{A_h} \eta_H + \rho_H(a) H_h.$$

This converges strongly in L^2 . The field variation is a pairing of $\dot{K}_h$ and K_h ; the metric variation is a bounded coefficient applied to $K_h \otimes K_h$. Both follow from Lemma A.2.

Higgs potential. On a bounded L^4 set,

$$\begin{aligned} \||H_h|^4 - |H|^4\|_1 &\leq C(\|H_h\|_4 + \|H\|_4)^3 \|H_h - H\|_4, \\ \||H_h|^2 H_h - |H|^2 H\|_{4/3} &\leq C(\|H_h\|_4 + \|H\|_4)^2 \|H_h - H\|_4. \end{aligned} \quad (\text{A.11})$$

The lower-degree terms satisfy the analogous estimates. These inequalities pass both the volume variation of the potential and its Higgs variation, including the finite parameter limits.

Dirac–Yukawa sector. Sobolev embedding on the four-dimensional compact chart gives strong L^4 convergence of both spinor variables from (B.10). The derivative terms have the form

$$C(e_h) \bar{\Psi}_h \partial \Psi_h, \quad C(e_h) (\partial \bar{\Psi}_h) \Psi_h.$$

They converge in L^1 , using the L^2 bounds for spinors and first derivatives. Spin-connection terms contain one L^2 coefficient $\omega(e_h)$ and the L^2 product $\bar{\Psi}_h \Psi_h$; they converge in L^1 by (A.7). Gauge

and Yukawa terms contain, respectively, $A_h \bar{\Psi}_h \Psi_h$ and an affine function of H_h multiplying $\bar{\Psi}_h \Psi_h$. Hölder with three L^4 factors gives convergence in $L^{4/3}$, hence in L^1 on the compact chart.

Varying $\Psi, \bar{\Psi}, A, H$ replaces one of these factors by its corresponding smooth test or derivative. Varying the metric differentiates $C(e_h)$ and $\omega(e_h)$. The additional spin term is $\dot{\omega}(e_h)[\dot{e}_h(k)]\bar{\Psi}_h \Psi_h$, again an L^2 – L^2 product. Thus the full metric variation, including the spin connection, converges uniformly on bounded C^1 test sets. This proves the matter assertion without omitting a fermionic stress term.

Gravity. Let $B(e)$ be the coframe bivector density of [Section A.5](#). On compactly supported variations, integration by parts changes the torsion-free Palatini density into

$$-dB(e) \wedge \omega(e) + B(e) \wedge \omega(e) \wedge \omega(e), \quad (\text{A.12})$$

with the chosen Einstein–Hilbert normalization. The volume term is algebraic in e . In the first variation of [\(A.12\)](#), each differentiated factor is in L^2 , each algebraic coefficient is in L^∞ , and the variation introduces at most k and ∂k . The coefficient and connection estimates of [Lemma A.2](#) therefore apply to every term. They also define the gravitational first variation at the limiting $H^1 \cap L^\infty$ metric. On smooth metrics it is the ordinary Einstein first variation, and its distributional extension is fixed by the integration-by-parts formula, as explained in [Section A.5](#).

Subtracting each finite multilinear expression one factor at a time gives a bound linear in d_K . The other factors are uniformly bounded by the packet hypotheses. A finite sum of these estimates proves [\(A.9\)](#). $\square$

Remark A.6 (Weak–strong gravity is a different route). The geometric condition [\(B.7\)](#) is a strong first-jet route, not a requirement of strong second derivatives. [Proposition A.7](#) gives an alternative Palatini passage from strong coframe products and weak, identified curvature. That alternative does not weaken the strong curvature condition in the *Yang–Mills stress*. The gravitational Palatini term is linear in curvature; the Yang–Mills metric variation is quadratic in curvature.

A.5 Palatini products, metric variations, and the spin-current alternative

A.5.1 The first-order gravitational distribution

Fix the orientation and curvature sign so that, for the torsion-free connection,

$$B_{ab}(e) \wedge R^{ab}(\omega(e)) = \text{Scal}(g) dV_g, \quad (\text{A.13})$$

where $B(e)$ is the appropriately normalized internal dual of $e \wedge e$. For a cutoff function χ equal to one near the support of a physical test,

$$\int \chi B \wedge R(\omega) = \int [-d(\chi B) \wedge \omega + \chi B \wedge \omega \wedge \omega]. \quad (\text{A.14})$$

Exterior Stokes proves the identity for smooth compactly supported data. The terms involving $d\chi$ do not contribute to a variation supported where $\chi = 1$.

For $e, e^{-1} \in L^\infty$ with $e \in H^1$, the Levi–Civita connection $\omega(e)$ is in L^2 . Every term on the right of [\(A.14\)](#) is integrable, and its derivative in the lift [\(2.6\)](#) is bounded by a constant times $\|k\|_{C^1}$. This defines the local Einstein first-variation distribution. Different choices of χ give the same functional on tests supported in their common unit region. On smooth metrics the distribution is $(G(g) + \Lambda g) dV_g / (2\kappa)$.

This definition keeps the derivative of the connection in divergence form. The need to specify a regularity class in which nonlinear curvature is a well-defined distribution is classical in general

relativity [50]; the representative used here is consistent with the stability framework for locally bounded metrics with locally square-integrable first derivatives and bounded inverse developed by LeFloch–Mardare [51]. It does not assert a pointwise curvature tensor or an unqualified product of a rough coefficient with an arbitrary distribution.

A.5.2 A weaker curvature passage

Proposition A.7 (Weak-curvature Palatini passage). *On a compact four-dimensional region suppose*

$$e_h \rightarrow e \text{ in } L^2, \quad \sup_h \|e_h\|_6 < \infty, \quad R_h \rightharpoonup R \text{ in } L^p, \quad p > \frac{3}{2}. \quad (\text{A.15})$$

Assume R is identified with the curvature of the limiting represented connection. Then $B(e_h) \rightarrow B(e)$ in $L^{p'}$, $p' = p/(p-1)$, and the Palatini pairings converge. The volume densities converge in L^1 . If the represented coframe variations converge strongly in L^2 , are uniformly bounded in L^6 , and have compatible supports, their coframe first variations also converge. For a full metric variation, any connection-Euler pairing introduced by the lift must additionally tend to zero, or be retained as part of the limiting connection equation.

Proof. Interpolation between strong L^2 and bounded L^6 gives strong L^q convergence for every $2 \leq q < 6$; the lower exponents follow by finite volume. Since $p > 3/2$, $2p' < 6$. The product estimate gives

$$\|e_h \wedge e_h - e \wedge e\|_{p'} \leq C\|e_h - e\|_{2p'}(\|e_h\|_{2p'} + \|e\|_{2p'}) \rightarrow 0.$$

Pair with weak L^p curvature. Taking $q = 4$ gives convergence of the four-coframe volume product in L^1 by a four-factor telescoping estimate. The same argument applies to $\dot{B}_h$, a sum of one coframe and one coframe-variation factor, and to the volume variation. A connection variation produces its own Euler pairing after integration by parts. The stated extra condition, rather than coframe compactness alone, is what controls that term. $\square$

The hypothesis identifying R is substantive. A weak limit of matrix-valued two-forms is not automatically the curvature of the limiting connection. Nor does the Palatini lemma apply to a quadratic gauge stress by replacing one factor with a Hodge coefficient.

A.5.3 Stationary elimination is not a universal torsion argument

The independent connection Euler equation for a Palatini–Holst action has the form

$$\mathcal{C}_e(T) = J_{\text{spin}}, \quad (\text{A.16})$$

where $\mathcal{C}_e$ is the Cartan map determined by the coframe and the Palatini–Holst coefficients. On a nondegenerate chart with a uniformly bounded inverse of that map,

$$\|T\| \leq C\|J_{\text{spin}} + r_{\text{conn}}\|.$$

Thus zero spin current and vanishing connection residual imply torsionlessness. With nonzero current, the inverse instead reconstructs contorsion. Substitution into the total action retains the resulting spin-current interaction [39, 40]. This is compatible with the connection alternative discussed in the Lorentzian construction [24], but differs from the metric action chosen in the main theorem.

There is also a separate normalization condition. A nonzero Holst coefficient alone does not imply a nonzero Einstein coefficient: on a torsion-free connection the Holst coframe equation reduces to the first Bianchi identity. The main action fixes a nonzero Palatini/Einstein coefficient $1/(2\kappa)$ before taking the limit. Pure Holst stationarity therefore cannot be used to infer (B.15).

B Strong-packet and regularity refinements

These results strengthen reduced closure on additional regular branches. Independent quartic integrability and spinor first-jet screens are optional sufficient conditions here, not premises of [Theorem 3.3](#). The sectorwise estimates used below are in [Section A.4](#).

Remark B.1 (Meaning of the classical limit). Throughout, “classical” distinguishes the non-quantized field theory from a quantum measure. At the reduced regularity the conclusion is a *distributional solution of the classical Euler system*, with the Einstein first variation defined by [Section A.5](#). Neither smoothness nor well-posedness of an initial-value problem follows from distributional stationarity alone. The regularity and evolution statements are given separately in [Corollary B.16](#) and [Theorem 5.1](#).

The Higgs potential requires a different endpoint norm. On a four-dimensional bounded region, an H^1 bound controls L^4 but does not make its unit ball compact in L^4 .

Proposition B.2 (A local quartic compactness certificate). *Let $H_h \rightharpoonup H$ in $H^1(K)$ on a bounded smooth four-dimensional region. Suppose $\Phi : [0, \infty) \rightarrow [0, \infty)$ is increasing, satisfies $\Phi(t)/t \rightarrow \infty$, and*

$$\sup_h \int_K \Phi(|H_h|^4) dV_0 < \infty. \quad (\text{B.1})$$

Then $H_h \rightarrow H$ strongly in $L^4(K)$. A uniform $L^{4+\delta}$ bound, $\delta > 0$, is sufficient.

Proof. Rellich compactness gives strong L^2 convergence and convergence in measure. For any $B > 0$,

$$\int_{\{|H_h|^4 > B\}} |H_h|^4 \leq \sup_{t > B} \frac{t}{\Phi(t)} \int_K \Phi(|H_h|^4).$$

The right side tends to zero uniformly in h . Hence the fourth powers are uniformly integrable. Convergence in measure and this uniform integrability give $\|H_h - H\|_4 \rightarrow 0$ by the elementary Vitali argument, applied after using $|H_h - H|^4 \leq 8(|H_h|^4 + |H|^4)$. Take $\Phi(t) = t^{1+\delta/4}$ for the last assertion. $\square$

B.1 A-priori routes to the action topology

The compact-screen formulation is useful at low regularity, but it should not be read as the only way to obtain the nonlinear action topology. On a regular branch, ordinary a-priori Sobolev estimates imply the required bosonic compactness directly.

Proposition B.3 (A-priori Sobolev bounds imply bosonic strong compactness). *Let $K \Subset M$ be a bounded smooth four-dimensional region. Suppose that for some $\sigma > 0$, after fixed local gauge and frame choices,*

$$\sup_h \left(\|e_h\|_{H^{2+\sigma}(K)} + \|A_h\|_{H^{1+\sigma}(K)} + \|H_h\|_{H^{1+\sigma}(K)} \right) < \infty, \quad \sup_h \|e_h^{-1}\|_{L^\infty(K)} < \infty. \quad (\text{B.2})$$

Assume also that the coefficient banks remain in a compact subset of the physical parameter region. Then every cutoff sequence has a subsequence such that

$$\begin{array}{ll} e_h \rightarrow e & \text{strongly in } H^1(K) \cap L^\infty(K), \\ A_h \rightarrow A & \text{strongly in } H^1(K) \text{ and hence in } L^4(K), \\ H_h \rightarrow H & \text{strongly in } H^1(K) \cap L^4(K), \\ F_{A_h} \rightarrow F_A & \text{strongly in } L^2(K), \\ D_{A_h} H_h \rightarrow D_A H & \text{strongly in } L^2(K). \end{array} \quad (\text{B.3})$$

Moreover H_h is uniformly bounded in $L^{4+\delta}(K)$ for some $\delta > 0$ depending only on σ and the dimension, so quartic uniform integrability follows automatically. Thus, on this regular branch, the bosonic part of the strong packet is obtained from a-priori boundedness rather than assumed precompactness or screen tails.

Proof. Rellich–Kondrachov gives compact embeddings $H^{2+\sigma}(K) \Subset H^1(K)$ and $H^{1+\sigma}(K) \Subset H^1(K)$. Since $2 + \sigma > 2 = d/2$, the first family is also precompact in $C^0(K)$ after lowering the Hölder exponent, hence in $L^\infty(K)$. The uniform inverse bound preserves the nondegenerate signature chart. Strong H^1 convergence of A_h gives strong L^4 convergence in four dimensions. Therefore

$$dA_h \rightarrow dA \quad \text{in } L^2, \quad A_h \wedge A_h \rightarrow A \wedge A \quad \text{in } L^2,$$

which proves the curvature convergence. The same argument, using

$$D_{A_h} H_h = dH_h + \rho_H(A_h) H_h,$$

gives strong L^2 convergence of the Higgs covariant derivative. Finally $H^{1+\sigma}$ embeds continuously into some $L^{4+\delta}$ with $\delta > 0$, yielding quartic uniform integrability. Finite-dimensional compactness handles the coefficients. $\square$

Proposition B.4 (Lorentzian Dirac stability produces spinor H^1 compactness). *Let $I \times \Sigma$ be a compact subslab with Σ closed. In a common local spin trivialization suppose the reconstructed spinors satisfy symmetric-hyperbolic systems*

$$A_h^0 \partial_t \Psi_h + A_h^j \partial_j \Psi_h + B_h \Psi_h = r_h, \quad (\text{B.4})$$

with the analogous dual system for $\bar{\Psi}_h$. Assume the matrices A_h^0 are uniformly positive definite, the systems are uniformly symmetric hyperbolic, and

$$\sup_h (\|A_h^\mu\|_{W^{1,\infty}(I \times \Sigma)} + \|B_h\|_{L_{t,x}^\infty} + \|\Psi_h\|_{L_t^\infty H_x^2} + \|\bar{\Psi}_h\|_{L_t^\infty H_x^2}) < \infty. \quad (\text{B.5})$$

Suppose the coefficients are Cauchy in $L^\infty(I \times \Sigma)$, the spinor and dual-spinor initial data are Cauchy in $L^2(\Sigma)$, and the residuals $r_h, \bar{r}_h$ are Cauchy in $L_t^2 L_x^2$. Then Ψ_h and $\bar{\Psi}_h$ are Cauchy in $H^1(I \times \Sigma)$. In particular, if $r_h, \bar{r}_h \rightarrow 0$, the limiting fields satisfy the Lorentzian Dirac–Yukawa and dual equations determined by the limiting coefficients.

Proof. For $w = \Psi_h - \Psi_j$, subtract the two first-order systems and use the h -system as the principal symmetric-hyperbolic operator. Uniform positivity of A_h^0 and the coefficient bounds in (B.5) give the standard L^2 energy inequality

$$\|w(t)\|_{L_x^2} \leq C_T \left[\|w(0)\|_{L_x^2} + \|r_h - r_j\|_{L_t^1 L_x^2} + \delta_{h,j} \right], \quad (\text{B.6})$$

where $\delta_{h,j} \rightarrow 0$ is the coefficient distance multiplied by uniform L^2 bounds for the first derivatives of Ψ_j . The spatial derivative bounds follow from the uniform H_x^2 estimate; the normal derivative is bounded by solving (B.4) for $\partial_t \Psi_j$, using uniform invertibility of A_j^0 and boundedness of the residuals. Hence Ψ_h is Cauchy in $C_t L_x^2$.

Interpolation on the closed three-manifold gives, uniformly in t ,

$$\|w(t)\|_{H_x^1} \leq C \|w(t)\|_{L_x^2}^{1/2} \|w(t)\|_{H_x^2}^{1/2},$$

so Ψ_h is Cauchy in $C_t H_x^1$. Finally solve (B.4) for $\partial_t \Psi_h$. Strong convergence of the spatial H^1 terms and coefficients, together with r_h Cauchy in $L_t^2 L_x^2$, gives strong convergence of $\partial_t \Psi_h$ in $L_t^2 L_x^2$. This is strong $H^1(I \times \Sigma)$ convergence. The dual system is identical. Passing to the limit in (B.4) proves the final assertion. $\square$

B.2 Primitive compactness certificates

The strong packet used below is a conclusion of a concrete compactness mechanism once the quantities entering the nonlinear first variation are controlled in their own topologies. The low-regularity formulation is deliberately modular: direct spacetime compact screens may be used, or spatial screens may be combined with the time regularity of [Proposition A.4](#). For spinors one may instead use the Lorentzian evolution route of [Proposition B.4](#); the regular Sobolev branch of [Theorem B.11](#) avoids compact screens altogether.

Definition B.5 (Classical compactness certificate). A finite-interface regulator sequence satisfies the classical compactness certificate on $K \Subset M$ if, after fixed local gauge and spin-frame identifications,

- (C1) the coframes are precompact in $L^\infty(K)$, uniformly nondegenerate, and bounded in $H^1(K)$; their first-derivative packet has a common compact screen in $L^2(K)$;
- (C2) the connections are precompact in $L^4(K)$, the curvatures are bounded in $L^2(K)$, and the curvature packet has a common compact screen in $L^2(K)$;
- (C3) the Higgs fields are bounded in $H^1(K)$, the covariant gradients are bounded in $L^2(K)$ and have a common compact screen, and there is an increasing Φ with $\Phi(t)/t \rightarrow \infty$ such that

$$\sup_h \int_K \Phi(|H_h|^4) dV_0 < \infty;$$

- (C4) for the spinors, either

- (C4a) $\Psi_h, \bar{\Psi}_h$ are precompact in $L^2(K)$ and bounded in $H^1(K)$, while their ordinary first-jet packets in the fixed local frames have common compact screens in $L^2(K)$; or
- (C4b) on a compact subslab containing K , the reconstructed Dirac and dual-Dirac systems satisfy the hypotheses of [Proposition B.4](#);

- (C5) the coefficient banks remain in a compact subset of the region $\kappa, g_j, \lambda_H > 0$.

A “common compact screen” means the hypotheses of [Lemma A.3](#); it may instead be verified by the spatial-tail and time-regularity conditions of [Proposition A.4](#).

B.3 The primitive convergence class

Definition B.6 (Classical strong packet). A finite-interface regulator sequence has a classical strong packet with limit $z = (e, A, H, \Psi, \bar{\Psi})$ if, on every $K \Subset M$ and in the fixed local bundle identifications,

$$e_h \rightarrow e \text{ in } L^\infty(K) \cap H^1(K), \quad \sup_h (\|e_h\|_\infty + \|e_h^{-1}\|_\infty) < \infty, \quad (\text{B.7})$$

$$A_h \rightarrow A \text{ in } L^4(K), \quad F_{A_h} \rightarrow F_A \text{ in } L^2(K), \quad (\text{B.8})$$

$$H_h \rightarrow H \text{ in } L^4(K), \quad D_{A_h} H_h \rightarrow D_A H \text{ in } L^2(K), \quad (\text{B.9})$$

$$\Psi_h \rightarrow \Psi, \quad \bar{\Psi}_h \rightarrow \bar{\Psi} \text{ in } H^1(K), \quad (\text{B.10})$$

$$\theta_h \rightarrow \theta, \quad \kappa_h, g_{j,h}, \lambda_{H,h} \text{ remain positive and bounded away from zero.} \quad (\text{B.11})$$

The limiting coframe is in the same orientation and signature chart. The curvatures in [\(B.8\)](#) are identified with the reconstructed connections, not independent weak tensor limits.

These sufficient conditions use screens for the actual curvature and covariant-gradient packets, the coframe noncollapse bound and, on the stated strong-packet branch, [Proposition B.2](#) for the Higgs amplitude. No optimality is claimed.

The Higgs conditions imply strong ordinary H^1 convergence because

$$dH_h = D_{A_h}H_h - \rho_H(A_h)H_h$$

and the second term converges strongly in L^2 by Hölder. For spinors, route (C4a) is a direct compactness certificate, whereas route (C4b) derives the same H^1 compactness from the Lorentzian Dirac evolution. No elliptic coercivity of a Lorentzian Dirac graph norm is used.

Theorem B.7 (Finite certificates produce the classical strong packet). *If a finite-interface regulator sequence satisfies [Definition B.5](#) on every $K \Subset M$, then every cutoff sequence has a subsequence and a field tuple $z = (e, A, H, \Psi, \bar{\Psi})$ for which the convergences of [Definition B.6](#) hold locally. The limiting curvature is F_A and the limiting Higgs derivative is $D_A H$; they are not independent tensor limits.*

Proof. On each compact chart, extract the uniformly convergent coframes and the L^4 -convergent connections. The common screens give strong L^2 convergence of the coframe derivatives, curvatures and covariant Higgs gradients. Passing to distributions in $dA_h = F_{A_h} - A_h \wedge A_h$ identifies the curvature. Rellich compactness and [Proposition B.2](#) give strong L^4 Higgs convergence; the identity $dH_h = D_{A_h}H_h - \rho_H(A_h)H_h$ identifies the gradient and gives strong H^1 convergence. Under (C4a), value compactness and the first-jet screens give strong H^1 spinor convergence; under (C4b), apply [Proposition B.4](#). Extract the coefficient bank and diagonalize over the atlas and compact exhaustion. Uniform inverse bounds preserve the signature chart. $\square$

B.4 Subsequential variational closure and cofinal uniqueness

Corollary B.8 (Strong-packet refinement of variational closure). *Let $(\mathfrak{I}_h)_{h \downarrow 0}$ be any finite-interface regulator sequence in the sense of [Definition 2.5](#). Assume it satisfies the classical compactness certificate of [Definition B.5](#) on every $K \Subset M$, the finite reconstruction is first-variation consistent,*

$$c_h(K) \longrightarrow 0, \tag{B.12}$$

and the physical common-action residual is controlled by [Proposition 2.6](#) with

$$L_h \sqrt{R_h} + e_h \longrightarrow 0. \tag{B.13}$$

Then every cutoff sequence $h_n \downarrow 0$ has a subsequence h_{n_j} and a field tuple $z = (e, A, H, \Psi, \bar{\Psi})$ such that $z_{h_{n_j}}$ converges to z in the classical strong-packet topology on compact subsets. Along this subsequence the reconstructed first variations converge to $D\mathcal{S}_\theta(z)$ and

$$D\mathcal{S}_\theta(z)[v] = 0 \quad (v \in \mathcal{V}_K). \tag{B.14}$$

Consequently the limiting fields solve the classical Yang–Mills, Higgs, and Dirac–Yukawa equations, and

$$G_{\mu\nu}(g) + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu}^{\text{SM}}(z) \quad \text{in } \mathcal{D}'(M). \tag{B.15}$$

More precisely, for the selected subsequence,

$$\|D\mathcal{S}_\theta(z)\|_{(\mathcal{V}_K^{r_0})^*} \leq L_{h_{n_j}} \sqrt{R_{h_{n_j}}} + e_{h_{n_j}} + c_{h_{n_j}}(K) + C_K d_K(z_{h_{n_j}}, z). \tag{B.16}$$

No summable adjacent-cutoff transport, continuum stress tensor, Euler equation, or candidate strong limit is assumed. Compactness produces accumulation points; consistency and stationarity identify every such strong-packet accumulation point as a classical Einstein–Standard-Model solution.

Proof. [Theorem B.7](#) supplies a strong-packet subsequence. [Proposition A.5](#) bounds its continuum first-variation difference by $C_K d_K$; [Proposition 2.6](#) bounds the finite stationarity term by $L_h \sqrt{R_h} + e_h$. Adding the consistency error gives (B.16). All three contributions vanish. Independent matter tests yield their Euler rows, and the inverse-metric test with (2.10) gives (B.15). The gravitational distribution is the first-order representative of [Section A.5](#). $\square$

Proposition B.9 (Summable cofinal transport). *Let $h_n \downarrow 0$ be a cofinal sequence satisfying the classical compactness certificate on every compact region, with adjacent cutoffs represented on the same local bundles. Suppose*

$$d_K(z_{h_{n+1}}, z_{h_n}) \leq \eta_{n,K}, \quad \sum_{n=1}^{\infty} \eta_{n,K} < \infty \quad (\text{B.17})$$

for every $K \in M$. Then (z_{h_n}) is Cauchy in each norm entering d_K and converges to a unique local strong-packet limit. If two admissible cofinal routes admit direct comparisons whose d_K error tends to zero after both cutoffs pass a common scale, their limits agree. In particular, if [Theorem B.7](#) supplies subsequential strong-packet limits and (B.17) holds, no subsequence choice remains.

Proof. For $m > n$, the triangle inequality bounds $d_K(z_{h_m}, z_{h_n})$ by $\sum_{j=n}^{m-1} \eta_{j,K} \rightarrow 0$. Completeness gives a unique norm limit; a subsequential strong packet from [Theorem B.7](#) supplies its curvature and covariant-derivative identifications. Passing the direct comparison between two routes to their limits proves route independence. $\square$

Corollary B.10 (Cofinal uniqueness and route independence). *Under the hypotheses of [Corollary B.8](#), let a cofinal sequence additionally satisfy the summable transport estimate (B.17). Then the entire cofinal sequence converges in the strong-packet topology to a single classical Einstein–Standard-Model solution. If two admissible cofinal routes satisfy the direct-comparison hypothesis of [Proposition B.9](#), their limits agree.*

Proof. Apply [Proposition B.9](#) and identify the unique accumulation point by [Corollary B.8](#). $\square$

Theorem B.11 (Regularity criterion for variational closure). *This is a sufficient regularity criterion. Let $(\mathfrak{I}_h)$ be a finite-interface regulator sequence and let $Q = I \times \Sigma \in M$ be a compact slab. Suppose that for some $\sigma > 0$, in fixed local gauges and frames,*

$$\sup_h \left(\|e_h\|_{H^{3+\sigma}(Q)} + \|A_h\|_{H^{3+\sigma}(Q)} + \|H_h\|_{H^{3+\sigma}(Q)} \right) < \infty, \quad \sup_h \|e_h^{-1}\|_{L^\infty(Q)} < \infty, \quad (\text{B.18})$$

and the coefficient banks remain in a compact physical set. Suppose the reconstructed spinors and dual spinors obey the Lorentzian systems of [Proposition B.4](#), with uniformly bounded $L_t^\infty H_x^2$ norms, precompact initial data in $L^2(\Sigma)$, and residuals tending to zero in $L_t^2 L_x^2$. Finally assume first-variation consistency and scaled stationarity, (B.12) and (B.13).

Then every cutoff sequence has a subsequence converging in the full classical strong packet on Q , and every such limit solves the classical Einstein–Standard-Model equations there. No compact-screen hypothesis and no a-priori spinor first-jet compactness is required on this regular branch.

Proof. [Proposition B.3](#) gives strong bosonic compactness. The stronger bounds (B.18) also give uniform coefficient convergence and the bounds required by [Proposition B.4](#). Extract Cauchy initial spinors and use the vanishing residuals to obtain their strong spacetime H^1 convergence. The resulting strong packet, consistency and scaled stationarity imply closure by [Corollary B.8](#). $\square$

Corollary B.12 (Renewal Geometry realization). *On the source-native Standard-Model branch of Proposition D.4, the compactness, consistency and scaled stationarity hypotheses of Corollary B.8 give a subsequential Einstein–Standard-Model limit. The additional hypotheses of Corollary B.10 give a unique route-independent limit. For the explicit local action, the finite physical-row, initial/gauge and leakage budgets of Theorem 5.1 instead yield quantitative regularity and stress convergence, with cofinal summability on the rate family of Corollary H.9.*

Proof. Combine the structural realization Proposition D.4 with Corollary B.8, and use Corollary B.10 for the optional transport conclusion. The quantitative statements follow from Theorem 5.1 and Corollary H.9. $\square$

Corollary B.13 (Exact finite critical branch). *Under the compactness and consistency hypotheses of Corollary B.8, suppose the finite configurations are exact critical points in all lifted physical directions,*

$$D\mathcal{S}_h(z_h^d)[\mathcal{I}_h v] = 0 \quad (v \in \mathcal{V}_K).$$

Then every strong-packet accumulation point satisfies (B.15) and all classical matter Euler equations without a separate source-residual hypothesis.

Proof. Exact criticality gives $\varepsilon_h(K) = 0$ in (2.14). The proof of Corollary B.8 then uses only $c_h(K) \rightarrow 0$ and $d_K(z_h, z) \rightarrow 0$. $\square$

Corollary B.14 (Distributional and negative-Sobolev stress convergence). *Assume the sectorwise consistency in (2.13). Define the operational stress density by*

$$\langle \mathbb{T}_h, k \rangle = -2D\mathcal{S}_{\text{SM},h}[\mathcal{I}_h(k, 0, 0, 0, 0)].$$

Along every strong-packet subsequence furnished by Corollary B.8, $\mathbb{T}_h \rightarrow T^{\text{SM}}\text{dV}_g$ in distributions. On each fixed compact test region it also converges in H^{-m} for every integer $m > r_0 + 2$, after using the fixed reference density and bundle frames.

Proof. Sectorwise consistency and Proposition A.5 give convergence in the dual C^{r_0} norm. Restrict the test functional to H_0^m and use $H_0^m \hookrightarrow C^{r_0}$ for $m > r_0 + 2$ in four dimensions. $\square$

Corollary B.15 (A realized random branch). *On a common probability space suppose the reconstructed fields satisfy the compactness and consistency hypotheses of Corollary B.8 and the cofinal-transport hypotheses of Corollary B.10 almost surely, with limiting field $z(\omega)$. Let the source estimate of Proposition 2.6 hold pathwise with deterministic L_h and $e_h \rightarrow 0$, and assume*

$$\sum_h L_h^2 \mathbb{E} \|a_h\|_{G_h^{-1}}^2 < \infty.$$

Then the limiting fields satisfy (B.15) and the matter Euler equations almost surely. It is enough to formulate the hypotheses on a countable dense test core with the displayed uniform bounds.

Proof. Tonelli gives $\sum_h L_h^2 \|a_h\|_{G_h^{-1}}^2 < \infty$ almost surely. Thus Proposition 2.6 makes the physical residual vanish. Apply Corollaries B.8 and B.10 on a countable determining core and extend by continuity. Independence is unnecessary. $\square$

B.5 Conditional regularity upgrade

Corollary B.16 (Regularity upgrade on a common slab). *Let a limit furnished by [Theorem 3.3](#) additionally satisfy, in fixed gauges on $I \times \Sigma$,*

$$(g, A, H) \in C_t H_x^s \cap C_t^1 H_x^{s-1} \cap C_t^2 H_x^{s-2}, \quad (\Psi, \bar{\Psi}) \in C_t H_x^{s-1} \cap C_t^1 H_x^{s-2}, \quad s \geq 5,$$

with a common nondegenerate Lorentzian chart. Then its distributional Euler identities are strong Sobolev field equations. In particular all second-order bosonic and first-order spinor rows vanish in $C_t H_x^{s-2}$. If the fields are smooth, these are the pointwise Einstein–Standard-Model equations.

Proof. Sobolev multiplication on the three-dimensional closed slices makes all coefficients and products in the Euler rows continuous in the indicated Sobolev spaces. The distribution represented by each such continuous row is zero by [Theorem 3.3](#); therefore the row is zero in that space. This is a regularity upgrade conditional on the displayed bounds, not a smoothing conclusion for a hyperbolic equation. $\square$

Remark B.17 (Stronger compactness and applications). The strong-packet criteria, explicit spinor evolution estimate, summable cofinal transport, exact-critical-point and realized-random refinements are proved in [Appendix B](#). They refine the reduced theorem when norm convergence of spinor first jets or a pathwise cofinal limit is needed. The structural realization [Proposition D.4](#) may be combined with [Theorem 3.3](#) once its analytic hypotheses are verified; that combination does not infer those hypotheses from the representation packet.

C Weak-limit estimates and diagnostic examples

This appendix proves the sectorwise passages used in the joint defect identity and the critical gauge-quotient theorem. Uniform coefficient convergence is used for measure-valued stresses; conservation has the additional connection and Noether hypotheses stated explicitly below.

C.1 Critical Higgs and Yang–Mills stress measures

We first weaken the matter compactness rather than silently deleting the resulting source. Work on a compact chart with smooth g_h converging uniformly to a continuous Lorentzian metric g , uniform inverse bounds and fixed signature, $A_h \rightarrow A$ in L^4 , and

$$H_h \rightharpoonup H \text{ in } H^1, \quad H_h \rightarrow H \text{ almost everywhere and in } L^q \ (q < 4). \quad (\text{C.1})$$

The latter convergence follows along a subsequence from Rellich compactness. These are positive comparison norms even though the action is Lorentzian. Uniform convergence of the metric, its inverse and the positive volume density is sufficient for all algebraic stress-measure passages in this subsection. The limiting coefficients have their continuous representatives, so their products with singular Radon measures are well defined. No convergence of metric derivatives is used for those passages. A separate C^1 hypothesis will be imposed when a covariant conservation identity is passed to the limit.

Lemma C.1 (Cubic Euler terms and quartic concentration). *Under [\(C.1\)](#),*

$$|H_h|^2 H_h \rightharpoonup |H|^2 H \text{ in } L^{4/3}, \quad D_{A_h} H_h \rightharpoonup D_A H \text{ in } L^2. \quad (\text{C.2})$$

After extraction on a fixed compact domain, there is a nonnegative Radon measure ν_H with

$$|H_h|^4 dV_{g_h} \rightharpoonup^* |H|^4 dV_g + \nu_H. \quad (\text{C.3})$$

On that compact domain, including its boundary, $\nu_H = 0$ is equivalent to strong L^4 convergence along the extracted sequence. The local version applies on smaller compact regions, without excluding escape through the edge of a larger chart.

Proof. The cubic sequence is bounded in the reflexive space $L^{4/3}$ and converges almost everywhere; boundedness and convergence in measure identify every weak subsequential limit with the displayed cubic. For the covariant derivative, write

$$A_h H_h - A H = (A_h - A) H_h + A(H_h - H).$$

The first term tends to zero in L^2 . For any L^2 test v , the second term pairs with $A^* v \in L^{4/3}$ and tends to zero by weak L^4 convergence of H_h . The ordinary derivative converges weakly in L^2 .

The fourth-power densities have bounded mass. Weak-star compactness and Fatou's lemma against nonnegative continuous tests give (C.3). On a compact domain, testing with one gives convergence of fourth-power integrals if $\nu_H = 0$. Weak L^4 convergence and uniform convexity then imply strong convergence. The converse follows from the product estimate (A.11) and uniform convergence of the volume density. $\square$

Let $\mathfrak{K}_H$ be the tensor-valued measure defect in the complete kinetic stress:

$$\begin{aligned} & \left(2 \operatorname{Re} \langle D_\mu H_h, D_\nu H_h \rangle - g_{h,\mu\nu} |D_{A_h} H_h|_{g_h}^2 \right) dV_{g_h} \rightharpoonup^* \\ & \left(2 \operatorname{Re} \langle D_\mu H, D_\nu H \rangle - g_{\mu\nu} |D_A H|_g^2 \right) dV_g + \mathfrak{K}_{H,\mu\nu}. \end{aligned} \quad (\text{C.4})$$

Such a subsequence exists by the positive L^2 bound on the covariant derivatives. No positivity of the Lorentzian tensor $\mathfrak{K}_H$ is claimed. Once the weak covariant-gradient limit has been identified, this stress passage uses no further convergence of the connection. In particular it applies to the critical quotient class below, where the connection is only strongly compact in subcritical L^q .

Theorem C.2 (Critical Higgs stress-defect identity). *For the minimal action (2.7), assume uniform metric convergence with continuous limit and uniform inverse bounds, the Higgs convergence (C.1), an identified weak gradient limit $D_{A_h} H_h \rightharpoonup D_A H$ in L^2 , and convergent parameters. Then*

$$T_h^H dV_{g_h} \rightharpoonup^* T^H dV_g + \mathfrak{S}_H, \quad \mathfrak{S}_{H,\mu\nu} = \mathfrak{K}_{H,\mu\nu} - \lambda_H g_{\mu\nu} \nu_H. \quad (\text{C.5})$$

If the remaining stress contributions and the gravitational first variation have their intended limits, and the operational metric residual and consistency error vanish, then

$$(\mathbb{G}(g) + \Lambda g) dV_g = \kappa (T^{\text{SM}} dV_g + \mathfrak{S}_H). \quad (\text{C.6})$$

For the conservation assertion assume additionally $g_h \rightarrow g$ in C^1 , that the total reconstructed stress densities are uniformly bounded as Radon measures and converge weakly star, and that their covariant divergences vanish distributionally in the limit. Then their limiting stress is conserved. If the regular limiting matter stress also satisfies its Noether identity (in particular for smooth limiting matter solutions), $\nabla^\mu \mathfrak{S}_{H,\mu\nu} = 0$. Distributional matter stationarity alone at the reduced regularity is not used to assert this extra Noether identity. Conservation does not imply $\mathfrak{S}_H = 0$.

Proof. The kinetic term is (C.4). In the potential, the quadratic density $|H_h|^2$ converges in L^1 , and the constant term converges with the volume density. The only additional potential contribution is the quartic measure (C.3), with sign fixed by (2.12). This proves (C.5). Test the metric equation with a compactly supported smooth k and pass each term to obtain (C.6).

The conservation statement follows from the separate transport lemma below, using the additional C^1 metric convergence. Subtract the limiting regular Noether identity only when it is defined and holds. A nonzero constant symmetric tensor on a flat chart already disproves the implication from vanishing divergence to a vanishing tensor. $\square$

Lemma C.3 (Covariant conservation under a strong connection limit). *Let smooth metrics g_h converge to g in C^1 on a compact chart, with uniform inverse bounds. Let symmetric covariant tensor densities T_h be uniformly bounded Radon measures and converge weakly star to T . If, for every compactly supported smooth vector field X ,*

$$\int \nabla_{g_h}^{(\mu} X^{\nu)} d\mathsf{T}_{h,\mu\nu} \longrightarrow 0, \quad (\text{C.7})$$

then $\int \nabla_g^{(\mu} X^{\nu)} d\mathsf{T}_{\mu\nu} = 0$. If $\mathsf{T} = T(z)dV_g + \mathfrak{S}$ and $T(z)dV_g$ separately has this identity, then $\mathfrak{S}$ has it as well.

Proof. Put $B_h(X) = \nabla_{g_h}^{(\mu} X^{\nu)}$. The coordinate formula for the covariant derivative, the inverse-metric bounds and C^1 convergence give $\|B_h(X) - B(X)\|_\infty \rightarrow 0$, with continuous $B(X)$. Thus

$$\left| \int (B_h(X) - B(X)) d\mathsf{T}_h \right| \leq \|B_h(X) - B(X)\|_\infty |\mathsf{T}_h|(K) \rightarrow 0.$$

The remaining fixed continuous coefficient passes through weak-star convergence. Subtraction proves the last assertion. Merely uniform metric convergence would not control $B_h(X)$ and is not sufficient for this step. $\square$

Proposition C.4 (Yang–Mills quadratic stress defect). *Work on a compact chart with smooth g_h converging uniformly to a continuous Lorentzian metric g , uniform inverse bounds and fixed signature, and assume an identified curvature limit*

$$\mathsf{F}_{A_h} \rightharpoonup \mathsf{F}_A \quad \text{in } L^2.$$

After extraction there is a symmetric tensor-valued finite Radon measure $\mathfrak{S}_{\text{YM}}$ such that

$$T_h^{\text{YM}} dV_{g_h} \rightharpoonup^* T^{\text{YM}} dV_g + \mathfrak{S}_{\text{YM}}. \quad (\text{C.8})$$

Strong L^2 convergence of F_{A_h} is sufficient for $\mathfrak{S}_{\text{YM}} = 0$. No strong L^4 convergence of A_h is needed for this algebraic stress-measure statement once its weak curvature limit is known to be F_A .

Proof. The positive reference L^2 bound on the curvature gives a uniform L^1 bound for every component of the quadratic stress density because the metric coefficients and volume densities converge uniformly. Weak-star compactness of finite measures therefore gives a subsequential tensor-valued limit. Subtracting the regular quadratic tensor built from F_A defines $\mathfrak{S}_{\text{YM}}$. If the curvatures converge strongly in L^2 , the quadratic product estimate (A.4) makes the difference of the stress densities tend to zero in L^1 . $\square$

Proposition C.5 (Matter equations survive the bosonic weak limit). *In the weak Yang–Mills and Higgs class above, suppose additionally that $e_h \rightarrow e$ in $L^\infty \cap H^1$ in compatible spin frames, with uniformly bounded inverse, and that $\Psi_h, \bar{\Psi}_h$ converge weakly in H^1 . If first-variation consistency and physical stationarity hold for all matter tests, the limiting Yang–Mills, Higgs and Dirac–Yukawa equations hold in distributions. The complete fermionic metric first variation has its intended limit without a separate stress-convergence assumption.*

Proof. In a gauge variation, $da + [A_h, a]$ converges strongly in L^2 and pairs with the weakly convergent curvature. In the Higgs kinetic variation, $D_{A_h} \eta_H$ converges strongly in L^2 . The gauge-current contribution pairs $D_{A_h} H_h$ with $\rho_H(a) H_h$; the latter converges strongly in L^2 by Rellich compactness. The cubic potential force converges weakly in $L^{4/3}$ by Lemma C.1. Thus all bosonic matter variations converge against fixed smooth tests, even though their metric variations may retain quadratic defects. The weak- H^1 fermion argument of Proposition 3.5 applies with $H_h \rightarrow H$ in L^2 and a uniform L^4 bound, as established there. This passes every fermionic matter and metric variation. The uniform C^1 test bound and a finite-net argument give convergence on the C^{r_0} test unit ball. Consistency and stationarity now imply the limiting matter Euler identities by Lemma A.1 restricted to matter tests. $\square$

C.2 Positive covariance measures

Lemma C.6 (Covariance measure of a weak quadratic packet). *Let K be a compact smooth domain, including its boundary, and let $Y_h \rightarrow Y$ in $L^2(K; E)$ for a fixed finite-rank real bundle with a smooth positive fiber metric. After extraction,*

$$Y_h \otimes Y_h dV_0 \rightharpoonup^* Y \otimes Y dV_0 + Q_Y, \quad |Y_h|^2 dV_0 \rightharpoonup^* |Y|^2 dV_0 + \mu_Y, \quad (\text{C.9})$$

where Q_Y is a positive-semidefinite symmetric matrix-valued Radon measure and $\mu_Y = \text{tr } Q_Y \geq 0$. The following conditions are equivalent along this subsequence:

$$\mu_Y = 0 \iff Q_Y = 0 \iff Y_h \rightarrow Y \text{ in } L^2(K; E). \quad (\text{C.10})$$

If $B_h \rightarrow B$ uniformly are continuous bilinear coefficient fields, then

$$B_h(Y_h, Y_h) dV_0 \rightharpoonup^* B(Y, Y) dV_0 + B : Q_Y, \quad |B : Q_Y| \leq C_B \mu_Y. \quad (\text{C.11})$$

Complex packets are treated by realification.

Proof. The quadratic entries are bounded in L^1 , so extract a common weak-star measure limit in a finite atlas. For a continuous real dual section a and a nonnegative continuous function φ , multiplication by $\sqrt{\varphi} a$ preserves weak L^2 convergence. Lower semicontinuity therefore gives

$$\int_K \varphi a^T Q_Y a \geq 0.$$

This proves positivity; taking the trace gives $\mu_Y \geq 0$. Writing the matrix measure relative to its trace, its density is positive semidefinite with trace one almost everywhere. In particular its Hilbert–Schmidt norm is at most one. This proves the domination in (C.11), and shows that trace zero is equivalent to matrix-measure zero. If $\mu_Y = 0$, testing the scalar measure convergence with the constant function one gives convergence of squared norms. Weak convergence and the Hilbert identity then imply strong convergence. The converse follows from the product estimate (A.4). Finally, the uniform coefficient error has total variation at most $\|B_h - B\|_\infty \sup_h \|Y_h\|_2^2$, which tends to zero; the fixed continuous coefficient B passes through weak-star convergence. $\square$

C.3 Equivariant tests and critical gauge compactness

Lemma C.7 (Equivariant physical tests). *For every smooth internal gauge transformation u ,*

$$\mathbf{n}_{u \cdot z}(u \cdot v) = \mathbf{n}_z(v). \quad (\text{C.12})$$

On every bounded weak packet for which $e, e^{-1} \in L^\infty$, $e \in H^1$, $A \in L^4$, $F_A, D_A H \in L^2$, $H \in H^1$, and $\Psi, \bar{\Psi} \in H^1$, the complete minimally coupled first variation satisfies

$$|D\mathcal{S}_\theta(z)[v]| \leq C_K \mathbf{n}_z(v). \quad (\text{C.13})$$

Proof. Internal gauge transformations act unitarily on every matter representation, and the covariant derivatives of the transformed tests transform homogeneously. This proves (C.12). For the second assertion, the Yang–Mills row pairs F_A with $D_A A$; the Higgs kinetic row pairs $D_A H$ with $D_A \eta_H + \rho_H(a)H$; and the quartic force is integrable by $H^1 \hookrightarrow L^4$ in four dimensions. The Dirac terms are controlled by one L^2 spinor derivative and bounded test coefficients. The metric variation of the spin connection has the form $P_1(e)k \partial e + P_2(e)\partial k$ and therefore lies in L^2 ; it pairs with the L^2 spinor bilinear. Finally the first-order Palatini representative of Section A.5 contains only bounded coframe coefficients, first coframe derivatives in L^2 , and $k, \partial k$. Summing the sector estimates proves (C.13). $\square$

Proposition C.8 (Critical Uhlenbeck quotient compactness). *Let $K' \Subset K$ and suppose*

$$\sup_h \|F_{A_h}\|_{L^2(K)} < \infty \quad (\text{C.14})$$

and Definition 4.5 holds. Then K' has a finite ball cover and cutoff-dependent local internal gauges in which

$$d^* A_h = 0, \quad \sup_h \|A_h\|_{W^{1,2}} < \infty, \quad (\text{C.15})$$

with the L^4 norms uniformly as small as desired after refining the cover. After extraction,

$$\begin{aligned} A_h &\rightharpoonup A && \text{in } W^{1,2}, \\ A_h &\rightarrow A && \text{in } L^q \quad (1 \leq q < 4), \\ F_{A_h} &\rightharpoonup F_A && \text{in } L^2. \end{aligned} \quad (\text{C.16})$$

Proof. Uniform integrability gives a radius, independent of the cutoff and the center, on which every curvature L^2 norm lies below the four-dimensional Uhlenbeck threshold. The local Coulomb theorem then yields

$$\|A_h\|_{W^{1,2}} + \|A_h\|_{L^4} \leq C \|F_{A_h}\|_{L^2}$$

on the corresponding smaller balls. Weak $W^{1,2}$ compactness and Rellich compactness give the first two limits. Since $A_h \rightarrow A$ strongly in L^2 , $A_h \wedge A_h \rightarrow A \wedge A$ in L^1 , while $dA_h \rightharpoonup dA$ in L^2 ; hence the weak curvature limit is F_A . The smallness of the local L^4 norm follows from the same small-energy estimate. $\square$

Lemma C.9 (Positive covariant graph control on a normalized chart). *Let B be a fixed bounded smooth four-dimensional chart, with a smooth positive-metric reference connection ∇^{ref} . Suppose a unitary internal representation has $\|A\|_4 \leq M_A$. For every smooth section u ,*

$$\|u\|_4 \leq C_B (\|u\|_2 + \|\nabla^{\text{ref}, A} u\|_2), \quad (\text{C.17})$$

$$\|u\|_{H^1} \leq C_B (1 + M_A) (\|u\|_2 + \|\nabla^{\text{ref}, A} u\|_2). \quad (\text{C.18})$$

The same estimates hold for the dual representation. The constants do not depend on the gauge transformation used to produce the displayed A .

Proof. Apply the scalar Sobolev inequality to $(|u|^2 + \epsilon^2)^{1/2}$ and then let $\epsilon \downarrow 0$. Metric compatibility and unitarity give the Kato inequality $|d|u|| \leq |\nabla^{\text{ref}, A} u|$, proving (C.17). This scalar inequality does not by itself control the full derivative of u . For that derivative use the exact identity

$$\nabla^{\text{ref}} u = \nabla^{\text{ref}, A} u - \rho(A)u, \quad \|\nabla^{\text{ref}} u\|_2 \leq \|\nabla^{\text{ref}, A} u\|_2 + C\|A\|_4\|u\|_4.$$

Together with the first estimate and the fixed smooth reference coefficients, this proves (C.18). No smallness of M_A is needed. $\square$

C.4 Full proof of critical gauge-quotient closure

Complete proof of Theorem 4.6. Extract the geometric and parameter subsequence and apply Proposition C.8. On each sufficiently small Coulomb ball the connection has uniformly small L^4 norm. The identities

$$dH_h = D_{A_h} H_h - \rho_H(A_h) H_h$$

and the four-dimensional Sobolev inequality give

$$\|H_h\|_4 \leq C(\|D_{A_h} H_h\|_2 + \|H_h\|_2) + C\|A_h\|_4\|H_h\|_4.$$

The final term is absorbed by the local small-energy choice, giving an ordinary H^1 bound for H_h . Rellich compactness gives strong L^q for $q < 4$ and almost-everywhere convergence. The weak limit of $D_{A_h} H_h$ is identified with $D_A H$ distributionally because $A_h \rightarrow A$ and $H_h \rightarrow H$ strongly in L^2 .

Apply Lemma C.9 to (4.12) in the normalized Coulomb chart. First Kato and scalar Sobolev give uniform L^4 spinor bounds; then subtracting $\rho(A_h)\Psi_h$ from the covariant derivative gives the full ordinary H^1 bound. Do the same in the dual representation. After extraction both spinors converge strongly in every L^q , $q < 4$, and their bilinears converge weakly in L^2 . This uses the positive internal comparison connection, not an elliptic estimate for the physical Lorentzian Dirac operator.

Fix now a smooth test in a normalized chart and pull it back through the cutoff-dependent gauge. Equation (C.12) keeps its test size uniformly bounded, so the finite first variation tends to zero without differentiating the gauge. In the Yang–Mills row, $D_{A_h} a \rightarrow D_A a$ strongly in L^2 for fixed smooth a and therefore pairs with the weak L^2 curvature. In the Higgs row, $D_{A_h} \eta_H \rightarrow D_A \eta_H$ strongly in L^2 , while the gauge-current term pairs the weak L^2 covariant gradient with the strongly convergent undifferentiated Higgs field. The cubic potential force is bounded in $L^{4/3}$ and converges almost everywhere, hence has the classical weak limit. For the Dirac rows, derivative terms use weak–strong pairing. For example, $A_h \bar{\Psi}_h \Psi_h \rightarrow A \bar{\Psi} \Psi$ and $H_h \bar{\Psi}_h \Psi_h \rightarrow H \bar{\Psi} \Psi$ strongly in L^1 , since all three factors converge strongly in L^3 . Terms with a smooth matter test require no larger exponents. Spin-connection terms pair a strongly convergent L^2 coefficient with a weakly convergent L^2 bilinear. Consequently the complete fermionic metric variation passes by the relaxed strong- L^2 coefficient version of Proposition 3.5.

Only the quadratic Yang–Mills/Higgs metric terms and the critical quartic potential can retain unresolved products. Their limits are precisely the positive covariance and quartic defects of Proposition C.4, Theorem C.2, and Lemma C.6. The coframes converge uniformly to their continuous representatives and strongly in H^1 . Thus the uniform algebraic metric-coefficient hypotheses of the stress-defect propositions hold, while strong first-jet convergence passes the first-order gravitational variation. This proves (4.14). No conservation of a measure-valued defect is

inferred from this step; such a conclusion would require the additional connection convergence and Noether hypotheses of [Lemma C.3](#). Finally, uniform integrability makes the quadratic covariance measures absolutely continuous; uniform integrability of $|H_h|^4$ together with convergence in measure gives strong L^4 convergence by Vitali, so the quartic concentration measure vanishes. $\square$

C.5 Concentration, oscillation and an alternative Euclidean branch

Remark C.10 (Zero defect versus the stronger Higgs higher-integrability endpoint). The reduced closure theorem and the exact zero-defect characterization require strong L^2 compactness of $D_{A_h} H_h$, together with the connection compactness used in [Proposition 3.4](#); they do not require a uniform $L^{8/3}$ bound on the Higgs covariant gradient. The latter is a strictly stronger reserve. For the explicit finite Higgs edge packet, [Proposition D.1](#) identifies it exactly with an $L^{4/3}$ bound on a positive companion kinetic density. The same proposition shows why ordinary quadratic screening cannot be silently upgraded to this endpoint.

Example C.11 (A critical bubble). On a four-dimensional chart choose $\varphi \in C_c^\infty(\mathbb{R}^4)$, a point x_0 , and a unit Higgs vector v . Set

$$H_\epsilon(x) = \epsilon^{-1} \varphi((x - x_0)/\epsilon) v.$$

Then $\|H_\epsilon\|_2 = O(\epsilon)$, while $\|dH_\epsilon\|_2$ and $\|H_\epsilon\|_4$ are independent of ϵ . The fields converge weakly in H^1 to zero, but

$$|H_\epsilon|^4 dx \rightharpoonup^* \left(\int_{\mathbb{R}^4} |\varphi|^4 dx \right) \delta_{x_0}.$$

These assertions follow directly by $x = x_0 + \epsilon y$. This example disproves a compactness inference from bounded critical energy; it does not assert that this bubble is a coupled stationary solution.

Example C.12 (Vanishing matter residual with conserved kinetic defect). On flat $(0, T) \times \mathbb{T}^3$, take $v_H = 0$, $A = 0$, $\Psi = 0$, a fixed unit Higgs vector v , and

$$H_n(t, x) = n^{-1} \sin(n(t + x^1)) v, \quad \ell = dt + dx^1.$$

Since ℓ is null, $\square H_n = 0$. The Higgs Euler residual is the cubic potential force and is $O(n^{-3})$ uniformly. The gauge current vanishes because the amplitude is real and has a fixed internal direction. Although $H_n \rightarrow 0$ strongly in L^4 ,

$$T_n^H = 2 \cos^2(n(t + x^1)) \ell \otimes \ell - \lambda_H n^{-4} \sin^4(n(t + x^1)) g \rightharpoonup \ell \otimes \ell.$$

Thus $\nu_H = 0$ but the kinetic defect is nonzero and conserved. The example concerns the matter system on a fixed background: its nonzero metric residual is exactly what prevents it from being a vacuum Einstein limit. It isolates the error in trying to derive stress convergence from matter stationarity or Bianchi conservation alone.

Proposition C.13 (A sufficient Euclidean defect-removal mechanism). *Let $V \hookrightarrow L^4$ be a real Hilbert space, and let the complete gauge-fixed Euclidean Higgs operators $\mathcal{A}_h : V \rightarrow V'$ satisfy*

$$\langle \mathcal{A}_h(u) - \mathcal{A}_h(v), u - v \rangle \geq c_1 \|u - v\|_V^2 + c_2 \|u - v\|_4^4, \quad c_1, c_2 > 0.$$

If $H_h \rightharpoonup H$ in V , $\mathcal{A}_h(H_h) = J_h \rightarrow J$ in V' , and $\mathcal{A}_h(H) \rightarrow J$ in V' , then $H_h \rightarrow H$ in $V \cap L^4$. If V controls the covariant H^1 norm, both Higgs defects vanish.

Proof. Use $(u, v) = (H_h, H)$. The right pairing is $\langle J_h - \mathcal{A}_h(H), H_h - H \rangle$, which tends to zero by strong dual convergence and weak boundedness. Each nonnegative term on the left tends to zero. $\square$

This is an alternative sufficient branch, not an assertion about the Lorentzian evolution operator. A positive quartic term alone does not establish the complete monotonicity estimate, particularly with a symmetry-breaking mass term or unfixed gauge directions.

C.6 Nonminimal Higgs curvature and the assembled improvement

The principal reduced theorem concerns the minimally coupled classical action. A retained nonminimal source must be added before taking variations. For the convention

$$\mathcal{S}_\xi = \xi \int f \text{Scal}(g) dV_g, \quad f = H^\dagger H, \quad (\text{C.19})$$

the compactly supported metric variation is

$$D_g \mathcal{S}_\xi[k] = \xi \int [f \mathbf{G}_{\mu\nu} + (g_{\mu\nu} \square_g - \nabla_\mu \nabla_\nu) f] k^{\mu\nu} dV_g. \quad (\text{C.20})$$

Consequently the stress contribution is minus twice the tensor in brackets multiplied by ξ . Formula (C.20) follows by varying $\sqrt{|g|}fR$ and integrating the two derivative terms in δR twice. Compact support is essential to omitting the boundary contribution; with a physical boundary the corresponding scalar–GHY term must be included.

Proposition C.14 (An assembled strong–weak improvement criterion). *Suppose $g_h \rightarrow g$ in $W^{1,\infty}$, $g_h \rightarrow g$ in $W^{2,p}$ for $1 < p < \infty$, with uniform inverse bounds and fixed signature, and $H_h \rightarrow H$ in $L^{2p'}$. Then*

$$f_h \mathbf{G}(g_h) + (g_h \square_{g_h} - \nabla^h \nabla^h) f_h \rightarrow f \mathbf{G}(g) + (g \square_g - \nabla \nabla) f \quad (\text{C.21})$$

in distributions, with the volume densities included in the pairing. If the minimal stress also converges and $\xi_h \rightarrow \xi$, the full improved metric variation converges.

Proof. Strong $L^{2p'}$ convergence gives $f_h \rightarrow f$ in $L^{p'}$. Curvature is linear in second metric derivatives with strongly convergent bounded coefficients, plus quadratic first-derivative terms. Hence $\mathbf{G}(g_h) \rightarrow \mathbf{G}(g)$ in L^p . This passes the first product. For the Hessian contribution, test the *assembled* expression and integrate both derivatives onto the smooth test and metric-density coefficients. The resulting coefficient has a weak L^p limit: its second metric derivatives occur linearly, and every lower-order coefficient converges strongly. Pairing with strong f_h in $L^{p'}$ passes the entire expression. This argument does not separately multiply uncontrolled distributions. $\square$

At the critical $p = 2$ threshold the Higgs input is strong L^4 , not merely a bounded H^1 sequence. If the assembled improvement has a subsequential distributional limit but not the value in (C.21), write its defect as $\mathfrak{J}$. Then the Higgs stress defect becomes

$$\mathfrak{S}_H = \mathfrak{K}_H - \lambda_H g \nu_H - 2\xi \mathfrak{J}. \quad (\text{C.22})$$

Existence of the assembled limit is an additional hypothesis; a critical Higgs norm alone does not supply it.

For completeness, if the scalar-curvature coefficient is $F(H) = 1/(2\kappa) + \xi f$ and is uniformly positive, the conformal change $\tilde{g} = (F/\chi_*)g$ gives a constant Einstein coefficient $\chi_* > 0$. For a real scalar-coordinate kinetic metric G_{AB}^0 , the transformed kinetic metric is

$$\tilde{G}_{AB} = \frac{\chi_*}{F} G_{AB}^0 + \frac{3\chi_*}{F^2} \partial_A F \partial_B F.$$

This identity follows from the four-dimensional conformal scalar-curvature formula and one integration by parts. With a boundary, the derivative term must be combined with the transformed GHY contribution. The change of frame reorganizes the principal coupling but supplies neither an effective-Planck lower bound nor hyperbolic smoothing. It therefore does not remove the compactness conditions of the main theorem.

D The local finite action and comparison reconstructions

We specify the local link action before its critical and quantitative estimates. The final subsection gives a distinct smooth comparison reconstruction. Its sampling statements are not substituted for consistency on arbitrary finite-energy link records.

D.1 The explicit local finite action and its quantitative estimates

This subsection specifies the finite action used in [Section 5](#). It is the local common-action representative built from the finite gravitational and Standard-Model link data, not a least-squares functional or a discretization chosen after the continuum equations are known. The consistency and leakage estimates below analyze this fixed action without adding counterterms or filtering its physical variables. All contractions are based at one lattice vertex. Let e_μ^a be an oriented invertible coframe and $q_{\lambda\nu}^a$ an independent first-jet coordinate. Define

$$\begin{aligned} G_{\lambda,\mu\nu}(e, q) &= \eta_{ab}(q_{\lambda\mu}^a e_\nu^b + e_\mu^a q_{\lambda\nu}^b), \\ \Gamma_{\mu\nu}^\rho(e, q) &= \frac{1}{2}g^{\rho\sigma}(G_{\mu,\nu\sigma} + G_{\nu,\mu\sigma} - G_{\sigma,\mu\nu}), \\ \Omega_\mu(e, q)^a{}_b &= e_\rho^a \Gamma_{\mu\nu}^\rho(e, q) e_b^\nu - q_{\mu\nu}^a e_b^\nu. \end{aligned} \tag{D.1}$$

At $q = \partial e$ this is the Levi-Civita connection. Put

$$\omega_{\mu,h}(x) = \Omega_\mu(e(x), \delta_h^+ e(x)), \quad L_\mu(x) = \exp(h\omega_{\mu,h}(x)), \quad U_\mu(x) = \exp(hA_\mu(x)).$$

For $\mu < \nu$ define the logarithmic plaquettes

$$\begin{aligned} F_{\mu\nu}^h(x) &= h^{-2} \log(U_\mu(x) U_\nu(x + he_\mu) U_\mu(x + he_\nu)^{-1} U_\nu(x)^{-1}), \\ R_{\mu\nu}^h(x) &= h^{-2} \log(L_\mu(x) L_\nu(x + he_\mu) L_\mu(x + he_\nu)^{-1} L_\nu(x)^{-1}). \end{aligned} \tag{D.2}$$

The logarithm is the analytic branch near the identity. Let

$$K_\mu^h = \frac{\rho_H(U_\mu)H(x + he_\mu) - H(x)}{h}, \quad V_\mu = \exp(h\sigma(\omega_{\mu,h}))\rho_S(U_\mu), \tag{D.3}$$

and

$$\nabla_\mu^h \Psi = \frac{V_\mu \Psi(x + he_\mu) - \Psi(x)}{h}, \quad \nabla_\mu^{h,\vee} \bar{\Psi} = \frac{\bar{\Psi}(x + he_\mu) V_\mu^{-1} - \bar{\Psi}(x)}{h}. \tag{D.4}$$

With $v(e) = \sqrt{-\det g}$ and the same coefficient normalization as [\(2.9\)](#), set

$$\begin{aligned} \mathcal{L}_{g,h} &= \frac{v(e)}{2\kappa} e_a^\mu e^{b\nu} (R_{\mu\nu}^h)^a{}_b - \frac{\Lambda}{\kappa} v(e), \\ \mathcal{L}_{\text{YM},h} &= -\frac{1}{4} v(e) g^{\mu\rho} g^{\nu\sigma} \langle F_{\mu\nu}^h, F_{\rho\sigma}^h \rangle g, \\ \mathcal{L}_{\text{H},h} &= -v(e) g^{\mu\nu} \text{Re} \langle K_\mu^h, K_\nu^h \rangle - v(e) V(H), \\ \mathcal{L}_{\text{D},h} &= v(e) \text{Re} \left\{ \frac{i}{2} [\bar{\Psi} \gamma^\mu(e) \nabla_\mu^h \Psi - (\nabla_\mu^{h,\vee} \bar{\Psi}) \gamma^\mu(e) \Psi] - \bar{\Psi} \mathcal{M}_Y(H) \Psi \right\}. \end{aligned} \tag{D.5}$$

The finite local common action is

$$S_h^{\text{loc}} = h^4 \sum_{x \in \mathcal{Q}_h} (\mathcal{L}_{g,h} + \mathcal{L}_{\text{YM},h} + \mathcal{L}_{H,h} + \mathcal{L}_{D,h}). \quad (\text{D.6})$$

It has all coframe components and temporal gauge links as variation variables; temporal gauge is imposed only on the evaluated records. The Cartan and spin links share the same coframe-derived connection.

D.1.1 A positive companion Higgs kinetic writer

The Lorentzian Higgs density in (D.5) is signed. For higher-integrability diagnostics it is useful to retain, on the *same* covariant Higgs edge packet $K_h^H = (K_\mu^h)_\mu$, a positive companion writer. This auxiliary writer does not alter the physical Lorentzian action. Let $r_h^{\mu\nu}(x)$ and $m_h(x) > 0$ be the positive one-form metric and cell mass obtained by sampling the fixed auxiliary Riemannian comparison data (r, dV_0) . On a compact nondegenerate chart assume the usual uniform comparison

$$c_r |\xi|^2 \leq r_h^{\mu\nu} \text{Re} \langle \xi_\mu, \xi_\nu \rangle \leq C_r |\xi|^2, \quad 0 < c_m \leq m_h/h^4 \leq C_m < \infty. \quad (\text{D.7})$$

Define the cellwise positive density

$$e_h^{H,+}(x) := r_h^{\mu\nu}(x) \text{Re} \langle K_\mu^h(x), K_\nu^h(x) \rangle, \quad (\text{D.8})$$

and the discrete positive norms

$$\|f_h\|_{L_h^p}^p := \sum_x m_h(x) |f_h(x)|^p. \quad (\text{D.9})$$

Proposition D.1 (Positive kinetic envelope and the $L^{8/3}$ endpoint). *Under (D.7) there are cutoff-independent constants $0 < c_0 \leq C_0 < \infty$ such that*

$$c_0 |K_h^H(x)|^2 \leq e_h^{H,+}(x) \leq C_0 |K_h^H(x)|^2 \quad (\text{D.10})$$

for every interior cell. Consequently

$$\|e_h^{H,+}\|_{L_h^{4/3}}^{1/2} \asymp \|K_h^H\|_{L_h^{8/3}}. \quad (\text{D.11})$$

On smooth bounded reconstructed families, the consistency estimate of [Proposition D.8](#) for K_h^H identifies this criterion with the corresponding local continuum bound for $D_{A_h} H_h$, up to the already controlled reconstruction error.

If $K_h^H \rightarrow K^H$ strongly in L_h^2 under a common reconstruction, then $e_h^{H,+} \rightarrow e^{H,+}$ strongly in L_{loc}^1 after reconstruction. This conclusion alone does not imply a cutoff-uniform $L^{4/3}$ bound for $e_h^{H,+}$ or an $L^{8/3}$ bound for K_h^H .

Proof. The pointwise comparison (D.10) is exactly the uniform positive definiteness of the sampled comparison Hodge form. Raising it to the power $4/3$, multiplying by m_h , and summing gives

$$\|e_h^{H,+}\|_{L_h^{4/3}}^{4/3} \asymp \sum_x m_h |K_h^H|^{8/3} = \|K_h^H\|_{L_h^{8/3}}^{8/3},$$

which is equivalent to (D.11). The smooth-family continuum statement follows from $K_h^H = D_A H + o(1)$ in the corresponding local norm and the uniform equivalence of the positive cell masses.

For the L^1 statement, use

$$||u|^2 - |v|^2| \leq |u - v|(|u| + |v|)$$

and Cauchy–Schwarz, together with the uniform coefficient convergence of r_h . The failure of an automatic $L^{4/3}$ upgrade is genuine. In a four-dimensional coordinate ball choose a nonzero $\varphi \in C_c^\infty(B_1)$ and set

$$Y_n(x) = n^{7/4} \varphi(n(x - x_0))v$$

for a fixed unit fiber vector v . Then $\|Y_n\|_2 = n^{-1/4} \|\varphi\|_2 \rightarrow 0$, while $\|Y_n\|_{8/3} = n^{1/4} \|\varphi\|_{8/3} \rightarrow \infty$. Thus $|Y_n|^2 \rightarrow 0$ strongly in L^1 although $\| |Y_n|^2 \|_{4/3} \rightarrow \infty$. Sampling a smooth version of this concentrating family on resolving meshes gives the same discrete conclusion. $\square$

Corollary D.2 (A positive amplitude–energy compiler). *If, on every compact interior chart,*

$$\sup_h \|e_h^{H,+}\|_{L_h^1} < \infty, \quad \sup_h \|e_h^{H,+}\|_{L_h^\infty} < \infty, \quad (\text{D.12})$$

then

$$\sup_h \|e_h^{H,+}\|_{L_h^{4/3}} < \infty, \quad \sup_h \|K_h^H\|_{L_h^{8/3}} < \infty. \quad (\text{D.13})$$

Any finite microscopic mechanism used to verify (D.12) must control this same Higgs kinetic writer; a bound for an unrelated positive observable is not a substitute.

Proof. Since $e_h^{H,+} \geq 0$,

$$\int (e_h^{H,+})^{4/3} \leq \|e_h^{H,+}\|_\infty^{1/3} \|e_h^{H,+}\|_1.$$

Apply [Proposition D.1](#). $\square$

Remark D.3 (Role in the present closure theorem). The criterion (D.13) is a stronger higher-integrability card, not a hidden hypothesis of [Theorems 3.3](#) and [4.2](#). Those results close the minimally coupled Higgs stress from strong L^2 compactness of the covariant gradient and the associated $H^1 \cap L^4$ convergence of the Higgs field. The $L^{8/3}$ reserve becomes relevant only in stronger weak-limit or compensated-compactness arguments that require additional integrability of the matter source. Establishing such a reserve from a particular microscopic source law is a separate selection problem.

D.2 Structural realization and representation packet

Proposition D.4 (Renewal realization of the finite interface). *The Renewal Geometry constructions [24, 25, 26] realize [Definition 2.1](#) on a selected Standard-Model branch. In particular, they provide a nonempty family of finite interfaces with the structural bundle content and common-action normalization required here. This proposition is a realization statement only; none of the compactness, first-variation consistency, stationarity, or PDE hypotheses of [Section 3](#) is inferred from those constructions.*

Proof. The three cited constructions provide respectively the Lorentzian carrier and gravitational normalization, the differential/spin realization, and the Standard-Model bundle, generation factor, Dirac/Yukawa incidence, positive source quotient and common action. These are precisely (I1)–(I5); the analytic hypotheses are imposed separately. $\square$

Remark D.5 (Structural provenance versus analytic input). In the structural gravity–matter construction [26], items (I2)–(I5) are organized in part by the double-centralizer relation

$$\mathcal{A}'_{\text{act},h} = \mathcal{M}_{\text{type},h}, \quad \mathcal{M}'_{\text{type},h} = \mathcal{A}_{\text{act},h}, \quad (\text{D.14})$$

and from the common-source Schur quotient

$$G_{i|g,h} = G_{i,h} - C_h G_{g,h}^{-1} C_h^* = S_{i,h}^* (I - P_{g,h}) S_{i,h}. \quad (\text{D.15})$$

These identities explain why the finite matter packet and the physical source normalization are not appended by hand. They are not used to prove spatial or temporal compactness, convergence of F_{A_h} or $D_{A_h} H_h$, spinor H^1 convergence, or any nonlinear stress limit. Those are analytic questions addressed independently below.

To specify the matter carrier independently of its structural origin, use $Q_{\text{em}} = T_3 + Y$ and the representations in Table 2. They give the usual quark, lepton and Higgs assignments [52]. The integer weight of the covering $U(1)$ factor is $6Y$; the fractions in the table therefore do not denote multivalued characters of $U(1)$.

Table 2: The fixed Standard-Model representation packet. Every fermion row carries a generation factor $\mathbb{C}^3$; the Higgs carries no generation factor. The neutral row is included only on the declared neutrino extension.

Field	SU(3)	SU(2)	Y	Spacetime chirality
Q_L	3	2	1/6	left
u_R	3	1	2/3	right
d_R	3	1	−1/3	right
L_L	1	2	−1/2	left
e_R	1	1	−1	right
ν_R (optional)	1	1	0	right
H	1	2	1/2	scalar

Lemma D.6 (Descent and covariant Yukawa blocks). *The representations in Table 2 descend to G_{SM} in (2.1). With $\tilde{H} = i\sigma_2 \bar{H}$ and generation matrices $Y_u, Y_d, Y_e \in M_3(\mathbb{C})$, the mass bilinear is assembled from*

$$\bar{Q}_L \tilde{H} Y_u u_R + \bar{Q}_L \tilde{H} Y_d d_R + \bar{L}_L \tilde{H} Y_e e_R + \text{Hermitian conjugates}. \quad (\text{D.16})$$

On the neutral extension one adds $\bar{L}_L \tilde{H} Y_\nu \nu_R$ and a constant symmetric Majorana block on the charge-conjugate doubled carrier. The resulting map $\mathcal{M}_Y(H)$ is affine in real Higgs coordinates and satisfies (2.5) on every bounded coefficient bank.

Proof. The covering homomorphism is

$$(g_3, g_2, z) \longmapsto (z^{-2} g_3, z^3 g_2).$$

Its kernel is generated by $(e^{2\pi i/3} I_3, -I_2, e^{\pi i/3})$. In each row the product of the color-center character, the weak-center character and z^{6Y} is one, proving descent. Each monomial in (D.16) has zero total hypercharge, and its color and weak indices are contracted invariantly. The same check applies to the optional neutrino coupling; the Majorana block is gauge neutral. The entries are linear in H or $\bar{H}$, apart from that constant block. Finite-dimensional operator bounds give (2.5). $\square$

No value of a generation matrix, gauge coupling or gravitational coefficient is inferred from this representation statement. Omitting the neutral row and its blocks gives the minimal packet; retaining them gives the declared neutral extension throughout the same analytic argument.

D.3 Concrete reconstruction and smooth first-variation consistency

This smooth comparison regulator verifies the reconstruction and first-variation consistency hypotheses of [Definition 2.5](#), with $c_h(K) = O(h)$ on bounded smooth families. It is distinct from the rough-link consistency analysis and does not rederive the structural interface.

D.3.1 Whitney maps and a signed Hodge realization

On a contractible coordinate region choose a quasiuniform shape-regular simplicial mesh of size h . Write C_h^k for oriented k -cochains. For an oriented simplex $\sigma = [i_0, \dots, i_k]$ with barycentric coordinates λ_i , define the Whitney form

$$w_\sigma = k! \sum_{j=0}^k (-1)^j \lambda_{i_j} d\lambda_{i_0} \wedge \dots \wedge \widehat{d\lambda_{i_j}} \wedge \dots \wedge d\lambda_{i_k}, \quad W_h c = \sum_{\sigma} c_{\sigma} w_{\sigma}. \quad (\text{D.17})$$

The de Rham map $R_h \alpha = (\int_{\sigma} \alpha)_{\sigma}$ is used for smooth test forms. It is not asserted to be a bounded point- or trace-evaluation map on arbitrary L^2 forms.

Lemma D.7 (Commuting reconstruction and positive comparison norm). *The maps satisfy $R_h W_h = I$ and $dW_h = W_h d_h$. If*

$$\|c\|_h^2 := \int_K |W_h c|_r^2 dV_0,$$

then W_h is an isometry onto its finite-element range. The reference mass matrix is positive definite. Shape regularity gives the usual cellwise scaling constants uniformly in h . Orthogonal projections P_h^k onto these ranges converge strongly to the identity on L^2 k -forms.

Proof. Integration of (D.17) over oriented simplices gives the Kronecker pairing with the cochain basis. Differentiating the barycentric formula and using $\sum_i \lambda_i = 1$ gives the coboundary identity; equivalently, integrate it and use Stokes on each simplex. The norm statement is its definition, and injectivity follows from $R_h W_h = I$. Scaling to a reference simplex and finite-dimensional norm equivalence give the uniform local mass bounds. Smooth forms are approximated by their Whitney interpolants in L^2 ; this follows by Taylor expansion on each cell and shape-regular scaling. Density of smooth forms and $\|P_h^k\| \leq 1$ give strong convergence of the projections. $\square$

This is the standard untwisted de Rham structure [15, 16]. Gauge curvature is treated separately: for a nonflat connection D_A^2 is curvature, and the ordinary commuting-complex identity is not a claim that $D_A^2 = 0$.

Let $g_h \rightarrow g$ uniformly with fixed signature and uniform inverse bounds. Using the fixed positive reference projections, define a discrete signed Hodge map by

$$W_h^{4-k} \star_h^k = P_h^{4-k} \star_{g_h} W_h^k. \quad (\text{D.18})$$

For $W_h^k c_h \rightarrow \alpha$ in L^2 ,

$$W_h^{4-k} \star_h^k c_h \longrightarrow \star_g \alpha \quad \text{in } L^2. \quad (\text{D.19})$$

Indeed, subtract $\star_g \alpha$, use uniform boundedness of $\star_{g_h}$, uniform convergence of its algebraic coefficients, and strong convergence of P_h^{4-k} . The same proof applies to a fixed smooth metric variation

because $\star_g[k]$ is algebraic in g, g^{-1}, k . The projections here are fixed by the positive reference metric, so no omitted derivative of a metric-dependent projection occurs.

This is a concrete signed-Hodge realization. It neither guarantees strong convergence of $W_h c_h$ nor excludes concentration of its quadratic products. Those are independent compactness requirements on the low-regularity branch; the regular a-priori branch of [Theorem B.11](#) derives them from stronger Sobolev control. Exact mass integration can be replaced by quadrature only after its action and first-variation errors have been estimated.

D.3.2 A finite smooth comparison reconstruction

The following construction suffices to populate the consistency statement; it is not a claim of a canonical interpolation for every renewal graph. On a regular coordinate mesh let $I_h f$ be the continuous piecewise-affine interpolant of nodal samples. Choose a fixed smooth mollifier ρ supported in the unit ball and put

$$\mathcal{J}_h f = \rho_h * I_h f, \quad \rho_h(x) = h^{-4} \rho(x/h). \quad (\text{D.20})$$

Work on a slightly larger region, so this local convolution creates no physical boundary condition. A fixed partition of unity handles a finite chart cover. The reconstruction is finite dimensional because $I_h f$ is determined by finitely many samples.

For f in a bounded C^3 set, reference-cell interpolation and convolution give

$$\|\mathcal{J}_h f - f\|_{W^{1,\infty}} \leq Ch \|f\|_{C^2}, \quad \|\mathcal{J}_h f\|_{W^{2,\infty}} \leq C \|f\|_{C^2}. \quad (\text{D.21})$$

For the second bound, differentiate the convolution of $\partial I_h f - \partial f$, whose L^∞ norm is $O(h)$; the derivative of ρ_h has L^1 norm $O(h^{-1})$. The remaining term is the convolution of $\partial^2 f$. The same estimates hold for a smooth parameter derivative $\dot{f}$. Nondegeneracy of a smooth reference coframe is preserved for small h by uniform approximation.

Use this map for coframe and matter comparison fields, with fixed local spin/gauge frames. For a smooth gauge connection let $U_\mu(x)$ denote backward parallel transport from $x + he_\mu$ to x . Define

$$\begin{aligned} U_{\mu\nu}(x) &= U_\mu(x) U_\nu(x + he_\mu) U_\mu(x + he_\nu)^{-1} U_\nu(x)^{-1}, \\ F_{\mu\nu}^h(x) &= h^{-2} \log U_{\mu\nu}(x), \\ K_\mu^h(x) &= h^{-1} (U_\mu(x) H(x + he_\mu) - H(x)). \end{aligned} \quad (\text{D.22})$$

The logarithm is evaluated on the identity chart, valid uniformly for sufficiently small h on the smooth bounded family. In [\(D.22\)](#) the fields may be the reconstructed smooth fields $\mathcal{J}_h f$.

For spinors use backward spin-gauge transport $\mathcal{U}_\mu$ and the centered covariant difference

$$\nabla_\mu^h \Psi(x) = \frac{\mathcal{U}_\mu(x) \Psi(x + he_\mu) - \mathcal{U}_\mu(x - he_\mu)^{-1} \Psi(x - he_\mu)}{2h}. \quad (\text{D.23})$$

The dual-spinor difference uses the dual transport. Insert these quantities into the symmetric density [\(2.8\)](#), the Higgs density, and the Yang–Mills density, and sum with the cell-volume/Hodge coefficients computed from the same coframe. Define the gravitational comparison action by the integrated first-derivative torsion-free density [\(A.12\)](#) on the same finite reconstruction. This is a finite-dimensional differentiable action; numerical integration, if used, must satisfy the stated derivative consistency.

Proposition D.8 (Smooth local action consistency). *On a bounded smooth coframe/gauge/matter family with a common inverse-coframe bound, the preceding comparison action and its physical variations satisfy, uniformly for $\|v\|_{C^{r_0}} \leq 1$,*

$$|D\mathcal{S}_{b,h}[\mathcal{I}_h v] - D\mathcal{S}_{b,\theta_h}(z_h)[v]| \leq C_K h, \quad b \in \{g, \text{SM}\}. \quad (\text{D.24})$$

The reconstruction converges in the strong packet. The estimates are local smooth consistency estimates; they do not assert stability on arbitrary finite-energy link configurations.

Proof. The integral equation for parallel transport and Taylor expansion along an edge give

$$U_\mu(x) = I + hA_\mu(x) + O(h^2).$$

Multiplying the four edge expansions cancels the terms linear in h and leaves

$$U_{\mu\nu}(x) = I + h^2 F_{A,\mu\nu}(x) + O(h^3).$$

The second-order contribution is $\partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$. Smoothness and the uniform identity chart therefore give $F^h = F_A + O(h)$. Covariant Taylor expansion similarly gives $K^h = D_A H + O(h)$ and $\nabla^h \Psi = \nabla^{e,A} \Psi + O(h)$.

These expansions also hold after one field or metric variation. To see this without interchanging an uncontrolled limit and derivative, differentiate the transport equation itself. If $U(t, s)$ is transport along a path and a is the connection variation, its derivative is a single integral of $U(t, u)a(u)U(u, s)$ with the fixed orientation sign. A second Taylor expansion gives the derivative of each displayed remainder, bounded by Ch after the same normalizations. For a metric variation the spin-connection derivative contains only the first jet of the coframe variation; (D.21) supplies the needed uniform bounds. The reconstruction of (2.6) differs from the same lift evaluated at the reconstructed metric by $O(h)$ in the norms entering the density.

The pointwise densities and their first variations are finite sums of smooth coefficient functions and products of these quantities. Their difference is $O(h)\|v\|_{C^{r_0}}$. A Riemann sum over cells of volume $O(h^4)$ has $O(h^{-4})$ terms on a fixed region, so the integrated error is $O(h)$, not an extensive error. The first-derivative gravitational expression obeys the same estimate directly from (D.21); exact integration eliminates its quadrature error. Differentiated consistent quadrature adds a further $O(h)$ term. This proves (D.24).

The reconstructed fields converge in $W^{1,\infty}$ on the smooth family, and their curvature and covariant derivatives converge by the same expansions. This implies every norm in the strong packet on a compact region. $\square$

The link-variable action has its usual finite gauge covariance. The comparison interpolation is only a means of comparing smooth fields and physical variations; arbitrary node-wise gauge transformations need not preserve that finite-dimensional interpolation chart. No exact lattice gauge identity is inferred merely from restriction to a non-invariant interpolation subspace. Equivariant reconstruction and nonlinear stability for arbitrary rough links are additional questions, distinct from the proved smooth estimate. Discretely gauge-invariant simplicial formulations provide another route, subject to their own consistency estimates [20].

Remark D.9 (Intrinsic finite action and analytical comparison devices). The local action combines the Cartan/gauge plaquettes, Higgs links, spin-gauge transport and calibrated relative normalization. Its structural provenance is Proposition D.4; the explicit representative is defined here. The auxiliary periodic box, trigonometric reconstruction and $P_{\leq K}$ in the leakage proof are comparison devices. They neither alter the finite Euler covector nor replace the physical record.

E Critical finite graphs and first-variation consistency

This appendix gives a first-jet proof of consistency for the unchanged local action of [Section D.1](#). Its purpose is qualitative closure at the critical four-dimensional endpoint. It is independent of the growing-band estimates used later for quantitative rates. We work on the periodic comparison box fixed in [Section 2.4](#); all conclusions may be restricted to an interior physical cylinder. Bundle components are expressed in one fixed compatible frame during the compactness argument. Internal site gauges are treated intrinsically below.

For a nodal array u_h set

$$\|u_h\|_{p,h}^p = h^4 \sum_x |u_h(x)|^p, \quad D_{\mu,h}^+ u_h(x) = \frac{u_h(x + he_\mu) - u_h(x)}{h}, \quad (\text{E.1})$$

with the evident modification for $p = \infty$, and let $\mathbf{R}_h^0 u_h$ denote the piecewise-constant reconstruction. Write $T_{\mu,m} u_h(x) = u_h(x + mhe_\mu)$. These are positive comparison norms; Lorentzian signs occur only in the physical action.

Lemma E.1 (Discrete translation compactness). *Let u_h be uniformly bounded in L_h^2 . If*

$$\lim_{\rho \downarrow 0} \limsup_{h \downarrow 0} \max_{\mu} \max_{0 < |m|h \leq \rho} \|T_{\mu,m} u_h - u_h\|_{2,h} = 0, \quad (\text{E.2})$$

then $\mathbf{R}_h^0 u_h$ is precompact in L^2 .

Proof. For $0 \leq \vartheta \leq 1$, direct integration on the two portions of each cell gives

$$\|\tau_{\vartheta he_\mu} \mathbf{R}_h^0 u_h - \mathbf{R}_h^0 u_h\|_2^2 = \vartheta \|T_{\mu,1} u_h - u_h\|_{2,h}^2.$$

An arbitrary displacement is an integer grid displacement plus a subcell remainder. Decomposing a four-dimensional displacement into coordinate parts therefore transfers (E.2) to a uniform continuum translation modulus for the piecewise-constant reconstructions. The Fréchet–Kolmogorov criterion yields the result. Equivalently, one may first mollify at fixed scale, approximate the mollified family by finitely many Fourier modes, and then remove the mollification using the translation estimate. $\square$

E.1 Critical finite graphs for the matter packets

Let $U_{\mu,h} = e^{hA_{\mu,h}}$ be the actual internal links. For any retained unitary representation ρ define

$$\mathcal{D}_{\mu,h}^U u_h = \frac{\rho(U_{\mu,h}) T_{\mu,1} u_h - u_h}{h}, \quad B_{\mu,h}^\rho = \frac{\rho(U_{\mu,h}) - I}{h}. \quad (\text{E.3})$$

For spinors the positive graph uses only the compact internal representation; the coframe-derived Lorentz spin transport remains in the physical Dirac term and is treated separately.

Lemma E.2 (Critical grid bounds and link coefficients). *There is a cutoff-independent constant such that*

$$\|u_h\|_{4,h} \leq C \left(\|u_h\|_{2,h} + \sum_{\mu} \|D_{\mu,h}^+ u_h\|_{2,h} \right), \quad (\text{E.4})$$

and, for a unitary internal graph,

$$\|u_h\|_{4,h} \leq C \left(\|u_h\|_{2,h} + \sum_{\mu} \|\mathcal{D}_{\mu,h}^U u_h\|_{2,h} \right). \quad (\text{E.5})$$

A family bounded in the discrete H^1 norm has a subsequence for which $R_h^0 u_h \rightarrow u$ strongly in L^2 , the forward differences converge weakly to ∂u in L^2 , and the trigonometric reconstructions are weakly compact in H^1 . If all forward differences converge strongly, then the trigonometric reconstructions converge strongly in H^1 and $R_h^0 u_h \rightarrow u$ strongly in L^4 .

If $R_h^0 A_h \rightarrow A$ strongly in L^4 , then

$$h\|A_h\|_{\infty,h} \rightarrow 0, \quad R_h^0 B_h^\rho \rightarrow \rho(A) \quad \text{strongly in } L^4. \quad (\text{E.6})$$

Consequently a bounded positive graph norm in (E.3), together with a bounded L_h^4 connection, controls the ordinary discrete H^1 norm.

Proof. On a reference four-cube, integration of the fourth power of the multilinear interpolant is a norm on the finite vector of vertex values. Scaling and summation reduce (E.4) to the continuum embedding $H^1 \hookrightarrow L^4$ in four dimensions. Pointwise unitarity gives the discrete Kato inequality $|D_{\mu,h}^+ u_h| \leq |\mathcal{D}_{\mu,h}^U u_h|$, proving (E.5). The compactness statements follow from Rellich compactness, discrete summation by parts and the sine-symbol bound for trigonometric interpolation. Strong derivative convergence and a smooth approximation of the limit give the endpoint L^4 convergence.

Strong L^4 convergence of the piecewise-constant connections makes their fourth powers uniformly integrable. A single cell has volume h^4 , hence $h\|A_h\|_{\infty,h} \rightarrow 0$. The exact identity

$$B_h^\rho = \int_0^1 e^{th\rho(A_h)} \rho(A_h) dt$$

and bounded-coefficient multiplication prove the second assertion in (E.6). Finally $D_h^+ u_h = \mathcal{D}_h^U u_h - B_h^\rho T u_h$ and Hölder give the ordinary first-difference bound. $\square$

Lemma E.3 (Identification of raw and trigonometric first jets). *On the odd periodic grids of side $L = 2\pi$, define $\|u_h\|_{1,h}^2 = \|u_h\|_{2,h}^2 + \sum_\mu \|D_{\mu,h}^+ u_h\|_{2,h}^2$. Then*

$$\|\mathcal{I}_h^{\text{trig}} u_h - R_h^0 u_h\|_2 \leq Ch \|D_h^+ u_h\|_{2,h}, \quad \|\mathcal{I}_h^{\text{trig}} u_h\|_{H^1} \leq C \|u_h\|_{1,h}. \quad (\text{E.7})$$

If $R_h^0 u_h \rightarrow u$ and $R_h^0 D_{\mu,h}^+ u_h \rightarrow v_\mu$ strongly in L^2 for every μ , then $v_\mu = \partial_\mu u$, $\mathcal{I}_h^{\text{trig}} u_h \rightarrow u$ strongly in H^1 , and $R_h^0 u_h \rightarrow u$ strongly in L^4 .

Proof. Use the normalized discrete Fourier coefficients with physical frequencies $\xi_\ell = 2\pi\ell/L$. For a fixed subcell displacement $y \in [0, h)^4$, discrete orthogonality gives

$$h^4 \sum_x |\mathcal{I}_h^{\text{trig}} u_h(x+y) - u_h(x)|^2 = L^4 \sum_\ell |\widehat{u}_h(\ell)|^2 |e^{i\xi_\ell \cdot y} - 1|^2.$$

Average over y . On the represented frequency cube, $|\xi_\ell| \leq C(\sum_\mu |(e^{ih\xi_{\ell,\mu}} - 1)/h|^2)^{1/2}$ by the sine bound. This proves both estimates in (E.7). Summation by parts against a fixed sampled smooth test identifies v_μ as $\partial_\mu u$.

For any strongly convergent raw L^2 array, its fixed discrete Fourier coefficients converge to those of the limit: replace the cellwise constant Fourier test by its continuum character, whose error tends uniformly to zero. Parseval and convergence of the total squared norm then give a vanishing high-frequency tail after first taking $h \downarrow 0$ and then the frequency threshold to infinity. Apply this observation to all first differences. The ratio of the trigonometric derivative symbol $i\xi_{\ell,\mu}$ to the forward symbol is uniformly bounded and tends to one at each fixed frequency. The finite-mode/tail argument therefore gives strong convergence of the trigonometric derivatives.

Finally choose smooth periodic $u^j \rightarrow u$ in H^1 and compare u_h with its sampled values $S_h u^j$. Discrete Sobolev bounds the raw L^4 difference by their discrete H^1 difference. For fixed j its limsup is at most $C\|u - u^j\|_{H^1}$. Let $j \rightarrow \infty$ and use continuous $H^1 \hookrightarrow L^4$ for the limit. This proves the raw endpoint statement without identifying raw arrays with trigonometric fields at a finite cutoff. $\square$

Proposition E.4 (Native Higgs graph compactness). *Suppose $R_h^0 A_h \rightarrow A$ strongly in L^4 , $\sup_h \|H_h\|_{2,h} < \infty$, and the literal Higgs-link packet $K_{\mu,h} = \mathcal{D}_{\mu,h}^U H_h$ satisfies $R_h^0 K_h \rightarrow K$ strongly in L^2 . After extraction,*

$$\begin{aligned} R_h^0 H_h &\rightarrow H && \text{strongly in } L^4, \\ R_h^0 D_h^+ H_h &\rightarrow dH && \text{strongly in } L^2, \\ \mathcal{I}_h^{\text{trig}} H_h &\rightarrow H && \text{strongly in } H^1, \quad K = D_A H. \end{aligned} \quad (\text{E.8})$$

Thus the literal covariant-gradient packet closes the quartic Higgs endpoint without a separate amplitude compactness or high-derivative hypothesis.

Proof. Lemma E.2 gives bounded ordinary discrete H^1 norms, strong L^2 compactness of the values, and bounded L_h^4 norms. Since one-step shifts have the same strong L^2 limit, the product $B_h^{\rho_H} T H_h$ converges strongly in L^2 to $\rho_H(A)H$; at the critical exponent this follows by first approximating A in L^4 by a bounded smooth coefficient. The exact graph identity $D_h^+ H_h = K_h - B_h^{\rho_H} T H_h$ then gives strong ordinary first-difference convergence. Discrete summation by parts identifies the limit derivative, and the final assertions of Lemma E.2 give strong H^1 and L^4 convergence. $\square$

For the literal logarithmic plaquette write

$$F_{\mu\nu}^h(A_h) = (D_{\mu,h}^+ A_{\nu,h} - D_{\nu,h}^+ A_{\mu,h}) + \mathcal{N}_{\mu\nu,h}(A_h). \quad (\text{E.9})$$

For the four slots $W_{\mu\nu,h}(A) = (A_\mu, T_{\mu,1} A_\nu, T_{\nu,1} A_\mu, A_\nu)$, the analytic logarithm has linear part $h(A_\mu + T_{\mu,1} A_\nu - T_{\nu,1} A_\mu - A_\nu)$. Subtracting that part and applying its integral second-order remainder gives, on a fixed small scaled logarithm chart,

$$\begin{aligned} |\mathcal{N}_{\mu\nu,h}(A)(x)| &\leq C|W_{\mu\nu,h}(A)(x)|^2, \\ \|\mathcal{N}_h(A) - \mathcal{N}_h(B)\|_{2,h} &\leq C(\|A\|_{4,h} + \|B\|_{4,h})\|A - B\|_{4,h}. \end{aligned} \quad (\text{E.10})$$

The same analytic expansion holds after one physical variation.

Proposition E.5 (Identification and variation of the literal curvature). *Suppose $R_h^0 A_h \rightarrow A$ strongly in L^4 and $R_h^0 F_h \rightarrow F$ strongly in L^2 . Then $F = F_A$. For each fixed smooth gauge test a ,*

$$R_h^0 (D_A F_h [S_h a]) \longrightarrow da + [A, a] \quad \text{strongly in } L^2. \quad (\text{E.11})$$

If the coframes converge strongly in H^1 on one compact nondegenerate chart, the complete finite Yang–Mills first variation, including its metric variation, converges to the continuum Yang–Mills first variation uniformly on bounded C^r test sets, $r \geq 2$.

Proof. By (E.6), the scaled slots eventually lie in the analytic plaquette domain. One-step shifts of A_h have the same strong L^4 limit. The quadratic plaquette envelope therefore gives $R_h^0 \mathcal{N}_h(A_h) \rightarrow A \wedge A$ strongly in L^2 . Testing (E.9) against a sampled smooth function and summing by parts identifies $F = dA + A \wedge A$. Differentiating the exact split in the nodal direction $S_h a$ yields (E.11). The gauge row is a strong L^2 –strong L^2 pairing. The metric row contains bounded coframe coefficients converging in measure multiplying the strongly convergent L^1 packet $F_h \otimes F_h$; Lemma A.2 passes it. Uniform C^1 bounds and a finite-net argument give the stated C^r -dual convergence. $\square$

For the spinor and dual-spinor blocks define the positive internal-link graph bound

$$\sup_h \left(\|\Psi_h\|_{2,h} + \|\mathcal{D}_h^U \Psi_h\|_{2,h} + \|\bar{\Psi}_h\|_{2,h} + \|\mathcal{D}_h^{U,\vee} \bar{\Psi}_h\|_{2,h} \right) < \infty. \quad (\text{E.12})$$

Proposition E.6 (Weak native spinor graphs and the complete Dirac variation). *Assume (E.12), strong L^4 connection convergence, the Higgs conclusions of Proposition E.4, strong H^1 coframe convergence on one compact nondegenerate chart, and the scaled spin-link margin (E.13). After extraction the spinors converge weakly in the reconstructed H^1 class, and the complete finite Dirac–Yukawa first variation converges to its continuum counterpart in $(\mathcal{V}_K^r)^*$ for every $r \geq 2$. This includes the full metric variation of the coframe-derived spin connection. Strong convergence of the spinor first differences is not required.*

Proof. The graph identity in Lemma E.2 gives ordinary discrete H^1 bounds. Hence the values converge strongly in L^2 after extraction, their first differences converge weakly in L^2 , and the undifferentiated spinor bilinears converge strongly in L^1 while remaining bounded in L^2 . The physical spin–gauge link can be written exactly as

$$\nabla_\mu^h \Psi_h = D_{\mu,h}^+ \Psi_h + B_{\mu,h} T_{\mu,1} \Psi_h,$$

with the analogous dual identity. The coefficient $B_{\mu,h}$ splits into the coframe-derived spin coefficient and the compact internal-link coefficient. Strong first-jet coframe convergence gives the former and its metric variation strongly in L^2 by Lemma A.2; strong L^4 connection convergence gives the latter by (E.6). Derivative terms therefore use weak–strong L^2 pairing, while link and Yukawa terms pair strongly convergent L^2 coefficients with weakly convergent L^2 bilinears. The same structures occur after metric, gauge, Higgs, spinor and dual-spinor variation. Uniform C^1 bounds and a finite C^1 net inside the C^r unit ball complete the proof. $\square$

E.2 The literal gravitational row at first-jet regularity

For the nodal coframes of Section D.1, write $\omega_{\mu,h} = \Omega_\mu(e_h, D_h^+ e_h)$. We say that the common logarithm-chart condition holds when the nodal coframes remain in one fixed compact nondegenerate chart and

$$\max_{x,\mu} h |\omega_{\mu,h}(x)| \leq c_* \quad (\text{E.13})$$

for a fixed c_* on which the four-exponential plaquette product and its logarithm are analytic along the relevant segments. This is a branch condition needed to define the literal logarithmic action; it is not a second-derivative bound.

Proposition E.7 (Literal gravitational consistency from strong first jets). *Suppose the actual nodal coframes satisfy*

$$R_h^0 e_h \rightarrow e \quad \text{in } L^2, \quad R_h^0 D_{\mu,h}^+ e_h \rightarrow \partial_\mu e \quad \text{strongly in } L^2, \quad (\text{E.14})$$

with values in one fixed compact nondegenerate coframe chart, and assume (E.13). Let an inverse-metric test k be lifted nodally by the symmetric lift (2.6). Then for every $r \geq 2$,

$$\sup_{\|k\|_{C^r} \leq 1} \left| DS_{g,h}^{\text{loc}}(e_h)[\mathcal{I}_h k] - D\mathcal{S}_{g,\theta}(e)[k] \right| \rightarrow 0. \quad (\text{E.15})$$

If the complete unfiltered reconstructed coframes converge strongly in H^1 to the same e and remain in the same compact chart, then the sectorwise consistency error relative to those reconstructed coframes tends to zero. No growing-band, C^5 , or strong curvature-discrepancy hypothesis is required.

Proof. Let

$$\Phi(Z_1, Z_2, Z_3, Z_4) = \log(e^{Z_1} e^{Z_2} e^{-Z_3} e^{-Z_4}), \quad \mathsf{L}Z = Z_1 + Z_2 - Z_3 - Z_4,$$

and define the exact quadratic remainder

$$\mathcal{Q}_h(W) = h^{-2}(\Phi(hW) - h\mathsf{L}W) = \int_0^1 (1-t) D^2\Phi(thW)[W, W] dt.$$

On the fixed scaled logarithm chart, $|\mathcal{Q}_h(W)| \leq C|W|^2$. If $W_h \rightarrow W$ and $Z_h \rightarrow Z$ strongly in L^2 , bounded-coefficient multiplication gives

$$\mathcal{Q}_h(W_h) \rightarrow \tfrac{1}{2} D^2\Phi(0)[W, W], \quad D\mathcal{Q}_h(W_h)[Z_h] \rightarrow D^2\Phi(0)[W, Z] \quad \text{in } L^1.$$

For the actual slots $W_{\mu\nu,h} = (\omega_{\mu,h}, T_{\mu,1}\omega_{\nu,h}, T_{\nu,1}\omega_{\mu,h}, \omega_{\nu,h})$, the literal Cartan plaquette has the exact split

$$R_{\mu\nu}^h = D_{\mu,h}^+ \omega_{\nu,h} - D_{\nu,h}^+ \omega_{\mu,h} + \mathcal{Q}_h(W_{\mu\nu,h}).$$

Discrete summation by parts moves the two difference terms onto the smooth Palatini coframe coefficient. Because $\Omega(e, p)$ is smooth in e and linear in p , (E.14) and Lemma A.2 give strong L^2 convergence of the literal connection, its finite shifts, the differentiated coframe coefficient, and all of their first physical variations. Differentiating the exact summed-by-parts action therefore leaves only products of strongly convergent L^2 arrays and the quadratic remainder terms above. All pass in L^1 and give precisely the first-derivative Palatini variation of Section A.5. The same expressions are bounded by $C\|k\|_{C^1}$; a finite-net argument gives (E.15). The final consistency statement follows by subtracting the continuum variation at the reconstructed coframe and applying Proposition 3.6 to the geometric sector. $\square$

Theorem E.8 (All-sector native consistency without a growing band). *Let $z_h^d = (e_h, A_h, H_h, \Psi_h, \bar{\Psi}_h)$ be actual nodal records of the unchanged local action (D.6). Assume:*

- (N1) *the coframes satisfy (E.14), the common compact chart and (E.13);*
- (N2) *$R_h^0 A_h \rightarrow A$ strongly in L^4 and the literal logarithmic curvature satisfies $R_h^0 F_h \rightarrow F$ strongly in L^2 ;*
- (N3) *$\sup_h \|H_h\|_{2,h} < \infty$ and $R_h^0 K_h \rightarrow K$ strongly in L^2 for the literal Higgs-link packet;*
- (N4) *the spinors satisfy (E.12);*
- (N5) *the coefficient banks converge in a compact physical parameter set.*

Then, after extraction, $F = F_A$, $K = D_A H$, the Higgs fields converge as in (E.8), and the spinors converge weakly in the reconstructed H^1 class. For every $r \geq 2$,

$$\sup_{\|v\|_{C^r} \leq 1} \left| DS_h^{\text{loc}}(z_h^d)[\mathcal{I}_h v] - D\mathcal{S}_\theta(z)[v] \right| \longrightarrow 0. \quad (\text{E.16})$$

The convergence holds sectorwise. If, in addition, the complete physical finite variation tends to zero on the corresponding unit test family, then z satisfies all minimally coupled torsion-free Einstein–Standard-Model Euler equations in distributions.

If the complete unfiltered reconstructed coframes stay in the same compact nondegenerate chart and the ordinary connection first differences converge strongly to ∂A , then the sectorwise consistency errors relative to the complete unfiltered reconstructed fields tend to zero. No independent matter-consistency or growing-band premise remains in this branch.

Proof. The gravitational sector is [Proposition E.7](#). The Yang–Mills identification and variation are [Proposition E.5](#); the Higgs sector is [Proposition E.4](#); and the complete fermionic sector is [Proposition E.6](#). Each sector has a cutoff-independent C^1 test bound, so their sum converges uniformly on the C^r test unit ball by a finite-net argument. Vanishing finite stationarity gives zero limiting first variation. For the final assertion, strong ordinary connection first-jet convergence identifies the complete reconstructed curvature and covariant Higgs gradient, while the reconstructed coframe convergence is covered by the reduced continuity theorem. Subtracting the native and reconstructed limits yields the sectorwise consistency errors. $\square$

The next appendix supplies the Wilson translation argument and the actual finite Coulomb normalization that discharge the packet hypotheses in [Theorem 4.7](#).

F Wilson compactness and same-record Coulomb normalization

The first result obtains scalar compactness before any gauge is chosen. A controlled representative then converts Wilson shifts to ordinary translations. The remaining estimates construct that representative from a small logarithmic seed and an exact site-gauge path.

F.1 Wilson compactness directly on the finite packets

Theorem F.1 (Direct native Wilson compactness). *Let Y_h be bounded in L_h^2 on the entire periodic comparison box and let its internal representation be unitary. Suppose*

$$\lim_{\rho \downarrow 0} \limsup_{h \downarrow 0} \Omega_h(Y_h; \rho) = 0. \quad (\text{F.1})$$

Then $|\mathbf{R}_h^0 Y_h|$ is precompact in L^2 and its square is uniformly integrable. This assertion requires no connection bound and holds in any site gauge. If, in addition, the links have coordinates $U_h = e^{hA_h}$ with $\sup_h \|A_h\|_{2,h} = M_A < \infty$, then $\mathbf{R}_h^0 Y_h$ itself is precompact in L^2 . No smooth link reconstruction or growing band is required.

Proof. Unitarity gives the pointwise inequality $|T_{\mu,m} Y_h| - |Y_h| \leq |W_{\mu,m}^U Y_h - Y_h|$. [Lemma E.1](#) therefore makes the scalar magnitudes precompact in L^2 . The squares are uniformly integrable by the finite-net argument of [Lemma G.14](#). With

$$q_h(R) = \|Y_h \mathbf{1}_{|Y_h| > R}\|_{2,h}, \quad \lim_{R \rightarrow \infty} \limsup_{h \downarrow 0} q_h(R) = 0,$$

the scalar assertion is complete before a logarithmic representative is used.

In a representative with the stated connection bound, let $P_{\mu,m,h}$ be the represented open-link product. Telescoping and unitarity imply

$$\|P_{\mu,m,h} - I\|_{2,h} \leq \sum_{j=0}^{|m|-1} \|\rho(U_\mu) - I\|_{2,h} \leq C|m|hM_A.$$

Here translated norms agree by periodicity; negative displacements use the inverse product. Split the translated packet at amplitude R . Since a unitary matrix minus the identity has operator norm at most two,

$$\|T_{\mu,m} Y_h - Y_h\|_{2,h} \leq \Omega_h(Y_h; \rho) + CR\rho M_A + 2q_h(R) \quad (|m|h \leq \rho). \quad (\text{F.2})$$

For fixed R , first take the limsup in h , then let $\rho \downarrow 0$, and only then let $R \rightarrow \infty$. The right-hand side tends to zero. Thus the ordinary integer-shift modulus vanishes. The exact subcell identity in [Lemma E.1](#) supplies arbitrary translations, proving vector L^2 precompactness. On buffered charts the same argument is local, with all translated endpoints kept in the buffer; the global periodic Hodge conclusions below use full-box screens. $\square$

F.2 Discrete Coulomb absorption and a finite same-record gauge

For a nodal Lie-algebra one-form put

$$\delta_h A_h = - \sum_{\mu=0}^3 D_{\mu,h}^- A_{\mu,h}, \quad D_{\mu,h}^- = \frac{I - T_{\mu,-1}}{h}. \quad (\text{F.3})$$

Lemma F.2 (Exact periodic discrete Hodge identity). *For every nodal one-form,*

$$\sum_{\mu,\nu} \|D_{\mu,h}^+ A_{\nu,h}\|_{2,h}^2 = \|d_h^+ A_h\|_{2,h}^2 + \|\delta_h A_h\|_{2,h}^2. \quad (\text{F.4})$$

Consequently $\|A_h\|_{1,h} \leq C(\|d_h^+ A_h\|_{2,h} + \|\delta_h A_h\|_{2,h} + \|A_h\|_{2,h})$ uniformly in h .

Proof. At one Fourier frequency let $d_\mu = (e^{i\ell_\mu h} - 1)/h$ on the side- 2π box. Exterior multiplication by d and contraction by $\bar{d}$ satisfy

$$\sum_{\mu < \nu} |d_\mu a_\nu - d_\nu a_\mu|^2 + \left| \sum_\mu \bar{d}_\mu a_\mu \right|^2 = \left(\sum_\mu |d_\mu|^2 \right) \left(\sum_\nu |a_\nu|^2 \right).$$

Parseval gives [\(F.4\)](#); adding the zero-mode L^2 norm gives the estimate. $\square$

Theorem F.3 (Critical discrete Coulomb compactness). *There is $\epsilon_* > 0$ such that, if*

$$\delta_h A_h = 0, \quad \|A_h\|_{4,h} \leq \epsilon_*, \quad \mathcal{R}_h^0 F_h(A_h) \rightarrow F \quad \text{strongly in } L^2, \quad (\text{F.5})$$

then, after extraction,

$$\begin{aligned} \mathcal{R}_h^0 A_h &\rightarrow A && \text{strongly in } L^4, \\ \mathcal{R}_h^0 D_h^+ A_h &\rightarrow \partial A && \text{strongly in } L^2, \\ \mathcal{I}_h^{\text{trig}} A_h &\rightarrow A && \text{strongly in } H^1, \quad F = F_A, \end{aligned} \quad (\text{F.6})$$

with $\delta A = 0$. The result is purely discrete; no continuum Coulomb theorem or high-derivative reserve is used.

Proof. The exact curvature split, its shifted-slot quadratic bound, and the Hodge identity give

$$\|A_h\|_{1,h} \leq C(\|F_h(A_h)\|_{2,h} + \|A_h\|_{4,h}^2 + \|A_h\|_{2,h}).$$

Hence A_h is bounded in the discrete H^1 norm. Extract raw strong L^2 convergence to A and weak convergence of the first differences. The one-step shifts have the same strong L^2 limit. The scaled slots hA_h are uniformly small because $h\|A_h\|_{\infty,h} \leq \|A_h\|_{4,h}$, and tend to zero in measure. The integral quadratic expansion therefore gives $\mathcal{R}_h^0 \mathcal{N}_h(A_h) \rightarrow A \wedge A$ in L^1 . Indeed its slot products converge strongly in L^1 , while its bounded analytic coefficients converge in measure. Discrete summation by parts now identifies $F = F_A$ and $\delta A = 0$.

Choose periodic mollifications A^j of A preserving $\delta A^j = 0$, with $A^j \rightarrow A$ in H^1 and $\|A^j\|_4 \leq \|A\|_4 \leq \epsilon_*$. Put $B_h^j = S_h A^j$. For each fixed j , and all sufficiently small h , $\|B_h^j\|_{4,h} \leq 2\epsilon_*$. Apply the discrete Hodge estimate and (E.10) to $A_h - B_h^j$:

$$\begin{aligned} \|A_h - B_h^j\|_{1,h} &\leq C(\|F_h(A_h) - F_h(B_h^j)\|_{2,h} + \|\delta_h B_h^j\|_{2,h} + \|A_h - B_h^j\|_{2,h}) \\ &\quad + 3C_N C_S \epsilon_* \|A_h - B_h^j\|_{1,h}. \end{aligned} \quad (\text{F.7})$$

Here C_N, C_S depend on the fixed logarithm chart and the periodic Sobolev/Hodge constants, not on h or j . Choose the threshold so that $3C_N C_S \epsilon_* < 1/2$ and absorb the last term. Fixed-smooth-field consistency, strong raw curvature convergence, and $\delta A^j = 0$ then imply

$$\limsup_{h \downarrow 0} \|A_h - B_h^j\|_{1,h} \leq C(\|F_A - F_{A^j}\|_2 + \|A - A^j\|_2).$$

Only after this limit is taken do we let $j \rightarrow \infty$. Strong H^1 convergence of A^j closes its quadratic curvature product, so the right-hand side tends to zero. Comparison with these fixed smooth arrays proves strong convergence of the raw first differences and of the raw L^4 values. Lemma E.3 supplies the trigonometric H^1 statement. No cutoff-dependent smooth comparison or estimate for its high derivatives is used. $\square$

For a small operator Z define

$$\mathcal{J}(Z) = \int_0^1 e^{tZ} dt, \quad \mathcal{K}(Z) = \frac{Z}{2} \coth\left(\frac{Z}{2}\right), \quad \mathcal{K}(0) = I. \quad (\text{F.8})$$

Lemma F.4 (Exact logarithmic site-gauge differential). *Use $U_\mu^q(x) = q_x U_\mu(x) q_{x+he_\mu}^{-1}$ and a fixed conjugation-invariant logarithm chart. Put $Z_\mu = h \operatorname{ad}_{A_\mu}$ and $m_\mu \xi = \frac{1}{2}(\xi + T_{\mu,1}\xi)$. For $q_x(t) = e^{t\xi_x}$,*

$$\left. \frac{d}{dt} A_\mu^{q(t)} \right|_{t=0} = -\mathcal{K}(Z_\mu) D_\mu^+ \xi - [A_\mu, m_\mu \xi]. \quad (\text{F.9})$$

In particular

$$\left. \frac{d}{dt} \frac{1}{2} \|A^{q(t)}\|_{2,h}^2 \right|_{t=0} = -\langle \delta_h A, \xi \rangle_{2,h}. \quad (\text{F.10})$$

These identities concern logarithmic link coordinates, not an additive approximation to a finite gauge transformation.

Proof. The differential of the exponential in left-trivialized coordinates is

$$\dot{U}_\mu U_\mu^{-1} = h \mathcal{J}(Z_\mu) \dot{A}_\mu.$$

For a site-gauge variation the same quantity is $\xi_x - e^{Z_\mu} \xi_{x+he_\mu}$. Using $\xi_x = m_\mu \xi - \frac{h}{2} D_\mu^+ \xi$ and $\xi_{x+he_\mu} = m_\mu \xi + \frac{h}{2} D_\mu^+ \xi$ gives

$$\mathcal{J}(Z)^{-1}(I - e^Z) = -Z, \quad \frac{1}{2} \mathcal{J}(Z)^{-1}(I + e^Z) = \mathcal{K}(Z),$$

which proves (F.9). On the real Lie algebra with its positive invariant metric, ad_A is skew-adjoint; $\mathcal{K}$ is an even analytic function and $\mathcal{K}(h \operatorname{ad}_A)^* A = A$. Moreover $\langle A, [A, \eta] \rangle = 0$. Pairing the differential with A and using $\delta_h = (D_h^+)^*$ therefore gives (F.10). $\square$

Define the mean-zero site space $\mathcal{Z}_h = \{\xi : h^4 \sum_x \xi_x = 0\}$, with norm $\|\xi\|_{\dot{H}_h^1} = \|D_h^+ \xi\|_{2,h}$ and dual norm

$$\|f\|_{\dot{H}_h^{-1}} = \sup_{\xi \in \mathcal{Z}_h, \|D_h^+ \xi\|_{2,h}=1} |\langle f, \xi \rangle_{2,h}|.$$

Periodic Poincaré and Sobolev give $\|\xi\|_{2,h} + \|\xi\|_{4,h} \leq C \|D_h^+ \xi\|_{2,h}$ on this space, with C independent of h . The logarithmic discrete Faddeev–Popov operator is

$$\mathcal{M}_A \xi = \delta_h \left(\mathcal{K}(h \text{ad}_{A_\mu}) D_\mu^+ \xi + [A_\mu, m_\mu \xi] \right)_\mu. \quad (\text{F.11})$$

Its range has zero mean because it is a discrete divergence.

Lemma F.5 (Uniform transverse inverse). *There is $r_{\text{FP}} > 0$, independent of h , such that $\|A\|_{4,h} < r_{\text{FP}}$ implies*

$$\langle \mathcal{M}_A \xi, \xi \rangle_{2,h} \geq \frac{1}{2} \|D_h^+ \xi\|_{2,h}^2 \quad (\xi \in \mathcal{Z}_h). \quad (\text{F.12})$$

Thus $\mathcal{M}_A : \mathcal{Z}_h \rightarrow \mathcal{Z}_h$ is invertible and

$$\|D_h^+ \mathcal{M}_A^{-1} f\|_{2,h} \leq 2 \|f\|_{\dot{H}_h^{-1}}. \quad (\text{F.13})$$

In particular, for any link array R ,

$$\|\mathcal{M}_A^{-1} \delta_h R\|_{2,h} + \|\mathcal{M}_A^{-1} \delta_h R\|_{4,h} + \|D_h^+ \mathcal{M}_A^{-1} \delta_h R\|_{2,h} \leq C \|R\|_{2,h}. \quad (\text{F.14})$$

The inverse depends smoothly on A on this finite-dimensional open set. Self-adjointness of $\mathcal{M}_A$ away from Coulomb gauge is not required.

Proof. On a fixed sufficiently small scaled chart, $\|\mathcal{K}(h \text{ad}_A) - I\| \leq Ch^2 |A|^2$. The four-dimensional cell mass gives $h \|A\|_{\infty,h} \leq \|A\|_{4,h}$. Consequently

$$|\langle (\mathcal{K} - I) D^+ \xi, D^+ \xi \rangle_{2,h}| \leq C \|A\|_{4,h}^2 \|D^+ \xi\|_{2,h}^2.$$

For the other perturbation, discrete Hölder, translation invariance and the mean-zero Sobolev estimate give

$$|\langle [A, m\xi], D^+ \xi \rangle_{2,h}| \leq C \|A\|_{4,h} \|m\xi\|_{4,h} \|D^+ \xi\|_{2,h} \leq C \|A\|_{4,h} \|D^+ \xi\|_{2,h}^2.$$

Choose r_{FP} so that $C(r_{\text{FP}} + r_{\text{FP}}^2) \leq 1/2$ and so that the scaled analytic chart is retained. This proves coercivity. The restriction to $\mathcal{Z}_h$ is injective, hence bijective because its domain and codomain have equal finite dimension. Pairing $\mathcal{M}_A \xi = f$ with ξ gives (F.13). Also $\|\delta_h R\|_{\dot{H}_h^{-1}} \leq \|R\|_{2,h}$, proving (F.14). Smooth dependence follows from the finite matrix inverse on its invertibility set. All displayed norm constants are independent of the number of sites. $\square$

Theorem F.6 (Finite critical Coulomb normalization). *There are $\epsilon_c, C_c > 0$, independent of h , such that any logarithmic seed B_h satisfying*

$$\|B_h\|_{4,h} + \|F_h(B_h)\|_{2,h} \leq \epsilon_c \quad (\text{F.15})$$

admits a finite site gauge q_h for which

$$A_{\mu,h} = h^{-1} \text{Log}(q_x e^{h B_{\mu,h}(x)} q_{x+h e_\mu}^{-1}), \quad \delta_h A_h = 0, \quad (\text{F.16})$$

and

$$\|A_h\|_{1,h} + \|A_h\|_{4,h} \leq C_c (\|B_h\|_{2,h} + \|F_h(B_h)\|_{2,h} + \|B_h\|_{4,h}^2). \quad (\text{F.17})$$

The threshold may be chosen so that $\|A_h\|_{4,h} \leq \epsilon_*$ in Theorem F.3. No bound on $\delta_h B_h$ or on the complete first differences of B_h is assumed.

Proof. All constants below refer to the fixed periodic box, invariant Lie-algebra metric and logarithm chart. They are independent of h . Choose $r > 0$ below the transverse threshold of [Lemma F.5](#) and below the scaled four-exponential logarithm threshold. Shrink it further in the absorption step below. For a fixed grid consider the auxiliary links $U_\mu^s = e^{shB_\mu}$, $0 \leq s \leq 1$.

Local gauge path. On the open set of pairs (s, q) for which

$$A_{s,\mu} = h^{-1} \text{Log}(q_x e^{shB_\mu} q_{x+he_\mu}^{-1}), \quad \|A_s\|_{4,h} < r,$$

define

$$\begin{aligned} R_{s,\mu} &= \mathcal{J}(h \text{ad}_{A_{s,\mu}})^{-1} \text{Ad}_{q_x} B_\mu, \\ \xi_s &= \mathcal{M}_{A_s}^{-1} \delta_h R_s \in \mathcal{Z}_h, \quad \dot{q}_x = \xi_s(x) q_x, \quad q_x(0) = I. \end{aligned} \quad (\text{F.18})$$

The right-hand side is a smooth tangent field on a finite product of compact Lie groups. The logarithms are defined at $(s, q) = (0, I)$, where $A_0 = 0$; finite-dimensional ODE existence gives a local solution. The exact chain rule of [Lemma F.4](#) gives

$$\dot{A}_s = R_s - (\mathcal{K}(h \text{ad}_{A_{s,\mu}}) D_\mu^+ \xi_s + [A_{s,\mu}, m_\mu \xi_s])_\mu, \quad \delta_h \dot{A}_s = 0.$$

Hence $\delta_h A_s = 0$ for as long as the path remains in the open set.

The physical L_h^2 estimate. The gauge tangent contributes zero when paired with A_s , by [\(F.10\)](#). The adjoint of $\mathcal{J}(h \text{ad}_{A_{s,\mu}})^{-1}$ fixes $A_{s,\mu}$. Therefore

$$\frac{1}{2} \frac{d}{ds} \|A_s\|_{2,h}^2 = \sum_\mu \langle A_{s,\mu}, \text{Ad}_{q_x} B_\mu \rangle_{2,h} \leq \|A_s\|_{2,h} \|B\|_{2,h}.$$

Apply this inequality first to $(\|A_s\|_{2,h}^2 + \epsilon^2)^{1/2}$ and then let $\epsilon \downarrow 0$. It follows that $\|A_s\|_{2,h} \leq s \|B\|_{2,h}$.

Curvature and the cutoff-independent first-exit bound. Exact conjugation covariance of the plaquette logarithm gives $\|F_h(A_s)\|_{2,h} = \|F_h(sB)\|_{2,h}$ on the declared chart. The exact split [\(E.9\)](#) yields

$$F_h(sB) = sF_h(B) - s\mathcal{N}_h(B) + \mathcal{N}_h(sB), \quad \|F_h(sB)\|_{2,h} \leq \|F_h(B)\|_{2,h} + C\|B\|_{4,h}^2.$$

Put $b_h = \|B\|_{2,h} + \|F_h(B)\|_{2,h} + \|B\|_{4,h}^2$. The discrete Hodge identity, Coulomb condition and quadratic plaquette bound then give, throughout this path,

$$\|A_s\|_{1,h} \leq C_H(b_h + \|A_s\|_{4,h}^2), \quad a_s := \|A_s\|_{4,h} \leq C_0 b_h + C_1 a_s^2. \quad (\text{F.19})$$

Choose r so that $C_1 r \leq 1/4$. While $a_s < r$, the last inequality gives $a_s \leq (4/3)C_0 b_h$. On the fixed-volume box $\|B\|_{2,h} \leq C\|B\|_{4,h}$, so the seed assumption makes $b_h \leq C(\epsilon_c + \epsilon_c^2)$. Choose ϵ_c so that $(4/3)C_0 b_h \leq r/2$. Thus the path cannot reach the boundary $a_s = r$. This choice also makes the right side smaller than the desired ϵ_* , after shrinking r if necessary.

Continuation without leaving the logarithm chart. The improved bound gives $h\|A_s\|_{\infty,h} \leq r/2$ and therefore places every normalized link strictly inside a common logarithm domain; all four-link products stay inside its corresponding plaquette domain. The seed links e^{shB} stay in their analytic domain by the same cell-mass bound and the small seed threshold. Moreover $\|R_s\|_{2,h} \leq C\|B\|_{2,h}$, since $\mathcal{J}^{-1}$ is uniformly bounded on this domain. By [\(F.14\)](#),

$$\|\xi_s\|_{2,h} + \|\xi_s\|_{4,h} + \|D_h^+ \xi_s\|_{2,h} \leq C\|B\|_{2,h}. \quad (\text{F.20})$$

For each fixed h this also bounds all site velocities in the finite ODE. Thus if the maximal path ended at some $s_0 \leq 1$, the group variables would have limits there. The strict logarithm and transverse-inverse margins would put the limiting point inside the smooth ODE domain, allowing continuation. This contradicts maximality unless the path reaches $s = 1$. No bound uniform in h on derivatives of the gauge itself is needed for this continuation.

At $s = 1$ the path gives the actual site gauge in (F.16). From $a_s \leq Cb_h$ and the first inequality in (F.19), with b_h small, we obtain (F.17). The conclusion is existence of a small Coulomb representative on this seed branch; it is not uniqueness among all site gauges and makes no assertion outside the logarithm chart. The auxiliary homotopy is a proof construction, not a modification of the observed record or of its physical dynamics. $\square$

Proposition F.7 (Rooted same-record gauge certificate). *Let $\mathfrak{T}_h$ be the fixed rooted spanning tree and define $B_h^\mathfrak{T}$ by (4.19). On its admissible logarithm chart the quantity (4.20) is invariant under finite site gauges. If $\mathfrak{c}_h^\mathfrak{T}(U_h) \leq \epsilon_c$, then the exact tree gauge followed by Theorem F.6 puts the original links in exact small discrete Coulomb gauge. Under this normalization:*

- (a) *every sector of the finite common action is unchanged;*
- (a) *the literal curvature and Higgs-gradient packets, the positive internal-link spinor graphs, and the Wilson moduli are unchanged in norm;*
- (a) *the norm of the complete action covector is unchanged in the positive physical mass metric whose link tangent is $\alpha_\mu = h^{-1}\dot{U}_\mu U_\mu^{-1}$.*

No derivative of the normalizing site gauge appears in these norm identities.

Proof. Endpoint cancellation along the tree gives $C_x(U^k) = k_o C_x(U) k_x^{-1}$ and hence $W_\mu^\mathfrak{T}(U^k) = k_o W_\mu^\mathfrak{T}(U) k_o^{-1}$. The logarithm is equivariant on the declared chart and the invariant inner product makes the certificate gauge invariant. The tree representative is therefore an exact gauge transform of the original links, not an approximation, and Theorem F.6 applies to it.

For the physical statements, plaquette and open-link endpoint cancellation give conjugation of the curvature packet at its base vertex. Direct substitution gives the corresponding unitary transformation of the Higgs-gradient and internal-link spinor packets. The coframe-derived spin transport commutes with the internal group, and the covariant Yukawa blocks transform in their prescribed representations; hence the full finite action is unchanged sector by sector. The physical link tangent transforms by $\alpha_\mu \mapsto \text{Ad}_{k_x} \alpha_\mu$ and the matter tangents unitarily. Thus the positive tangent map is an isometry, and differentiating the exact action invariance proves invariance of the dual covector norm. $\square$

G Determinant normalization and covariant Wilson transfer

G.1 The determinant sector: flux, harmonic coordinates, and exact assembly

We work on the periodic comparison box of side $L = 2\pi$, with $h = L/n$ and positive cell masses h^4 . Keeping L in the formulas distinguishes physical differences from lattice-unit differences. The Abelian flux and flat-holonomy decomposition is classical [53, Section 7]. The proofs below record its physical normalization, its critical compactness estimates, and its application to the same Standard-Model links and physical variations. No change to the Lorentzian common action is involved.

G.1.1 The determinant screen and the integer flux

For $u_h = \chi(U_h)$ let $p_{\mu\nu,h}$ be its oriented plaquette and put

$$\phi_{\mu\nu,h} = \text{Arg } p_{\mu\nu,h} \in (-\pi, \pi], \quad f_{\mu\nu,h} = h^{-2} \phi_{\mu\nu,h}. \quad (\text{G.1})$$

Let d_h denote the forward exterior derivative on the periodic cubical complex, $\delta_h = d_h^*$ its positive-mass adjoint, and $\Delta_h = d_h \delta_h + \delta_h d_h$ the nonnegative Hodge Laplacian. On one-forms δ_h is (F.3). For an anti-Hermitian $X = (X_3, X_2) \in \mathfrak{g}_{\text{SM}}$ define

$$\ell_c(X) = \frac{1}{i} \text{Tr } X_2, \quad \Pi_{\text{ss}} X = X - Z_c \ell_c(X). \quad (\text{G.2})$$

Both maps are bounded and adjoint equivariant, with ℓ_c adjoint invariant and $\ell_c(Z_c) = 1$.

Lemma G.1 (Determinant compactness, admissibility, and flux). *Suppose the literal full curvature has bounded L_h^2 norm and vanishing adjoint Wilson modulus. Then, for all sufficiently small h ,*

$$f_h = \ell_c(\mathbb{F}_h), \quad \omega_h(f; \rho) \leq C \Omega_h(\mathbb{F}_h; \rho), \quad \max_{x, \mu < \nu} |\phi_{\mu\nu,h}(x)| \longrightarrow 0, \quad (\text{G.3})$$

where ω_h is the ordinary discrete translation modulus. In particular, $\mathbb{R}_h^0 f_h$ is precompact in L^2 and $d_h f_h = 0$ exactly for sufficiently small h . Its coordinate-plane fluxes are transverse-base-point-independent integers:

$$h^2 \sum_{x \in \mathbb{T}_{\mu\nu,h}^2} f_{\mu\nu,h}(x) = 2\pi m_{\mu\nu,h}, \quad \bar{f}_{\mu\nu,h} := L^{-4} h^4 \sum_x f_{\mu\nu,h}(x) = \frac{2\pi}{L^2} m_{\mu\nu,h}. \quad (\text{G.4})$$

The integer vector obeys $2\pi|m_h| \leq \|f_h\|_{2,h}$, and hence is constant along a subsequence. None of these conclusions uses a bound on the logarithms of the original links.

Proof. The scalar-magnitude part of Theorem F.1 gives uniform integrability of $|\mathbb{R}_h^0 \mathbb{F}_h|^2$ without a connection bound. A cell has mass h^4 , so

$$\max_x h^4 |\mathbb{F}_h(x)|^2 \longrightarrow 0.$$

Thus the full scaled curvature $h^2 \mathbb{F}_h$ tends uniformly to zero. Since $\chi(e^X) = e^{i\ell_c(X)}$, the principal determinant argument is $h^2 \ell_c(\mathbb{F}_h)$ eventually. Applying the invariant functional ℓ_c to an adjoint Wilson difference gives exactly the corresponding ordinary difference of f_h . Lemma E.1 therefore proves its compactness and the first two assertions.

The product of the six oriented Abelian plaquettes around a cube is one. Their signed arguments sum to 2π times an integer; when $\max |\phi_h| < \pi/3$, the absolute value of that sum is less than 2π . The integer is zero, giving $d_h f_h = 0$. The product over a periodic coordinate plane is also one, which gives its integer flux. Summing the real cube identity over a plane shows that its flux is unchanged by a transverse shift. Averaging all parallel planes gives the second identity in (G.4). Finally $L^2 |\bar{f}_h| \leq \|f_h\|_{2,h}$ proves the integer bound. The finite set of permitted integer vectors admits a constant subsequence. $\square$

G.1.2 A uniformly controlled transverse potential

Lemma G.2 (Strong stability of the periodic discrete Hodge inverse). *For a closed, zero-mean real two-form f_h , set*

$$a_h^0 = \delta_h \Delta_h^\dagger f_h, \quad (\text{G.5})$$

where $\Delta_h^\dagger$ vanishes on the constant modes. Then

$$\begin{aligned} d_h a_h^0 &= f_h, & \delta_h a_h^0 &= 0, & \overline{a_h^0} &= 0, \\ \|D_h^+ a_h^0\|_{2,h} &= \|f_h\|_{2,h}, & L^{-1} \|a_h^0\|_{2,h} + \|a_h^0\|_{4,h} &\leq C \|f_h\|_{2,h}. \end{aligned} \quad (\text{G.6})$$

If $R_h^0 f_h \rightarrow f$ strongly in L^2 , then $R_h^0 a_h^0 \rightarrow a^0$ strongly in L^4 and $R_h^0 D_h^+ a_h^0 \rightarrow \partial a^0$ strongly in L^2 , where $a^0 = \delta \Delta^\dagger f$. On the odd grids used for trigonometric reconstruction, $\mathcal{I}_h^{\text{trig}} a_h^0 \rightarrow a^0$ strongly in H^1 .

Proof. The Hodge operators commute with $\Delta_h^\dagger$. Since f_h is closed and has no harmonic part, $d_h \delta_h \Delta_h^\dagger f_h = f_h$ and $\delta_h^2 \Delta_h^\dagger f_h = 0$. The discrete Hodge identity gives the exact derivative norm. On a Fourier mode with $\xi_\ell = 2\pi\ell/L$ the symbols are $d_{h,\mu} = (e^{ih\xi_\ell, \mu} - 1)/h$ and $\lambda_h = \sum_\mu |d_{h,\mu}|^2$. The smallest nonzero λ_h is bounded below by cL^{-2} , independently of n . Poincaré and the scaled discrete Sobolev inequality therefore give (G.6).

The multiplier of $\delta_h \Delta_h^\dagger$ is bounded by $C/|\xi_\ell|$, and that of each of its full first differences is uniformly bounded. Both converge at each fixed nonzero frequency. Strong raw L^2 convergence gives convergence of every fixed discrete Fourier coefficient and of the total Parseval mass. Consequently

$$\lim_{R \rightarrow \infty} \limsup_{h \downarrow 0} L^4 \sum_{|\xi_\ell| > R} |\hat{f}_h(\ell)|^2 = 0.$$

Apply the multiplier bounds to this tail and pass the finitely many low modes first. This proves strong L^2 convergence of a_h^0 and of all its first differences. Lemma E.3 then gives the L^4 and reconstructed H^1 conclusions. This proof uses no rate of decay of the tail and no growing frequency band. $\square$

Proposition G.3 (Zero-flux Abelian normalization with unrestricted flat holonomy). *Let periodic $U(1)$ links have $\max |\phi_h| < \pi/3$ and zero flux. The potential (G.5), together with constants $c_{\mu,h} \in [-\pi/L, \pi/L]$, determines a periodic site gauge s_h such that*

$$u_{\mu,h}^{s_h} = e^{iha_{\mu,h}}, \quad a_h = a_h^0 + c_h, \quad \delta_h a_h = 0. \quad (\text{G.7})$$

The constants retain the four cycle holonomies of the flat residual. For a screened bounded-curvature sequence, these coordinates are strongly precompact in raw L^4 , their first differences are strongly precompact in L^2 , and their trigonometric reconstructions are strongly precompact in H^1 . Moreover $h\|a_h\|_{\infty,h} \rightarrow 0$; thus the coordinates are eventually principal. No smallness of $\|f_h\|_{2,h}$, $\|a_h\|_{4,h}$, or the cycle phases is required.

Proof. Zero flux gives $\bar{f}_h = 0$ by (G.4). The plaquette of $e^{iha_h^0}$ is exactly $e^{ih^2 d_h a_h^0} = e^{i\phi_h}$. Hence $r_{\mu,h} = u_{\mu,h} e^{-iha_{\mu,h}^0}$ is flat. Its four cycle products $w_{\mu,h}$ are independent of the base point, since moving a cycle across a strip multiplies it by a product of trivial plaquettes. Choose $Lc_{\mu,h} = \text{Arg } w_{\mu,h}$ using one fixed endpoint convention. The ratio $r_h e^{-ihc_h}$ is flat with trivial cycle products. Its ordered product from a fixed root to a vertex defines a periodic, path-independent site phase s_h with $s_h(x + he_\mu) = s_h(x) r_{\mu,h}(x) e^{-ihc_{\mu,h}}$. This proves (G.7) exactly.

For sequences, [Lemma G.1](#) gives strong curvature compactness and [Lemma G.2](#) gives strong potential compactness. Extract also the bounded four-vector c_h . Uniform integrability of the fourth powers of the raw potentials implies $\max_x h^4 |a_h(x)|^4 \rightarrow 0$, proving the resolved-link assertion. The compactness is a statement about the entire family as $h \downarrow 0$: otherwise a violating sequence would have the convergent subsequence just constructed. $\square$

Remark G.4 (Nonzero flux is not removed by screening). For fixed flux m , the fluctuation potential $\delta_h \Delta_h^\dagger(f_h - 2\pi m/L^2)$ still satisfies the preceding bounds and compactness statement. The remaining flux must be kept in a background or transition description. For example, for one pair $\mu < \nu$, lattice indices $k \in \{0, \dots, n-1\}^4$ and integer m , the periodic links

$$v_\nu(k) = e^{2\pi i m k_\mu / n^2}, \quad v_\mu(k) = \begin{cases} e^{-2\pi i m k_\nu / n}, & k_\mu = n-1, \\ 1, & k_\mu \neq n-1, \end{cases}$$

with all other components one have constant physical curvature $f_{\mu\nu} = 2\pi m/L^2$ on every plaquette, including the seam, for large n . Their curvature translation modulus is zero. Conversely, links $e^{i h a_h}$ with one periodic real potential and $h \|a_h\|_\infty \rightarrow 0$ have zero flux eventually: the principal plaquette argument equals the real curl of $h a_h$, and summing that curl over a coordinate plane telescopes to zero. Thus the global hypothesis in [Corollary G.8](#) cannot be dropped on regularity grounds. No global trivialization of a nonzero-flux sector is claimed.

G.1.3 Assembly in the actual Standard-Model group

Proposition G.5 (Exact determinant normalization and semisimple remainder). *Let the full curvature screen hold and the determinant flux vanish. Construct s_h, a_h by [Proposition G.3](#). At each site choose $t_h(x) \in \mathbb{R}$ with $e^{i t_h(x)} = s_h(x)$ and define*

$$g_h(x) = e^{t_h(x) Z_c}, \quad U'_h = U_h^{g_h}, \quad V_{\mu,h} = e^{-h a_{\mu,h} Z_c} U'_{\mu,h}. \quad (\text{G.8})$$

Then $V_h \in G_{\text{ss}}$ exactly and, for sufficiently small h ,

$$F_h(V_h) = \Pi_{\text{ss}} F_h(U'_h). \quad (\text{G.9})$$

The adjoint Wilson transport of U'_h equals that of V_h . Its semisimple curvature screen is therefore bounded by the original full screen, up to a fixed norm constant. A change of the site lifts t_h changes V_h only by a central G_{ss} site gauge, and does not change its rooted certificate. All transformations applied to matter and to the common action are transformations of the combined G_{SM} record.

Proof. By [\(G.11\)](#), $\chi(e^{t Z_c}) = e^{i t}$, so $\chi(V_{\mu,h}) = e^{-i h a_{\mu,h}} \chi(U'_{\mu,h}) = 1$. Centrality gives the exact plaquette relation

$$P_{U',\mu\nu} = e^{h^2 f_{\mu\nu,h} Z_c} P_{V,\mu\nu}.$$

All scaled full-curvature arguments tend uniformly to zero by [Lemma G.1](#). The commuting central factor and the unique local logarithm then give $F_h(V_h) = F_h(U'_h) - Z_c f_h$, which is [\(G.9\)](#). Central factors act trivially in the adjoint representation, so the open-link transports and their projected curvature differences have the asserted relation.

The central one-parameter subgroup has period 12π , not 2π :

$$z_6 = e^{2\pi Z_c} = (e^{-2\pi i/3} I_3, -I_2) \in G_{\text{ss}}, \quad z_6^6 = I. \quad (\text{G.10})$$

Changing $t_h(x)$ to $t_h(x) + 2\pi k_x$ thus multiplies $g_h(x)$ by the central site element $z_6^{k_x}$. The corresponding change of V_h is an actual G_{ss} gauge transformation; apply [Proposition F.7](#) for certificate invariance. A semisimple normalization q_h of V_h acts on the full record by $q_h g_h$, with its central factor retained. Exact covariance preserves all sector actions, the full physical covector norm, and the full matter Wilson and graph norms. In particular no fractional hypercharge is regarded as a single-valued character of the determinant circle, and V_h alone is never substituted for the matter connection. $\square$

Lemma G.6 (Flat semisimple normalization without a small-holonomy assumption). *If V_h is a flat periodic $\text{SU}(3) \times \text{SU}(2)$ link field, it is gauge equivalent to constant links $e^{hb_{\mu,h}}$ with commuting, uniformly bounded $b_{\mu,h} \in \mathfrak{g}_{\text{ss}}$. After extraction these constant coordinates converge, and their first differences and curvatures vanish identically. Their cycle holonomies need not be close to the identity.*

Proof. Flatness implies that the based coordinate-cycle holonomies commute. In each $\text{SU}(N)$ factor, simultaneously diagonalize them and choose real eigenangles for each holonomy. Their sum is an integer multiple $2\pi m$; subtracting $2\pi m$ from one eigenangle preserves its exponential and gives a traceless anti-Hermitian logarithm. All angles remain bounded by a constant depending only on N . Dividing the four logarithms by L gives commuting $b_{\mu,h}$ with the prescribed cycle holonomies. Comparison of path transports for the two flat link fields gives a site gauge: plaquette moves preserve the comparison, and their equal based cycle holonomies make it periodic. This argument compares two flat connections; it does not assume that the pointwise ratio of arbitrary non-Abelian connections is flat. The bounded finite-dimensional tuple $(b_{\mu,h})_\mu$ has a convergent subsequence. Constant conjugations needed for diagonalization are also actual internal gauges. $\square$

Example G.7 (Non-small determinant curvature with compact gauge data). Let M_0, c be arbitrary fixed real numbers, and on the side- L grid set

$$a_{1,h}(x) = M_0 \sin(2\pi x^0/L) + c, \quad a_{\mu,h} = 0 \ (\mu \neq 1), \quad U_{\mu,h} = e^{hZ_c a_{\mu,h}}.$$

The divergence is zero and the determinant flux vanishes. Its only nonzero determinant curvature component is $M_0 D_{0,h}^+ \sin(2\pi x^0/L)$, which converges strongly in L^2 and has a vanishing translation modulus. Its norm need not be small, and the flat cycle phase contributed by c is unrestricted. The remainder of (G.8) is flat. Thus the gauge and compactness conditions of [Corollary G.8](#) include bounded families excluded by a small full-curvature certificate. Arbitrary site gauges do not change these conclusions. This example verifies the range of the gauge criterion only; it does not assert coupled finite Einstein–Standard-Model stationarity.

G.2 Determinant-resolved closure

The smallness condition in [Theorem 4.7](#) can be restricted to the semisimple factor. For the group in (2.1), put

$$\chi(U_3, U_2) = \det U_2 = (\det U_3)^{-1}, \quad G_{\text{ss}} = \ker \chi = \text{SU}(3) \times \text{SU}(2). \quad (\text{G.11})$$

The Abelian curvature of $\chi(U_h)$ is its principal plaquette argument divided by h^2 . Its six integer coordinate-plane fluxes are defined in (G.4). The full curvature Wilson screen supplies both compactness of this scalar curvature and eventual admissibility of its arguments; no independent determinant screen is imposed. On zero flux, [Propositions G.3](#) and [G.5](#) construct from the links a real co-closed determinant potential a_h and an exactly defined semisimple remainder $V_h \in G_{\text{ss}}$.

The potential retains its four harmonic coordinates. All matter fields continue to use the combined G_{SM} links.

Corollary G.8 (Determinant-resolved finite Wilson closure). *Assume (F1), (F3), (F4) and (F5) of Theorem 4.7. In place of (F2), assume that the determinant flux is zero for all sufficiently small h and that the semisimple remainder constructed above satisfies*

$$\|B_h^{\mathfrak{T}}(V_h)\|_{4,h} + \|F_h(V_h)\|_{2,h} \leq \epsilon_c \quad (\text{G.12})$$

for a predeclared rooted tree and admissible logarithms. Here ϵ_c is the threshold of Theorem F.6 for G_{ss} . Then all the convergence and variational conclusions of Theorem 4.7 hold after actual G_{SM} site gauges. The full logarithmic connection is co-closed and has the form

$$A_h = b_h + Z_c a_h, \quad Z_c = \left(-\frac{i}{3}I_3, \frac{i}{2}I_2\right), \quad \delta_h b_h = \delta_h a_h = 0. \quad (\text{G.13})$$

Only b_h needs to lie in the small critical ball. Neither the determinant curvature norm, the determinant potential L_h^4 norm, nor its flat cycle holonomies is required to be small.

Alternatively, (G.12) may be replaced by exact flatness of V_h . In that case its semisimple cycle holonomies also need not be small. Zero determinant flux is a topological restriction of the single periodic-coordinate conclusion, not a consequence of the curvature screen.

Proof. Lemma G.1 first extracts the determinant curvature without selecting a gauge. On zero flux, Proposition G.3 constructs a_h with strong L^4 and first-difference L^2 compactness. The exact group split of Proposition G.5 yields V_h with its semisimple curvature Wilson screen inherited from the original full record. Under (G.12), Proposition F.7 supplies b_h in small discrete Coulomb gauge. Theorem F.1 gives strong semisimple curvature compactness, and Theorem F.3 gives strong L^4 and full first-difference L^2 convergence of b_h . When V_h is flat, the bounded constant representatives of Lemma G.6 give these conclusions directly, without smallness.

Since Z_c is central, adding $Z_c a_h$ preserves the Coulomb condition and introduces no mixed curvature commutator. The normalized actual links are exactly $e^{hb_h} e^{hZ_c a_h} = e^{hA_h}$. Their logarithms are resolved: $h\|a_h\|_{\infty} \rightarrow 0$, and the semisimple part is either uniformly in the small logarithm chart or bounded and constant. The full A_h is thus bounded in L_h^4 and converges strongly there, although its norm need not be small. The full literal curvature is the sum of the semisimple curvature and $Z_c f_h$.

The unchanged Higgs Wilson screen now gives strong Higgs-gradient compactness by Theorem F.1, which requires only a bounded full connection L_h^2 norm. Apply Propositions E.4 and E.6 and then Theorem E.8. Both applied normalizations are site transformations of G_{SM} , so the complete physical covector norm is unchanged. Assumption (F5) therefore gives all limiting Euler identities. The reconstructed consistency and stress conclusions follow from the same strong-packet products as in the preceding theorem; no small-Coulomb theorem is applied to the full, possibly large, A_h . $\square$

Remark G.9 (Logical scope of the finite certificates). The order of the argument is gauge normalization, compactness of the literal packets, identification of the limiting connection and covariant derivatives, and convergence of the complete first variation. Physical stationarity is used only in the last step. Scalar-magnitude compactness, which precedes gauge normalization in the determinant argument, follows from unitarity alone. The semisimple certificate is still an independent finite hypothesis: the result is not a non-Abelian curvature-only gauge construction. The geometric and spinor conditions are unchanged. Neither this corollary nor Theorem 4.7 uses a verified Cauchy tube, a source-drift law, or the high-order reader bounds needed for the separate quantitative shadow theorem.

G.3 Gauge-intrinsic test budgets and Wilson-screen transfer

This subsection gives a regular reconstruction route to the equivariant budgets of [Theorem 4.6](#) and compares finite Wilson shifts with continuum parallel transport. Its regularity-dependent transfer estimates are not premises of the native closure proof in [Appendix E](#) and [Section G.1](#). In the equivariant-budget argument a normalized continuum test is pulled back to the original representation, with exact covariance expressed by (4.8). This test-based argument is distinct from the actual finite site-gauge normalizations used in those appendices.

G.3.1 Equivariant consistency of the explicit local action

For a metric bundle connection $\mathbb{D}$ on a finite direct sum of the coframe, adjoint, Higgs and spinor test bundles, the ordinary cellwise quadrature estimate has the following covariant form.

Lemma G.10 (Covariant cell-average quadrature). *Let $\mathbb{D}$ be a smooth metric bundle connection on the buffered chart. For a cell $C_x = x + [0, h]^4$, use parallel transport along a fixed family of paths inside the cell, each of length at most Ch , and define*

$$(\mathbb{Q}_h^\mathbb{D} v)(x) = h^{-4} \int_{C_x} P_{x \leftarrow y}^\mathbb{D} v(y) \, dy. \quad (\text{G.14})$$

All integrals and norms in this lemma use coordinate Lebesgue measure; a fixed smooth comparison density can be incorporated into the Euler density X . For a smooth X with bounded covariant derivative,

$$\left| h^4 \sum_x \langle X(x), \mathbb{Q}_h^\mathbb{D} v(x) \rangle - \int \langle X, v \rangle \, dx \right| \leq Ch \|\mathbb{D}X\|_\infty \|v\|_1. \quad (\text{G.15})$$

Moreover $\|\mathbb{Q}_h^\mathbb{D} v\|_{\infty, h} \leq \|v\|_\infty$ and $\|\mathbb{Q}_h^\mathbb{D} v\|_{2, h} \leq \|v\|_2$. The formulas are equivariant under unitary bundle changes of frame.

Proof. Metric compatibility rewrites the difference as

$$\sum_x \int_{C_x} \langle P_{y \leftarrow x}^\mathbb{D} X(x) - X(y), v(y) \rangle \, dy.$$

The covariant fundamental theorem of calculus along the chosen path bounds the first factor by $Ch\|\mathbb{D}X\|_\infty$. Integrating proves (G.15). The two norm estimates follow from unitarity and Jensen's inequality. Endpoint transformation of parallel transport proves equivariance. $\square$

The averaging in (G.14) is applied only to physical tests, not to the observed fields or the finite action. It avoids point evaluation on a four-dimensional $W^{1,2}$ test class; a nodal quadrature estimate cannot be inferred from that norm alone. The native C^r consistency theorem uses the separately specified smooth nodal test lift and does not require this averaging construction.

Let $B \geq 1$ be a derivative scale and suppose a reconstructed field tuple satisfies

$$\|\partial^\alpha z\|_{L^\infty(Q')} \leq C_q B^{|\alpha|}, \quad |\alpha| \leq q, \quad q \geq 5, \quad hB \leq c_{\text{res}}. \quad (\text{G.16})$$

The complete continuum Euler density then has differential degree at most two in the bosonic rows and one in the Dirac rows, so on the fixed chart

$$\|\mathcal{E}_0(z)\|_{L^2} \leq CB^2, \quad \|\mathbb{D}_z \mathcal{E}_0(z)\|_{L^\infty} \leq CB^3. \quad (\text{G.17})$$

For the physical coframe lift of a test v ,

$$\|\tilde{v}_z\|_{L^\infty} \leq C\mathbf{n}_z(v), \quad \|\mathbb{D}_z\tilde{v}_z\|_{L^2} \leq CB\mathbf{n}_z(v). \quad (\text{G.18})$$

The second estimate follows because differentiating the symmetric coframe lift introduces at most one first coframe derivative, while the matter components are controlled directly by (4.8).

For the gauge links it is convenient to use physical left-link tangents. If $U_\mu = e^{hA_\mu}$ and $\dot{U}_\mu U_\mu^{-1} = h\alpha_\mu$, let

$$\|V_h\|_{0,h,\text{link}}^2 := h^4 \sum_x \left(|\dot{e}|^2 + |\alpha|^2 + |\eta_H|^2 + |\eta_\Psi|^2 + |\eta_{\bar{\Psi}}|^2 \right) \quad (\text{G.19})$$

with the fixed positive coefficient metrics, and let σ_h^{link} be the dual norm of DS_h^{loc} in this metric. On every compact logarithm chart the differential of the exponential map is uniformly invertible, so σ_h^{link} is uniformly equivalent to the logarithmic-coordinate row norm used in (5.3).

Proposition G.11 (Equivariant native first-variation budgets). *Under (G.16), the unchanged local action of Section D.1 and the covariant cell-average physical test lift $\mathcal{I}_h^{\text{cov}}$ defined below satisfy*

$$|DS_h^{\text{loc}}[\mathcal{I}_h^{\text{cov}}v] - D\mathcal{S}_\theta(z)[v]| \leq ChB^3\mathbf{n}_z(v), \quad (\text{G.20})$$

$$|DS_h^{\text{loc}}[\mathcal{I}_h^{\text{cov}}v]| \leq C\sigma_h^{\text{link}}\mathbf{n}_z(v). \quad (\text{G.21})$$

Thus the budgets (4.9)–(4.10) may be chosen with

$$c_h^{\text{eq}} \leq ChB^3, \quad \varepsilon_h^{\text{eq}} \leq C\sigma_h^{\text{link}}. \quad (\text{G.22})$$

The link-row norm is exactly invariant under finite site gauge transformations.

Proof. Express the complete raw Euler covector in the physical coframe, logarithmic-link and matter coordinates of Proposition H.1. Let $\tilde{v}_z$ be the corresponding continuum physical-coordinate test, including its symmetric coframe lift. Choose the positive comparison connection $\mathbb{D}_z$ acting on these coordinate bundles and define the finite coordinate test by $Q_h^{\mathbb{D}_z}\tilde{v}_z$. Its gauge-coordinate component a_h is converted to the actual link tangent by $h^{-1}\dot{U}U^{-1} = \mathcal{J}(h\text{ad}_A)a_h$. This specifies $\mathcal{I}_h^{\text{cov}}$ as a linear physical test lift; no finite field is replaced by a cell average.

Pair the $O(hB^3)$ pointwise Euler-row consistency estimate with this averaged test. Jensen and (G.18) bound the contribution by $ChB^3\mathbf{n}_z(v)$. For the remaining comparison apply Lemma G.10 to $X = \mathcal{E}_0(z)$ and $\tilde{v}_z$. Equation (G.17) gives $\|\mathbb{D}_z X\|_\infty \leq CB^3$, and on the fixed box $\|\tilde{v}_z\|_1 \leq C\mathbf{n}_z(v)$. Thus this term has the same bound, proving (G.20) uniformly even when the ordinary derivatives of the pulled-back test are large.

The averaged coordinate lift has mass norm at most $C\mathbf{n}_z(v)$. The logarithmic-to-left-link Jacobian is uniformly bounded on the fixed scaled logarithm chart, so the same bound holds in the physical link/matter mass metric. Cauchy–Schwarz proves (G.21). Exact site-gauge invariance of the finite action and the isometry of its physical tangent map preserve the complete dual link-row norm. Uniform invertibility of the same Jacobian proves the claimed equivalence with the logarithmic-coordinate row norm. This last invariance is a statement about the finite covector, independent of the auxiliary test averaging. $\square$

G.3.2 Finite Wilson packets and continuum covariant translations

For a continuum connection A and a coordinate displacement se_μ , let $P_{\mu,s}^A(x)$ denote parallel transport from $x + se_\mu$ back to x and put

$$\mathcal{T}_{\mu,s}^A Y(x) := \rho(P_{\mu,s}^A(x))Y(x + se_\mu). \quad (\text{G.23})$$

This operation is exactly internal-gauge covariant.

Proposition G.12 (Finite packet and open-link transfer). *Assume (G.16) and $hB_h^2 \rightarrow 0$. For the literal packets (4.21),*

$$\max_x |\mathbb{F}_h(x) - F_{A_h}(x)| + \max_x |\mathbb{K}_h(x) - D_{A_h} H_h(x)| \leq ChB_h^2, \quad (\text{G.24})$$

and for $|m|h \leq \varrho_0$,

$$\left\| \rho(\mathcal{U}_{\mu,m}(x)) - \rho(P_{\mu,mh}^{A_h}(x)) \right\| \leq C\varrho_0 h B_h. \quad (\text{G.25})$$

Consequently, if a literal packet Y_h^{d} obeys

$$\lim_{\varrho \downarrow 0} \limsup_{h \downarrow 0} \Omega_h(Y_h^{\text{d}}; \varrho) = 0, \quad (\text{G.26})$$

then its continuum packet Y_h satisfies, on every smaller chart,

$$\lim_{\varrho \downarrow 0} \limsup_{h \downarrow 0} \max_{\mu} \sup_{|s| \leq \varrho} \|\mathcal{T}_{\mu,s}^{A_h} Y_h - Y_h\|_{L^2} = 0. \quad (\text{G.27})$$

Proof. The plaquette expansion of Lemma H.3 and the amplitude-preserving scaling of Lemma H.4 give

$$U_{\alpha\beta}(x) = I + h^2 F_{A_h, \alpha\beta}(x) + O(h^3 B_h^2),$$

which yields the curvature part of (G.24) after applying the analytic logarithm. Covariant Taylor expansion of the Higgs link gives the second part.

For one edge, Duhamel's formula compares the exact parallel-transport ODE with the frozen coefficient entering $U_\mu(x)$; the error is $O(h^2 B_h)$. Unitary factors do not amplify operator norm, so telescoping over $|m|$ edges gives (G.25).

For a lattice displacement $s = mh$, replace both endpoint continuum packets by the finite readers and continuum transport by the open finite link. The resulting errors are $O(hB_h^2)$ and $O(\varrho h B_h) \|Y_h^{\text{d}}\|_2$; the remaining term is exactly the Wilson difference in (4.17). For arbitrary s choose m with $|s - mh| \leq h$. The growing-band reserve controls the final sub-cell displacement by $O(hB_h^2)$. Taking first $h \downarrow 0$ and then $\varrho \downarrow 0$ proves (G.27). $\square$

Lemma G.13 (Translation Kato inequality). *For a unitary bundle representation,*

$$||Y(x + se_\mu)| - |Y(x)|| \leq |\mathcal{T}_{\mu,s}^A Y(x) - Y(x)|. \quad (\text{G.28})$$

Thus a vanishing covariant L^2 translation modulus controls the ordinary translation modulus of the scalar magnitude $|Y|$.

Proof. Parallel transport preserves the fiber norm, and the result is the reverse triangle inequality. $\square$

Lemma G.14 (Compact scalar magnitude implies energy uniform integrability). *If a family is relatively compact in $L^2(K)$ on a finite-measure region, then the squares of its elements are uniformly integrable:*

$$\lim_{|E| \downarrow 0} \sup_f \int_E |f|^2 = 0. \quad (\text{G.29})$$

Proof. Choose a finite L^2 net. Absolute continuity of the integral for the finitely many net elements and $\int_E |f|^2 \leq 2\|f - f_j\|_2^2 + 2 \int_E |f_j|^2$ give the claim. $\square$

Proposition G.15 (Covariant screen compactness at the critical endpoint). *Let Y_h be bounded in $L^2(B)$ and satisfy (G.27). Then $|Y_h|$ is relatively compact in L^2_{loc} and $|Y_h|^2$ is uniformly integrable locally. If, in addition, after local gauge normalization*

$$\sup_h \|A_h\|_{L^4(B)} \leq M_A, \quad (\text{G.30})$$

then Y_h itself is precompact in $L^2_{\text{loc}}(B)$.

Proof. By Lemma G.13, the scalar magnitudes have a vanishing ordinary translation modulus. Their L^2 boundedness and the boundedness of the chart allow the Fréchet–Kolmogorov criterion, proving relative L^2 compactness. Apply Lemma G.14 to obtain uniform integrability of $|Y_h|^2$.

It remains to remove the parallel transport. The transport ODE gives

$$\|\rho(P_{\mu,s}^{A_h}(x)) - I\| \leq C \min\{1, L_{h,s}(x)\}, \quad L_{h,s}(x) = |s| \int_0^1 |A_h(x + \theta se_\mu)| d\theta. \quad (\text{G.31})$$

Minkowski gives $\|L_{h,s}\|_4 \leq |s|M_A$. For $E_{h,s,\delta} = \{L_{h,s} > \delta\}$, Chebyshev gives $|E_{h,s,\delta}| \leq M_A^4 |s|^4 \delta^{-4}$. On the complement, the square of the transport error is bounded by $C\delta^2 |Y_h|^2$; on the exceptional set use the uniform bound on the unitary transport and the uniform integrability just proved. Hence

$$\sup_h \|(\rho(P_{\mu,s}^{A_h}) - I)Y_h(\cdot + se_\mu)\|_2 \longrightarrow 0 \quad (s \rightarrow 0). \quad (\text{G.32})$$

Combining this with the covariant screen gives an ordinary vector translation modulus for Y_h . Fréchet–Kolmogorov now yields local strong L^2 precompactness. $\square$

Lemma G.16 (Critical Coulomb Hodge absorption). *On a fixed small-energy Coulomb ball B suppose A_h, A are co-closed, $A_h \rightarrow A$ in $L^2(B)$, their $L^4(B)$ norms are uniformly sufficiently small, and $F_{A_h} \rightarrow F_A$ in $L^2(B)$. Then, for every $B' \Subset B$,*

$$A_h \rightarrow A \quad \text{strongly in } L^4(B'). \quad (\text{G.33})$$

Proof. Choose $\chi \in C_c^\infty(B)$ with $\chi = 1$ on B' and put $a_h = A_h - A$. Apply the compactly supported Hodge and Sobolev estimates to χa_h , extended by zero. Since $d^*a_h = 0$, differentiation of the cutoff produces only terms bounded by $C_\chi \|a_h\|_2$. The curvature identity therefore gives

$$\|\chi a_h\|_4 \leq C\|F_{A_h} - F_A\|_2 + C_\chi \|a_h\|_2 + C(\|A_h\|_4 + \|A\|_4)\|\chi a_h\|_4.$$

Choose the small-energy threshold to make the last coefficient less than one half. Absorb that term and pass to the limit. No unrestricted Hodge estimate up to the boundary of B is being used. $\square$

H Quantitative consistency and physical forcing estimates

This appendix proves the two finite-action estimates used in [Section 5](#). Their assumptions concern the full reconstructed record and the comparison head on the same buffered chart. The qualitative Wilson theorem does not use this growing-band route.

H.1 Consistency and forcing statements

Proposition H.1 (Growing-band consistency of the unchanged finite action). *Let y be a smooth physical field tuple on Q' with uniformly bounded head amplitudes and inverse metric, and assume*

$$\|\partial^\alpha y\|_\infty \leq C_q K^{|\alpha|}, \quad |\alpha| \leq q, \quad q \geq 5, \quad hK \leq c_{\text{res}}. \quad (\text{H.1})$$

Then the local finite action of [Section D.1](#) satisfies

$$\max_{x \in Q'_h} |E_h^{\text{raw}}(\mathcal{S}_h y)(x) - \mathcal{S}_h \mathcal{E}_0(y)(x)| \leq ChK^3, \quad (\text{H.2})$$

where $\mathcal{E}_0$ is the complete minimally coupled torsion-free Einstein–Standard-Model Euler density in the same physical coordinates. The literal Cartan plaquette curvature also obeys

$$\|I_h^0 R_h^{\text{raw}}(\mathcal{S}_h y) - \text{Riem}(g)\|_{L^2(Q),+} \leq ChK^3. \quad (\text{H.3})$$

The constants depend on the bounded chart but not on h or K .

Proof. The proof is given in [Section H.3](#). Its central point is an amplitude-preserving rescaling $\xi = Kx$, $\rho = hK$, $\nu = K^{-1}$. The exact local density splits as K^2 times a uniformly regular bosonic stencil plus K times a uniformly regular Dirac stencil. First-variation consistency at normalized mesh ρ is $O(\rho)$, giving $O(hK^3)$ after restoring the physical derivative factors. No field amplitude is divided by K and the metric–spinor chain is included. $\square$

Theorem H.2 (Leakage-tolerant physical forcing estimate). *Assume the complete record and its low-frequency comparison head lie in the fixed calibrated field-value chart on the buffered box, with uniform amplitude and inverse-metric margins. Assume $hK \leq c_{\text{res}}$, $\tau_{h,m}(K) \leq \tau_*$, and $\varepsilon_{c,h} \leq 1$, with τ_* sufficiently small for these margins. For the complete, unfiltered reconstruction $z_h = \mathcal{I}_h^{\text{trig}} u_h$,*

$$\epsilon^{\text{phys}}(z_h) := \|\mathcal{R}_B(z_h)\|_{L^2(Q)} + \|\mathcal{R}_D(z_h)\|_{L^2(Q)} \leq C\varepsilon_{0,h}, \quad (\text{H.4})$$

$$\mathcal{Y}_k(z_h) := \|\mathcal{R}_B(z_h)\|_{L_t^2 H_x^k(Q)} + \|\mathcal{R}_D(z_h)\|_{L_t^2 H_x^{k+1}(Q)} \leq CF_{h,k}, \quad (\text{H.5})$$

$$\|I_h^0 R_h^{\text{raw}}(u_h) - \text{Riem}(g_h)\|_{L^2(Q),+} \leq ChK^3. \quad (\text{H.6})$$

Here $\mathcal{R}_B$ collects the Einstein, Yang–Mills and Higgs physical Euler rows, while $\mathcal{R}_D$ contains the Dirac and dual-Dirac rows. No filtering is applied to the field entering these residuals or to the finite action.

Proof. [Section H.4](#) proves the finite-order tail transfer. The low-frequency proof field has the same chart margins and a complex spatial strip of width a/K ; [Lemma H.7](#) upgrades its small L^2 source to the differentiated source norm with only the logarithmic factor in (5.4). The full measured tail is then restored at cost $K^{k+2}\tau_{h,m}$. The zero-order estimate follows from [Proposition H.1](#) and [Lemma H.5](#) applied to the full reconstructed field, whose first five derivatives satisfy (H.1) by the tail bounds. The same cellwise comparison gives the Cartan estimate. $\square$

H.2 Regular plaquette expansion

Lemma H.3 (Regularity of the plaquette map). *For bounded matrices X, Y, a, b the map*

$$\Phi(h; X, Y, a, b) = h^{-2} \log(e^{hX} e^{h(Y+ha)} e^{-h(X+hb)} e^{-hY})$$

extends analytically through $h = 0$ with $\Phi(0) = a - b + [X, Y]$. All fixed-order derivatives are uniformly bounded on compact argument sets for sufficiently small h .

Proof. The product inside the logarithm is $I + h^2(a - b + [X, Y]) + O(h^3)$. Since the logarithm is analytic near the identity and the linear term vanishes, the removable quotient may be written as an integral of the second h derivative. This also proves uniform bounds for parameter derivatives. $\square$

H.3 Amplitude-preserving scaling

Lemma H.4 (Differential-order scaling). *Under (H.1), put $\rho = hK$, $\nu = K^{-1}$ and $\tilde{y}(\xi) = y(\xi/K)$, without rescaling the field amplitudes. The exact density (D.5) has the form*

$$\mathcal{L}_h(y) = K^2 \tilde{\mathcal{L}}_{\rho, \nu}^B(\tilde{y}) + K \tilde{\mathcal{L}}_{\rho, \nu}^D(\tilde{y}), \quad (\text{H.7})$$

where both normalized stencil densities and their required finite jet derivatives are uniformly regular for small ρ and $0 \leq \nu \leq 1$. The normalized Euler rows carry the same factors K^2 and K .

Proof. The algebraic Cartan reader is linear in a first difference, so $\omega_h = K\tilde{\omega}_\rho$ and $e^{h\omega_h} = e^{\rho\tilde{\omega}_\rho}$. The Cartan plaquette therefore has factor K^2 . For the gauge plaquette use Lemma H.3 with arguments νA ; the removable quotient shows $F^h = K\tilde{F}_{\rho, \nu}$, so the Yang–Mills density has factor K^2 . The Higgs link is

$$K_\mu^h = K \frac{e^{\rho\nu\rho_H(A_\mu)} H(\xi + \rho e_\mu) - H(\xi)}{\rho},$$

which is a regular first-jet function times K . The spin link is $e^{\rho(\sigma(\tilde{\omega}_\rho) + \nu\rho_S(A))}$, so each Dirac difference has factor K and the bilinear density has factor K . Potential, mass and cosmological coefficients appear with nonnegative powers of ν inside the normalized densities. Since the varied head amplitudes are not rescaled, differentiation of the finite sum leaves exactly the factors in (H.7). $\square$

Proof of Proposition H.1. At normalized mesh ρ , Taylor expansion of the regular shifted-jet densities and their first variations gives an $O(\rho)$ Euler error with constants uniform in ν . Restoring the factors of Lemma H.4 gives $C(K^2 + K)\rho \leq ChK^3$. The same normalized plaquette expansion gives $O(\rho)$ for Cartan curvature, which scales by K^2 , proving (H.3). The metric row differentiates the Cartan reader and the spin link throughout; no connection is frozen in this argument. $\square$

Lemma H.5 (From nodal Euler bounds to a physical source norm). *Let f be a C^1 vector-valued function on the buffered slab Q' and let $I_h^0 \mathbf{S}_h f$ be its piecewise-constant reconstruction on cells meeting $Q \Subset Q'$. For all sufficiently small h ,*

$$\|f\|_{L^2(Q)} \leq C\|\mathbf{S}_h f\|_{0, h; Q'} + Ch\|\partial f\|_{L^\infty(Q')}. \quad (\text{H.8})$$

For the complete physical Euler density of a field satisfying (H.1), the last term is $O(hK^3)$. Thus a nodal consistency estimate and a finite mass-norm Euler bound give a continuum L^2 source estimate without an assumption that the nonlinear Euler density is band limited.

Proof. On a cell based at x , the fundamental theorem of calculus gives $|f(y) - f(x)| \leq Ch\|\partial f\|_\infty$. Square, integrate over the fixed-volume union of cells, and use the exact mass normalization of the piecewise-constant reconstruction. The bosonic Euler density has differential degree at most two and the Dirac density at most one; differentiating once and applying (H.1) bounds its first derivative by CK^3 on the fixed coefficient chart. Bounded invertible algebraic changes from Euler densities to physical rows preserve the asserted L^2 estimate. $\square$

H.4 Measured spectral leakage

For the proof only, write $u = P_{\leq K}u + r$ and $z_h^{\text{lo}} = \mathcal{I}_h^{\text{trig}} P_{\leq K}u_h$. The complete record remains u_h .

Lemma H.6 (Finite-order tail bounds). *For $0 \leq j \leq m = k + 2$,*

$$\max_{|\alpha| \leq j} K^{-|\alpha|} \|\partial^\alpha(z_h - z_h^{\text{lo}})\|_\infty \leq C_j \tau_{h,j}(K). \quad (\text{H.9})$$

For sufficiently small hK and $\tau_{h,m}$ the full and low-frequency fields stay in the same nondegenerate chart. Moreover

$$\|E_h^{\text{raw}}(u_h) - E_h^{\text{raw}}(\mathcal{S}_h z_h^{\text{lo}})\|_{0,h;Q'} \leq CK^2 \tau_{h,2}, \quad (\text{H.10})$$

$$\|\mathcal{R}_B(z_h) - \mathcal{R}_B(z_h^{\text{lo}})\|_{L_t^2 H_x^k} + \|\mathcal{R}_D(z_h) - \mathcal{R}_D(z_h^{\text{lo}})\|_{L_t^2 H_x^{k+1}} \leq CK^{k+2} \tau_{h,m}. \quad (\text{H.11})$$

The corresponding zero-order physical source difference is bounded by $CK^2 \tau_{h,2}$.

Proof. Differentiate the finite Fourier tail and use $|\ell^\alpha| \leq K^{|\alpha|}(1 + |\ell|_1/K)^j$. Smooth algebraic dependence of the metric, inverse metric, frame and finite representation coefficients on the field heads gives (H.9). The finite Euler row is, after the exact shifted-first-jet reduction in Lemma H.4, a smooth function of finitely many shifted jets through second order with overall bosonic factor at most K^2 ; the mean-value theorem gives (H.10). The continuum bosonic Euler rows have differential degree at most two and the Dirac rows degree one. Differentiate their difference k and $k + 1$ spatial times, respectively; every product contains at least one tail factor and at most $k + 2$ total field derivatives. Equation (H.9) gives (H.11). $\square$

Lemma H.7 (Logarithmic derivative upgrade for a band-limited source). *Let $f(t, x)$ extend holomorphically in the three spatial variables to width a/K , be bounded there by A , and satisfy $\|f\|_{L^2(Q)} \leq \epsilon \leq 1$. Then for every fixed integer $j \geq 1$,*

$$\|f\|_{L_t^2 H_x^j(Q)} \leq C_j \epsilon K^j \left[1 + \log \left(2 + \frac{C_j A K^{j+3}}{\epsilon} \right) \right]^j. \quad (\text{H.12})$$

When $\epsilon = 0$ the right side is interpreted as zero.

Proof. Spatial contour shifting gives $|\widehat{f}(t, \ell)| \leq A e^{-a|\ell|_1/K}$. For a frequency threshold B , Plancherel bounds the low frequencies by $C_j(1 + B)^j \epsilon$, while the exponentially small tail is bounded by $C_j A K^{j+3} e^{-aB/(2K)}$. Choosing B proportional to $K[1 + \log(2 + C_j A K^{j+3}/\epsilon)]$ proves the claim. $\square$

Proof of Theorem H.2. The full field satisfies the growing C^5 reserve by Lemma H.6; hence Proposition H.1 and (5.3), followed by the nodal-to-continuum estimate Lemma H.5, give (H.4) for the unfiltered field. For the differentiated source, (H.10) first transfers the small finite row to the low-frequency field at cost $K^2 \tau_{h,2}$. By Bernstein estimates the low-frequency field changes by at most a fixed constant times $K|\text{Im } x|$ in a complex spatial strip. Taking its width a/K with a small

relative to the declared field-value and inverse-metric margins preserves the analytic branches of every algebraic coefficient. Complexification is in the real field coordinates, including real and imaginary spinor and Higgs components. Thus the bosonic and Dirac source amplitudes in this strip are bounded by CK^2 and CK . [Lemma H.5](#) supplies the needed continuum L^2 bound for these nonlinear sources before the logarithmic upgrade. Apply [Lemma H.7](#) with $j = k$ and $j = k + 1$. Finally restore the measured full tail by [\(H.11\)](#). This is exactly the budget [\(5.4\)](#). The Cartan assertion is the second estimate of [Proposition H.1](#) applied to the full field. $\square$

Proposition H.8 (A sufficient nodal-accuracy condition). *Let u_h^b have Fourier support in $|\ell|_\infty \leq K$ and let $u_h = u_h^b + e_h$ with $\max_x |e_h(x)| \leq e$, preserving the affine field-chart constraints. Then*

$$\tau_{h,j}(K) \leq C_j e h^{-2} (hK)^{-j}. \quad (\text{H.13})$$

In particular, with $m = k + 2$, the sufficient tolerance

$$e \leq c h^{m+3} K^{m+2} \quad (\text{H.14})$$

implies $\tau_{h,m} = O(hK^2)$ and, on the rate families of [Corollary H.9](#), $\tau_{h,2} = O(hK)$.

Proof. Parseval for the normalized four-dimensional discrete Fourier transform gives $\sum_\ell |\hat{e}_h(\ell)|^2 \leq e^2$. There are $O(h^{-4})$ Fourier modes, so Cauchy–Schwarz bounds the absolute coefficient sum by $Ch^{-2}e$. On the grid spectrum $(1 + |\ell|_1/K)^j \leq C_j (hK)^{-j}$, which proves [\(H.13\)](#). Substitute [\(H.14\)](#) for the final assertion. $\square$

H.5 An explicit cofinal rate

Corollary H.9 (An explicit cofinal rate). *Suppose $\sigma_h = O(h)$, $K_h \asymp h^{-\beta}$ with $0 < \beta < 1/(k+4)$, and*

$$\tau_{h,2} = O(hK_h), \quad \tau_{h,k+2} = O(hK_h^2). \quad (\text{H.15})$$

Then

$$F_{h,k} = O\left(h^{1-\beta(k+4)}(1 + |\log h|)^{k+1}\right), \quad \epsilon^{\text{phys}}(z_h) = O(h^{1-3\beta}).$$

For example,

$$k = 4, \quad K_h \asymp h^{-1/18}, \quad i_{h,4} + \gamma_{h,4} = O(h^{1/2}) \quad (\text{H.16})$$

with [\(H.15\)](#) gives $O(h^{1/2})$ state, curvature and Standard-Model stress convergence; the strong forcing is $O(h^{5/9}(1 + |\log h|)^5)$ and the zero-order physical and Cartan residuals are $O(h^{5/6})$. Along a dyadic sequence the adjacent state, curvature and stress differences are summable, so cofinal transport is a conclusion on this branch.

Proof. The hypotheses give $\varepsilon_{c,h} = O(hK_h^3)$ and $L_{h,k} = O(1 + |\log h|)$. Insert these bounds in [\(5.4\)](#) and apply [Theorem 5.1](#). For $k = 4$, $\beta = 1/18$, the exponents are $5/9$ and $5/6$ as stated; comparison of adjacent records through the same reference gives dyadic summability. $\square$

I Initial-constraint retraction and shadow refinements

The exact constraint reduction and verified Cauchy tube are defined in [Section 5.3](#). We first solve the retained exact constraint map, then distinguish evaluation and complementary-correction errors, and finally derive the full-record forcing interpolation.

I.1 Exact and validated initial correction

Proposition I.1 (Uniform initial-constraint retraction). *Assume [Definition 5.3](#). Suppose*

$$A_h := D_\lambda \Phi_h(x_h, 0), \quad \sigma_{\min}(A_h) \geq \gamma_* > 0, \quad (\text{I.1})$$

and, on $|\lambda| \leq r_*$,

$$\|D_\lambda \Phi_h(x_h, \lambda) - A_h\| \leq \frac{\gamma_*}{2}. \quad (\text{I.2})$$

If

$$c_h := |\Phi_h(x_h, 0)| \leq \beta_h, \quad 2\gamma_*^{-1}\beta_h \leq r_*, \quad (\text{I.3})$$

there is a unique zero λ_h of the reduced exact map in the closed ball of radius $2\gamma_*^{-1}\beta_h$. The corrected datum $d_h = \iota_h(x_h, \lambda_h)$ belongs to $\mathcal{C}_k$ and satisfies

$$\|d_h - \mathcal{U}(\hat{z}_h)(0)\|_{H^k} \leq 2B_*\gamma_*^{-1}\beta_h. \quad (\text{I.4})$$

Only the initial comparison datum is corrected; the observed spacetime record is not projected or filtered. Membership in the regular Cauchy tube is a separate condition in the shadow theorem.

Proof. Write $\Phi_h(x_h, \lambda) = \Phi_h(x_h, 0) + A_h\lambda + N_h(\lambda)$. Then $N_h(0) = 0$ and $\|N_h(\lambda) - N_h(\mu)\| \leq (\gamma_*/2)|\lambda - \mu|$ on the ball. The frozen-inverse map

$$\mathcal{T}_h(\lambda) = -A_h^{-1}\Phi_h(x_h, 0) - A_h^{-1}N_h(\lambda)$$

is 1/2-Lipschitz. On $|\lambda| \leq 2\gamma_*^{-1}\beta_h$ it satisfies

$$|\mathcal{T}_h(\lambda)| \leq \gamma_*^{-1}\beta_h + \frac{1}{2}|\lambda| \leq 2\gamma_*^{-1}\beta_h.$$

Banach contraction gives a unique fixed point in this ball, and invertibility of A_h identifies it with a zero of the exact reduced map. The exact factorization, not the contraction alone, now gives $\mathfrak{C}_k(d_h) = j_h\Phi_h(x_h, \lambda_h) = 0$. Finally (5.9) and (5.11) give (I.4). All constants are independent of m_h when the displayed physical norm margins are uniform. $\square$

Corollary I.2 (Validated constraint evaluation with an explicit error budget). *Assume the exact family and factorization of [Definition 5.3](#). Let $\tilde{\Phi}_h$ be an evaluated C^1 reduced map and put $\tilde{A}_h = D_\lambda \tilde{\Phi}_h(x_h, 0)$. Suppose, on the correction ball,*

$$\begin{aligned} \sigma_{\min}(\tilde{A}_h) &\geq \gamma_*, \\ \sup_\lambda \|D_\lambda \tilde{\Phi}_h(x_h, \lambda) - \tilde{A}_h\| &\leq \gamma_*/4, \\ \sup_\lambda \|D_\lambda \Phi_h(x_h, \lambda) - D_\lambda \tilde{\Phi}_h(x_h, \lambda)\| &\leq \gamma_*/4, \\ |\Phi_h(x_h, 0) - \tilde{\Phi}_h(x_h, 0)| &\leq \zeta_{0,h}. \end{aligned} \quad (\text{I.5})$$

Then the retraction and displacement conclusions of [Proposition I.1](#) hold with the effective residual bound

$$\beta_h^{\text{eff}} = |\tilde{\Phi}_h(x_h, 0)| + \zeta_{0,h}, \quad 2\gamma_*^{-1}\beta_h^{\text{eff}} \leq r_*. \quad (\text{I.6})$$

In every shadow estimate using this evaluated map, β_h is replaced by β_h^{eff} . In particular an exact zero of $\tilde{\Phi}_h$ does not remove $\zeta_{0,h}$.

Proof. The two derivative bounds give $\sup_\lambda \|D_\lambda \Phi_h(x_h, \lambda) - \tilde{A}_h\| \leq \gamma_*/2$. Use the frozen matrix $\tilde{A}_h$ in the preceding contraction proof, applied to the exact map Φ_h . The initial residual has norm at most β_h^{eff} . The fixed point therefore solves the exact constraints, with displacement at most $2B_*\gamma_*^{-1}\beta_h^{\text{eff}}$. $\square$

Remark I.3 (Initial normalization and complementary corrections). The base point in (5.9) is the actual initial jet of the normalized record. Fixing the initial hypersurface and the first differential of a coordinate transformation is not used to identify all metric or spinor first jets before and after that transformation. If a preliminary complementary solve instead supplies a base point with

$$\|\iota_h(x_h, 0) - \mathcal{U}(\hat{z}_h)(0)\|_{H^k} \leq \zeta_h^{\text{comp}},$$

the triangle inequality replaces the initial bound by $\zeta_h^{\text{comp}} + 2B_*\gamma_*^{-1}\beta_h^{\text{eff}}$. That full quantity must enter the shadow budget. The formulas stated below use the exact-basepoint case (5.9); any nonzero complementary correction requires this explicit additional charge.

I.2 Residual interpolation

Proposition I.4 (Full-record residual interpolation). *Assume (5.12)–(5.13) and let the complete finite Euler-row norm satisfy $\sigma_h \leq s_h$. Then on a common smaller cylinder*

$$\varepsilon_h := \|\mathcal{R}_B(z_h)\|_{L^2} + \|(r_D, \bar{r}_D)(z_h)\|_{L^2} \leq C_R(s_h + h). \quad (\text{I.7})$$

Put

$$\eta_{h,k} := C_R \left(\varepsilon_h^{1-k/s} + \varepsilon_h^{1-(k+1)/s} \right). \quad (\text{I.8})$$

Then

$$\|\mathcal{R}_B(z_h)\|_{H^k(Q)} + \|(r_D, \bar{r}_D)(z_h)\|_{H^{k+1}(Q)} \leq \eta_{h,k}. \quad (\text{I.9})$$

For $\varepsilon_h \leq 1$, $\eta_{h,k} \leq C_R \varepsilon_h^{1-(k+1)/s}$.

Proof. The uniform reader reserve permits Proposition H.1 and Lemma H.5 to be applied with a fixed derivative scale, which gives (I.7). The bosonic Euler rows have differential order at most two and the Dirac rows order at most one. Equation (5.13) therefore gives a cutoff-independent H^s bound for the differentiated residuals. Sobolev/Fourier interpolation between this bound and (I.7) gives exponents $1 - k/s$ for the bosonic H^k residual and $1 - (k+1)/s$ for the Dirac H^{k+1} residual. The second exponent is smaller and dominates when $\varepsilon_h \leq 1$. $\square$

I.3 Harmonic defects and an illustrative rate

Corollary I.5 (Exact shadows with a monitored harmonic defect). *Retain the exact-reduction, regular Cauchy-tube and physical-forcing hypotheses of Theorem 5.4, but allow the normalized record $\hat{z}_h$ to be only approximately harmonic. Keep exact temporal internal gauge and the common foliation margins. Put*

$$\begin{aligned} \hat{\gamma}_{h,k} &= \|C(\hat{g}_h)\|_{L_t^2 H_x^{k+1}} + \|\partial_t C(\hat{g}_h)\|_{L_t^2 H_x^k}, \\ \mathfrak{d}_h^C &= 2B_*\gamma_*^{-1}\beta_h + C_R\sqrt{T}(\eta_{h,k} + \hat{\gamma}_{h,k}). \end{aligned} \quad (\text{I.10})$$

If the enlarged budget lies within the same stability radius, the exact solution with corrected initial datum d_h satisfies (5.15) with $\mathfrak{d}_h^C$ in place of $\mathfrak{d}_h$. A validated constraint-evaluation error or a nonzero complementary initial correction is included as in Corollary I.2 and Remark I.3.

Proof. Generate the exact solution from the same corrected compatible datum. Use (J.16) with the additional forcing in (J.11). Cauchy–Schwarz in time charges it by $C_R\sqrt{T}\hat{\gamma}_{h,k}$. The first-exit and curvature/stress arguments are unchanged. No projection of the approximate metric onto exact harmonic gauge is used. $\square$

Corollary I.6 (A transparent data-centred rate). *Take $k = 4$, $s = 10$, $q \geq 15$ and fix one certified reader-energy radius R . If*

$$s_h = O(h), \quad \beta_h = O(h^{1/2}), \quad (\text{I.11})$$

then

$$\varepsilon_h = O_R(h), \quad \eta_{h,4} = O_R(h^{1/2}), \quad \mathfrak{d}_h = O_R(h^{1/2}), \quad (\text{I.12})$$

and the exact physical shadow differs from the normalized finite record by $O_R(h^{1/2})$ in the norms of (5.15).

Proof. For $k = 4$ and $s = 10$, the bosonic interpolation exponent is $3/5$ and the Dirac exponent is $1/2$. Substitute (I.11) into (I.7)–(5.14). $\square$

J Coupled stability and constraint propagation

J.1 Coupled Einstein–matter actual-jet stability

This subsection proves the geometric-analysis stability estimate used in Theorem 5.1. Its variables are actual derivatives and curvatures of the reconstructed fields, so the argument is a coupled Einstein–matter estimate rather than an evolution of auxiliary jets followed by an identification step.

Use one common spacelike foliation and its adapted orthonormal frame

$$e_0 = N^{-1}(\partial_t - \beta^j \partial_j), \quad e_a = e_a^j \partial_j \quad (a = 1, 2, 3), \quad 0 < N_- \leq N \leq N_+. \quad (\text{J.1})$$

Here the spatial frame is a fixed smooth algebraic choice on the positive spatial-metric chart. In particular the e_a , $a = 1, 2, 3$, are tangent to the Cauchy slices. This choice is needed when counting derivatives of the Dirac residual. The lapse, shift and frame coefficients have the uniform margins of the regular chart; harmonic coordinates are not imposed on the approximate metric. Choose Clifford multiplication c_a satisfying $c_a c_b + c_b c_a = -2\eta_{ab}I$. Let ∇ be the torsion-free spin connection twisted by the retained gauge representation and write

$$\mathcal{R}_{ab}^\nabla = \mathcal{R}_{ab}^S + \rho(F_{ab}), \quad \mathcal{R}_{ab}^S = \frac{1}{4} \sum_{c,d} \epsilon_c \epsilon_d R_{abcd} c_c c_d.$$

Lemma J.1 (Twisted half-Ricci contraction). *With the preceding conventions,*

$$\sum_b \epsilon_b c_b \mathcal{R}_{ba}^\nabla = \frac{1}{2} \sum_b \epsilon_b \text{Ric}_{ab} c_b + \sum_b \epsilon_b c_b \rho(F_{ba}). \quad (\text{J.2})$$

Proof. At one point choose a normal orthonormal frame. Insert $\mathcal{R}_{ba}^S = \frac{1}{4} R_{bacd} c^c c^d$ and use $c_b c_c c_d = c_{[b} c_c c_{d]} - \eta_{bc} c_d + \eta_{bd} c_c - \eta_{cd} c_b$. The fully alternating term vanishes by the first Bianchi identity, the last contraction vanishes by antisymmetry, and the remaining two contractions give one half of the Ricci contraction. The gauge curvature acts on the twisting factor and contributes the last term directly. $\square$

Put

$$r_D = \mathcal{D}_{g,A}\Psi - \mathcal{M}_Y(H)\Psi, \quad X_a = \nabla_{e_a}\Psi.$$

Proposition J.2 (Residual-sensitive tangential spinor prolongation). *For each frame index a ,*

$$(\mathcal{D}^{T^*}X)_a = \mathcal{M}X_a + \mathcal{M}_H[D_aH]\Psi + \nabla_a r_D + \frac{1}{2}\text{Ric}_{ab}c^b\Psi + c^b\rho(F_{ba})\Psi. \quad (\text{J.3})$$

Moreover the normal jet is algebraic,

$$X_0 = c_0 c_i X_i - c_0 \mathcal{M}\Psi - c_0 r_D, \quad (\text{J.4})$$

so the tangential prolonged equations require spatial derivatives of r_D but no time derivative of r_D .

Proof. Commute ∇_a with the twisted Dirac operator and use [Lemma J.1](#). The covariant product rule differentiates the affine mass map to $\mathcal{M}_H[D_aH]$. The Dirac equation itself gives [\(J.4\)](#). In a moving frame, the normal covector component enters the tangential covector connection only algebraically, so substituting [\(J.4\)](#) introduces no normal derivative of the residual. $\square$

Let

$$\mathcal{E}_{ab} = G_{ab} + \Lambda g_{ab} - \kappa T_{ab}^{\text{SM}}, \quad \mathcal{E}_{ab}^{\text{tr}} = \mathcal{E}_{ab} - \frac{1}{2}g_{ab} \text{tr}_g \mathcal{E}.$$

Then

$$\text{Ric}_{ab} = \kappa \left(T_{ab}^{\text{SM}} - \frac{1}{2}g_{ab} \text{tr}_g T^{\text{SM}} \right) + \Lambda g_{ab} + \mathcal{E}_{ab}^{\text{tr}}. \quad (\text{J.5})$$

The complete torsion-free Standard-Model stress is algebraic in the actual first jets

$$(g, F, H, DH, \Psi, \bar{\Psi}, X_a, \bar{X}_a)$$

and contains no derivative of F , DH or X . After [\(J.4\)](#) it is a smooth algebraic function of the tangential actual-jet state and is at most linear in the Dirac residual.

For an arbitrary metric in this foliated coordinate chart, in exact temporal internal gauge $A_0 = 0$, define for every coordinate metric component $u = g_{\mu\nu}$, $p_u = e_0 u$, $q_{u,a} = e_a u$, and

$$E_a = F(e_0, e_a), \quad B_a = -\frac{1}{2}\varepsilon_{abc}F(e_b, e_c), \quad \Pi = D_{e_0}H, \quad Q_a = D_{e_a}H.$$

The realified actual-jet state is

$$\mathcal{U} = (g_{\mu\nu}, p_u, q_{u,a}, A_i, E_a, B_a, H, \Pi, Q_a, \Psi, (X_a)_{a=1}^3, \bar{\Psi}, (\bar{X}_a)_{a=1}^3). \quad (\text{J.6})$$

The dual spinor and its tangential covariant jets are included explicitly. On the physical duality branch these are redundant blocks fixed by the adjoint identification; otherwise the dual equation is retained as an additional symmetric-hyperbolic block. All state norms include the retained dual variables. Let r^A , r_H and r_D denote the physical Yang–Mills, Higgs and Dirac residuals.

Lemma J.3 (Exact harmonic-defect forcing identity). *For a smooth metric in the common coordinate chart put $C^\mu = g^{\alpha\beta}\Gamma_{\alpha\beta}^\mu$ and $C_\nu = g_{\nu\mu}C^\mu$. Define*

$$\mathcal{H}_{\mu\nu}(g, C) = \nabla_{(\mu}C_{\nu)} - \frac{1}{2}g_{\mu\nu}\nabla_\alpha C^\alpha. \quad (\text{J.7})$$

Then the reduced tensor of [\(J.21\)](#) satisfies $\hat{\mathbf{G}} = \mathbf{G} - \mathcal{H}$. In four dimensions its residual obeys

$$(\hat{\mathbf{G}} + \Lambda g - \kappa T)_{\mu\nu}^{\text{tr}} = \mathcal{E}_{\mu\nu}^{\text{tr}} - \nabla_{(\mu}C_{\nu)}. \quad (\text{J.8})$$

There is a smooth expression $\mathcal{Q}_{\mu\nu}(g, \partial g)$, quadratic in first metric derivatives, such that every smooth field tuple satisfies

$$-\frac{1}{2}g^{\alpha\beta}\partial_\alpha\partial_\beta g_{\mu\nu} + \mathcal{Q}_{\mu\nu}(g, \partial g) = \kappa(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}\text{tr}_g T) + \Lambda g_{\mu\nu} + \mathcal{E}_{\mu\nu}^{\text{tr}} - \nabla_{(\mu}C_{\nu)}. \quad (\text{J.9})$$

Thus failure of harmonic gauge contributes only C and its first derivatives to the metric wave source.

Proof. The first identity is the definition of the reduced tensor. In dimension four, $\text{tr}_g \mathcal{H} = -\nabla_\alpha C^\alpha$, so trace reversal of $\mathcal{H}$ is exactly $\nabla_{(\mu} C_{\nu)}$. This proves (J.8). To verify the derivative count, expand

$$\text{Ric}_{\mu\nu} = \partial_\alpha \Gamma_{\mu\nu}^\alpha - \partial_\nu \Gamma_{\mu\alpha}^\alpha + \Gamma_{\alpha\lambda}^\alpha \Gamma_{\mu\nu}^\lambda - \Gamma_{\nu\lambda}^\alpha \Gamma_{\mu\alpha}^\lambda.$$

After subtracting $\nabla_{(\mu} C_{\nu)}$, the second-derivative part is $-\frac{1}{2}g^{\alpha\beta}\partial_\alpha\partial_\beta g_{\mu\nu}$. Every remaining term comes from a differentiated inverse metric or a product of Christoffel symbols, hence is smooth in g, g^{-1} and quadratic in ∂g . This defines $\mathcal{Q}$. Combine the resulting reduced Ricci identity with (J.5). This is the usual harmonic reduction [54], here retained as an identity with its nonzero gauge defect rather than restricted to $C = 0$. $\square$

Proposition J.4 (Gauge-defect-aware block-symmetric actual-jet equation). *On a compact nondegenerate foliated chart, every smooth actual field tuple in exact temporal internal gauge, with both spinor rows retained, satisfies*

$$\begin{aligned} \partial_t \mathcal{U} + \sum_{j=1}^3 \mathcal{A}^j(g) \partial_j \mathcal{U} &= \mathcal{F}(\mathcal{U}) + \mathcal{B}(\mathcal{U})(\mathcal{E}, r^A, r_H, r_D, \bar{r}_D) \\ &\quad + \sum_{j=1}^3 \mathcal{Q}^j(g) \partial_j(r_D, \bar{r}_D) + \mathfrak{G}(\mathcal{U}; C, \partial C), \end{aligned} \quad (\text{J.10})$$

$$\mathfrak{G}(\mathcal{U}; C, \partial C) = \mathcal{J}_C(\mathcal{U})C + \mathcal{J}_C^0(g) \partial_t C + \sum_{j=1}^3 \mathcal{J}_C^j(g) \partial_j C. \quad (\text{J.11})$$

Here $C = (C^\mu(g))$ is the actual harmonic defect, not an independent state variable. The matrices $\mathcal{A}^j$ are symmetric in the fixed positive component norm after realification; the other coefficient maps are smooth on bounded actual-jet sets. No derivative of $\mathcal{E}$, r^A or r_H and no time derivative of $(r_D, \bar{r}_D)$ occurs. If $C = 0$, the gauge-forcing term vanishes and the exactly harmonic system is recovered.

Proof. Use (J.9) for each metric component. First-order reduction in the adapted frame gives symmetric wave blocks for (p_u, q_u) . The additional source $-\nabla_{(\mu} C_{\nu)}$ is a linear combination of $\partial_\alpha C^\lambda$ with coefficients algebraic in g , and of C^λ with coefficients algebraic in $g, \partial g$. Multiplication by the lapse and frame matrices of the wave reduction therefore gives exactly (J.11). It changes no principal matrix and introduces no second derivative of C .

The Yang–Mills equation and Bianchi identity give the symmetric Maxwell blocks for (E, B) . The Higgs equation and covariant commutator identity give symmetric wave blocks for (Π, Q) . Temporal gauge gives $\partial_t A_i = F_{0i}$, an algebraic head equation in this state. Metric and Higgs head equations follow from their defining first jets. The Dirac head has Hermitian normal-frame principal matrices $-c_0 c_a$; realification makes them real symmetric. Transposing the dual row gives the transposed principal matrices, symmetric in the dual positive component norm, so it has the same energy estimate. Passing from the adapted frame to coordinates multiplies each wave, Maxwell or Dirac block by the common real lapse/frame coefficients and adds the scalar shift contribution $-\beta^j I$, preserving symmetry.

For the tangential spinor jets use Proposition J.2. The contracted curvature there is the physical Ricci tensor, so (J.5) eliminates it without assuming harmonic gauge and without differentiating stress. Substitute the algebraic normal jet (J.4) throughout. Since $e_a = e_a^j \partial_j$ for $a = 1, 2, 3$, its residual derivative is

$$\nabla_a r_D = e_a^j \partial_j r_D + \text{a connection coefficient times } r_D,$$

and similarly for the dual row. The normal covector component enters the tangential prolonged connection only algebraically. Hence no $\partial_t r_D$ is introduced. Frame commutators and spin-connection coefficients depend only on the actual first metric jets. The off-shell stress is algebraic in those jets and at most linear in the normal Dirac residual. Consequently all residual terms have the form displayed in (J.10), with no differentiated bosonic residual. This constructs the same symmetric principal blocks for exactly and approximately harmonic records. $\square$

Lemma J.5 (Derivative-counted complete forcing bound). *Fix $k \geq 4$ and an actual-jet H^k ball with the stated chart and foliation margins. Let $\mathcal{F}_{\text{err}}$ denote the sum of the residual and gauge-forcing terms on the right of (J.10). Then*

$$\begin{aligned} \|\mathcal{F}_{\text{err}}\|_{H_x^k} &\leq C_M (\|\mathcal{R}_B\|_{H_x^k} + \|\mathcal{R}_D\|_{H_x^{k+1}} \\ &\quad + \|C\|_{H_x^{k+1}} + \|\partial_t C\|_{H_x^k}). \end{aligned} \quad (\text{J.12})$$

Proof. The state determines $g, \partial g$, so every zeroth-order coefficient in (J.10) has a bounded H^k norm on this ball. Smooth composition and the H^k algebra estimate on the three-dimensional closed slices control products with the undifferentiated residuals. The matrices multiplying $\partial_j r_D$ and $\partial_\alpha C$ depend only on g ; these products require respectively $r_D \in H^{k+1}$, $C \in H^{k+1}$ and $\partial_t C \in H^k$. Lowering the index of C adds only products of C with first metric jets. The normalized Euler densities and physical rows are related by smooth invertible algebraic coefficient maps on this chart. Summing these estimates proves the bound. $\square$

Theorem J.6 (Coupled actual-jet stability and first-exit closure). *Let z_* be a regular exact Einstein–Standard-Model solution on $Q = [0, T] \times \mathbb{T}^3$ in the preceding gauges with $\mathcal{U}_* \in C_t H_x^{k+1}$, $k \geq 4$. Let $\hat{z}$ be a smooth actual field tuple on the same slab and foliated chart, in exact temporal internal gauge. Both spinor and dual-spinor blocks are included. Harmonic coordinates are required for z_* but not for $\hat{z}$. Define*

$$d = \|\hat{\mathcal{U}}(0) - \mathcal{U}_*(0)\|_{H^k} + \|\mathcal{R}_B(\hat{z})\|_{L_t^2 H_x^k} + \|\mathcal{R}_D(\hat{z})\|_{L_t^2 H_x^{k+1}} + \|C(\hat{g})\|_{L_t^2 H_x^{k+1}} + \|\partial_t C(\hat{g})\|_{L_t^2 H_x^k}.$$

There are constants $d_, C_* > 0$, depending only on the reference and fixed chart margins, such that $d \leq d_*$ implies*

$$\|\hat{\mathcal{U}} - \mathcal{U}_*\|_{C_t H_x^k} \leq C_* d, \quad (\text{J.13})$$

$$\|\text{Riem}(\hat{g}) - \text{Riem}(g_*)\|_{L_t^2 H_x^{k-1}} + \|T^{\text{SM}}(\hat{z}) - T^{\text{SM}}(z_*)\|_{L_t^2 H_x^k} \leq C_* d. \quad (\text{J.14})$$

No all-time H^k bound for $\hat{\mathcal{U}}$ is assumed. Existence of $\hat{z}$ on the slab is assumed.

Proof. Choose a radius $\rho > 0$ such that the H^k tube of radius ρ around the reference state stays inside the common nondegenerate foliated chart. This is possible by $H^k(\mathbb{T}^3) \hookrightarrow C^1$ and the fixed reference margins. On an interval on which $\|\hat{\mathcal{U}} - \mathcal{U}_*\|_{H^k} < \rho$, put $W = \hat{\mathcal{U}} - \mathcal{U}_*$ and subtract the two identities of Proposition J.4. The reference has zero physical and harmonic residuals. With the approximate-state principal matrices on the left,

$$\partial_t W + \mathcal{A}^j(\hat{g}) \partial_j W = \mathcal{F}(\hat{\mathcal{U}}) - \mathcal{F}(\mathcal{U}_*) + (\mathcal{A}^j(g_*) - \mathcal{A}^j(\hat{g})) \partial_j \mathcal{U}_* + \mathcal{F}_{\text{err}}.$$

The first term is locally Lipschitz in H^k . The second is bounded by $C\|W\|_{H^k}$ because the reference has one additional spatial derivative. For $|\alpha| \leq k$ the commutator estimate

$$\|[\partial^\alpha, a] \partial_j W\|_2 \leq C_k (\|\nabla a\|_\infty \|W\|_{H^k} + \|a\|_{H^k} \|\nabla W\|_\infty)$$

and $H^k \hookrightarrow W^{1,\infty}$ control the differentiated principal terms. Symmetry and integration by parts on the closed slice give a positive energy $E_k^{1/2} \simeq \|W\|_{H^k}$ with

$$\begin{aligned} \frac{d}{dt} E_k^{1/2} &\leq C E_k^{1/2} + C b_k(t), \\ b_k &= \|\mathcal{R}_B\|_{H^k} + \|\mathcal{R}_D\|_{H^{k+1}} + \|C\|_{H^{k+1}} + \|\partial_t C\|_{H^k}. \end{aligned} \quad (\text{J.15})$$

At a zero of the energy use $(E_k + \epsilon^2)^{1/2}$ and pass to the limit. The forcing estimate here is precisely [Lemma J.5](#); no exactly harmonic equation has been substituted for the approximate metric.

Integration and Gronwall give

$$\sup_{0 \leq t \leq T} \|W\|_{H^k} \leq C_T \left(\|W(0)\|_{H^k} + \int_0^T b_k(t) dt \right) \leq C_* d \quad (\text{J.16})$$

as long as the tube condition holds. Choose d_* so that the right side is less than $\rho/2$ and the initial state is inside this smaller tube. Continuity then excludes a first exit before T , proving [\(J.13\)](#). Existence of the smooth approximate record on the slab is used here; no uniform all-time Sobolev bound for it is assumed.

The equation bounds $\partial_t W$ in $L_t^2 H_x^{k-1}$ by the same state and forcing quantities. Recover ordinary first metric derivatives algebraically from (g, p, q) , and differentiate these identities once, using the metric first-jet rows for the time derivative. All second metric derivatives then differ by $O(d)$ in $L_t^2 H_x^{k-1}$. The coordinate curvature formula, linear in these second derivatives and quadratic in first derivatives, proves the curvature estimate. For the complete stress, normal-spinor elimination gives smooth algebraic maps with

$$T^{\text{SM}}(z) = \mathcal{T}_0(\mathcal{U}) + \mathcal{T}_1(\mathcal{U})(r_D, \bar{r}_D).$$

Consequently

$$\|T^{\text{SM}}(\hat{z}) - T^{\text{SM}}(z_*)\|_{L_t^2 H_x^k} \leq C \left(\sqrt{T} \|W\|_{C_t H_x^k} + \|(r_D, \bar{r}_D)\|_{L_t^2 H_x^k} \right) \leq C_* d.$$

This includes the off-shell normal-spinor contribution and proves [\(J.14\)](#). $\square$

J.2 Common-slab metric stability

We now describe a way of obtaining geometric convergence from a reduced evolution. This result is intentionally local in physical time. It assumes a common hyperbolic slab and a strongly controlled *complete* source; it does not manufacture that source from the algebraic duality. The energy argument belongs to the standard symmetric-hyperbolic framework [\[55, 54\]](#).

Let $s \geq 5$ be an integer and let smooth metrics g_h on $[0, T] \times \Sigma$ solve, in one background-harmonic coordinate/frame realization,

$$g_h^{\alpha\beta} \partial_\alpha \partial_\beta g_h = \mathcal{N}(g_h, \partial g_h) + \mathcal{S}_h. \quad (\text{J.17})$$

Fixed background-connection terms are included in $\mathcal{N}$. Its dependence on first derivatives is the usual smooth quadratic one on a compact nondegenerate metric chart. Here $\mathcal{S}_h$ is a tensor source, not the action $\mathcal{S}_h$.

Theorem J.7 (Common-slab metric stability). *Assume uniform hyperbolicity with a common time function, uniform inverse-metric bounds, and*

$$\sup_h \left(\|g_h\|_{L_t^\infty H_x^s} + \|\partial_t g_h\|_{L_t^\infty H_x^{s-1}} \right) < \infty. \quad (\text{J.18})$$

Suppose the initial data are Cauchy in $H^{s-1} \times H^{s-2}$ and the sources are Cauchy in $L_t^1 H_x^{s-2}$. Then

$$\begin{aligned} g_h &\longrightarrow g && \text{in } C_t H_x^{s-1}, \\ \partial_t g_h &\longrightarrow \partial_t g && \text{in } C_t H_x^{s-2}, \\ \text{Riem}(g_h) &\longrightarrow \text{Riem}(g) && \text{in } L_t^1 H_x^{s-3}. \end{aligned} \quad (\text{J.19})$$

In particular, compatible coframes chosen on one compact frame chart converge in the geometric topology of (B.7). A vanishing reduced-equation residual can be included in $\mathcal{S}_h$ with the same source norm.

Proof. Subtract the equations for h and j , and put $w = g_h - g_j$. Use the principal wave operator with coefficient g_h on the left. The forcing consists of $\mathcal{S}_h - \mathcal{S}_j$, the difference of the quadratic first-derivative terms, and

$$(g_j^{\alpha\beta} - g_h^{\alpha\beta}) \partial_\alpha \partial_\beta g_j.$$

At one derivative below the a priori bound, $H^{s-2}(\Sigma)$ is an algebra. Spatial second derivatives of g_j and mixed time-space derivatives are uniformly bounded in H^{s-2} . Solving (J.17) for $\partial_t^2 g_j$ gives an $L_t^1 H^{s-2}$ bound with size controlled by $1 + \|\mathcal{S}_j\|_{L_t^1 H^{s-2}}$. Consequently the forcing, apart from $\mathcal{S}_h - \mathcal{S}_j$, has H^{s-2} norm bounded by

$$a_{h,j}(t) (\|w\|_{H^{s-1}} + \|\partial_t w\|_{H^{s-2}}), \quad \sup_{h,j} \int_0^T a_{h,j}(t) dt < \infty.$$

The differentiated wave energy $E_{h,j}$ is uniformly equivalent to the square of the displayed difference norm. Its square root obeys the integrated estimate

$$E_{h,j}(t)^{1/2} \leq C \left(E_{h,j}(0)^{1/2} + \int_0^t \|\mathcal{S}_h - \mathcal{S}_j\|_{H^{s-2}} du \right) \exp \left(C \int_0^t (1 + a_{h,j}(u)) du \right). \quad (\text{J.20})$$

For completeness, commute spatial derivatives through the principal operator, use the differentiated normal derivative as energy multiplier, and integrate on the closed Σ . The uniformly spacelike slices and bounded lapse and shift give a positive wave energy; commutators are bounded by the coefficient norms in (J.18) and the Sobolev product estimate above. Apply the scalar differential inequality to $(E_{h,j} + \delta)^{1/2}$ and let $\delta \downarrow 0$ to obtain (J.20). This form uses the L^1 source norm directly; it does not replace it by an unjustified squared-source bound.

Cauchy initial and source data prove the first two limits. The reduced equation then gives convergence of $\partial_t^2 g_h$ in $L_t^1 H^{s-3}$. All other second derivatives converge in $C_t H^{s-3}$, and the inverse metrics and quadratic first-derivative terms converge by Sobolev product estimates. The coordinate formula for curvature gives the last limit. Smooth dependence of a locally fixed coframe on the metric gives the coframe assertion. $\square$

The loss of one derivative in the difference estimate avoids treating a quasilinear coefficient difference times a second derivative as if it had top-order Lipschitz regularity. Likewise, L^1 control of the source yields the stated L^1 -in-time curvature conclusion, not continuity in time of every curvature component. Stronger conclusions require stronger source and multiplier estimates.

J.3 The Ward residual forces the harmonic constraint

For a harmonic gauge covector C_μ define

$$\hat{G}_{\mu\nu} = G_{\mu\nu} - \nabla_{(\mu} C_{\nu)} + \frac{1}{2} g_{\mu\nu} \nabla^\alpha C_\alpha. \quad (\text{J.21})$$

Suppose a smooth reconstructed field satisfies

$$\hat{G}_h + \Lambda_h g_h = \kappa_h T_h + r_h, \quad \mathcal{W}_{h,\nu} = \kappa_h \nabla_h^\mu T_{h,\mu\nu} + \nabla_h^\mu r_{h,\mu\nu},$$

with constant parameters on spacetime. The source $\mathcal{W}_h$ includes every retained sector and the reduced-equation residual.

Proposition J.8 (Subsidiary equation and constraint stability). *With the curvature convention fixed by (J.21),*

$$\square_{g_h} C_{h,\nu} + \text{Ric}(g_h)_\nu{}^\mu C_{h,\mu} = -2\mathcal{W}_{h,\nu}. \quad (\text{J.22})$$

On a common closed spatial slab with uniform wave-energy coefficients and a uniform $L_t^1 L_x^\infty$ bound for the curvature potential,

$$\|(C_h, \nabla C_h)\|_{L_t^\infty L_x^2} \leq C_T \left(\|(C_h, \nabla C_h)(0)\|_{L_x^2} + \|\mathcal{W}_h\|_{L_t^1 L_x^2} \right). \quad (\text{J.23})$$

Vanishing initial constraint data and Ward residual therefore imply $C_h \rightarrow 0$ in the subsidiary energy norm.

Proof. Take the covariant divergence of the reduced Einstein equation. Contracted Bianchi removes the Einstein tensor and the constant cosmological term. The identity

$$2\nabla^\mu \nabla_{(\mu} C_{\nu)} - \nabla_\nu \nabla^\alpha C_\alpha = \square C_\nu + \text{Ric}_\nu{}^\mu C_\mu$$

gives (J.22). For the estimate, use the positive wave energy of a one-form relative to the common time foliation, including its L^2 term. Integration by parts has no spatial boundary flux. Coefficient derivatives and the curvature potential multiply the energy by an integrable function, and the forcing pairs with the time derivative. The square-root energy inequality and Gronwall give (J.23). $\square$

This proposition is not a consequence of the finite commutant identity. Exact classical gauge and relabeling invariance provide Noether identities, but their continuum use requires the whole source and the physical test realization. In an evolution-based construction, the subsidiary residual must be monitored along with the reduced equation. On the variational branch of Corollary B.8, by contrast, independent physical variations already impose the full Euler system. Conservation is asserted at the regularity at which the corresponding covariant distributional identity is defined, not by multiplying rough distributions formally.

A physical timelike boundary adds a boundary energy flux to (J.23). Maximal dissipation for the main system does not automatically preserve the subsidiary constraint, and dissipation alone does not identify a boundary domain of the same closed action. These are separate extensions; the present closed-spatial-cylinder result avoids, rather than solves, that boundary problem.

K Controlled comparison and generated finite families

These are conditional nonemptiness and calibration constructions. Smooth sampling verifies consistency on prescribed smooth fields; nondegenerate finite shadowing additionally requires a uniform linearized inverse. The generated evolution below instead starts from constrained initial data and retains its separate initial-solver hypotheses.

K.1 Smooth calibration and finite critical-point shadowing

Proposition K.1 (Smooth sampling is a consistency realization). *Let z_* be a smooth classical configuration on a compact slab with a fixed nondegenerate coframe chart. On the prescribed shape-regular reconstruction and local action of [Section D.3](#), smooth sampling gives $z_h \rightarrow z_*$ in the strong packet and*

$$c_h(K) \leq C_K h. \quad (\text{K.1})$$

If z_ solves the classical Euler equations, then on the same smooth family*

$$\varepsilon_h(K) \leq C'_K h. \quad (\text{K.2})$$

Under the nondegeneracy hypotheses of [Proposition K.2](#), these approximate critical points are shadowed by exact finite critical points.

Proof. Apply [Proposition D.8](#) for strong-packet convergence and $c_h(K) = O(h)$. If z_* is critical, [Proposition A.5](#) gives $\varepsilon_h(K) \leq c_h(K) + C_K d_K(z_h, z_*) = O(h)$. The final assertion is [Proposition K.2](#). $\square$

Proposition K.2 (Nondegenerate critical-point shadowing). *Let z_* be a smooth classical critical configuration and x_h^0 its sampled finite configuration in a gauge-fixed finite-dimensional physical slice X_h . Represent the finite Euler covector on that slice by a map $F_h : X_h \rightarrow X_h$ using a fixed positive slice inner product. Suppose, on a ball $B_r(x_h^0)$ and uniformly for small h ,*

$$(S1) \quad \|F_h(x_h^0)\| \leq C_0 h;$$

$$(S2) \quad DF_h(x_h^0) \text{ is invertible and } \|DF_h(x_h^0)^{-1}\| \leq M;$$

$$(S3) \quad \|DF_h(x) - DF_h(y)\| \leq L\|x - y\|;$$

$$(S4) \quad \text{the reconstruction is locally Lipschitz, } d_K(\mathcal{R}_h x, \mathcal{R}_h y) \leq C_R \|x - y\|.$$

Then, for sufficiently small h , there is a unique exact finite critical point $x_h^ \in B_r(x_h^0)$ in the Newton ball*

$$\|x_h^* - x_h^0\| \leq 2MC_0 h, \quad (\text{K.3})$$

and its reconstruction satisfies $d_K(\mathcal{R}_h x_h^, z_*) = O(h)$. Hence a nondegenerate smooth Einstein–Standard-Model solution generates a nearby exact finite stationary branch whenever the finite linearized physical Euler operator has a cutoff-uniform inverse and the second variation is uniformly controlled.*

Proof. Put $B_h = DF_h(x_h^0)^{-1}$ and define the frozen-Newton map

$$\mathcal{T}_h(x) = x - B_h F_h(x).$$

Let $\rho_h = 2MC_0 h$. Taylor's formula and (S3) give, for $x, y \in B_{\rho_h}(x_h^0)$,

$$\|\mathcal{T}_h(x) - \mathcal{T}_h(y)\| \leq ML\rho_h \|x - y\|.$$

For small h , $ML\rho_h < 1/2$. The same Taylor estimate gives

$$\|\mathcal{T}_h(x) - x_h^0\| \leq MC_0 h + \frac{1}{2} ML\rho_h^2 \leq \rho_h,$$

again for small h . Thus $\mathcal{T}_h$ is a contraction of the Newton ball into itself and has a unique fixed point there. Since B_h is invertible, the fixed-point equation is $F_h(x_h^*) = 0$. The reconstruction estimate follows from (S4), (K.3), and the $O(h)$ sampling convergence of [Proposition K.1](#). $\square$

If the positive slice norm used to represent F_h is uniformly equivalent to the physical dual norm induced by the test lifts, condition (S1) follows from (K.2). The substantive additional hypothesis is then the cutoff-uniform invertibility and local Lipschitz control of the linearized physical Euler map.

This establishes a nonempty classical comparison class once a smooth solution is specified, and it upgrades that class to exact finite stationarity on every locally nondegenerate branch satisfying Proposition K.2. It is not a proof that every renewal history selects this discretization or that every classical solution has a uniformly invertible physical linearization. The following explicit reduction supplies a backreacting comparison field without appealing to a general coupled existence theorem.

Proposition K.3 (A nonvacuum Einstein–Higgs comparison family). *Let $\Sigma = \mathbb{T}^3$, $\kappa, \lambda_H > 0$, and take*

$$g = -dt^2 + a(t)^2 \delta_{ij} dx^i dx^j, \quad H = \frac{\phi(t)}{\sqrt{2}} h_0, \quad A = 0, \quad \Psi = \bar{\Psi} = 0, \quad (\text{K.4})$$

where h_0 is a fixed unit Higgs vector and ϕ is real. Put

$$U(\phi) = \lambda_H (\phi^2/2 - v_H^2)^2.$$

For initial data $a_0 > 0$, ϕ_0 , π_0 , and $\mathcal{H}_0$ satisfying

$$3\mathcal{H}_0^2 = \Lambda + \kappa \left(\frac{1}{2}\pi_0^2 + U(\phi_0) \right), \quad (\text{K.5})$$

the system

$$\dot{a} = a\mathcal{H}, \quad \dot{\phi} = \pi, \quad \dot{\pi} = -3\mathcal{H}\pi - U'(\phi), \quad \dot{\mathcal{H}} = -\frac{\kappa}{2}\pi^2 \quad (\text{K.6})$$

has a smooth local solution with $a > 0$, and (K.4) solves the torsion-free Einstein–Standard-Model equations. Choosing $\pi_0 \neq 0$ gives a genuinely time-dependent matter stress and backreaction. Smooth sampling yields the finite residual conclusion of Proposition K.1.

Proof. The right side of (K.6) is smooth, so local ordinary differential equation existence and uniqueness apply. Since $a(t) = a_0 \exp(\int_0^t \mathcal{H})$, the scale factor stays positive on its existence interval. The Higgs kinetic term is $-\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$, giving energy density $\rho = \frac{1}{2}\pi^2 + U$ and pressure $p = \frac{1}{2}\pi^2 - U$. Its Euler equation is the third equation of (K.6).

The gauge current vanishes: H and $\dot{H}$ have the same fixed internal direction and a real relative coefficient, whereas all infinitesimal unitary generators are skew-adjoint. All fermionic equations are satisfied by zero spinors. Differentiating the Friedmann residual gives

$$\frac{d}{dt} \left(3\mathcal{H}^2 - \Lambda - \kappa \left(\frac{1}{2}\pi^2 + U \right) \right) = 6\mathcal{H} \left(-\frac{\kappa}{2}\pi^2 \right) - \kappa \left(\pi(-3\mathcal{H}\pi - U') + U'\pi \right) = 0.$$

Thus the constraint persists. Together with $\dot{\mathcal{H}} = -(\kappa/2)(\rho + p)$ it gives both independent Einstein equations for the flat homogeneous metric. If $\pi_0 \neq 0$, then $\dot{\mathcal{H}}(0) < 0$, so the example is not merely a vacuum metric relabeled by an internal carrier. On a smaller compact existence slab all required smoothness and noncollapse bounds hold, and Proposition K.1 applies. $\square$

Corollary K.4 (A nonempty calibration class for the local action). *Let z_* be a smooth solution on a compact slab, with compatible fixed gauges and a smooth periodic extension to the buffered comparison box that remains in the chosen field-value chart. Nodal sampling followed by trigonometric reconstruction yields records for S_h^{loc} with $\sigma_h = O(h)$ and with the budgets of Theorem 5.1 tending to zero for a suitable $K_h \rightarrow \infty$. Thus the quantitative hypotheses are nonempty. A backreacting Einstein–Higgs example is supplied by Proposition K.3, after a regular gauge change on a smaller slab.*

Proof. Smooth Fourier decay and the aliasing formula give uniform H^q bounds and convergence of each fixed number of reconstructed derivatives. Fixed-scale consistency yields $\sigma_h = O(h)$ because the reference Euler row vanishes; derivative convergence controls the initial and harmonic mismatches. Choose $K_h \asymp h^{-\beta}$ with $0 < \beta < 1/(k+4)$. The bound $\tau_{h,j} \leq C_q K_h^{2-q}$ from (M.5), for sufficiently large $q > k+5$, makes both hK_h^{k+4} and K_h^{k+5-q} vanish, including the logarithmic factors in the forcing budget. This constructs a calibration family, not a microscopic selector. $\square$

K.2 Generated finite dynamics from constrained data

This subsection gives an independent nonemptiness construction for the controlled finite class. It is local in time and deliberately high-regularity; sharp thresholds are not claimed. Its role is not to replace the native source theorem of Theorem L.6, but to show that the quantitative hypotheses admit a fully finite realization generated from constrained finite initial data. The construction uses the positive symmetric actual-jet system of Section J.1 and the standard conformal strategy for the Einstein constraints [56, 57].

Assumption K.5 (Compatible elliptic and conformal initial solver). The seed family lies on fixed gauge and conformal slices for which the Gauss and vector-momentum sources are orthogonal to the corresponding elliptic kernels. Bounded inverses on these compatible subspaces are supplied, with cutoff-uniform estimates at every Sobolev order used below. The chosen conformal weights and matter data give the scalar equation (K.18), with the source bound (K.17). The finite solvers preserve compatibility and have the approximation estimates in (K.7). A constraint-compatible analytic core with the same gauge completion is available for the physical-identification argument. On this core the constrained physical Cauchy problem admits locally unique analytic solutions with continuation controlled by the declared Sobolev bounds and field-chart margins.

The following construction is conditional on this initial-solver and physical-Cauchy interface. The principal variational closure and the deterministic a-posteriori stability theorems do not use this supplementary existence input. It does not assert invertibility of an unrestricted Gauss or momentum operator on a torus. In particular, a real standing-spinor profile alone is not a certificate of vanishing gauge charge or of compatibility with all elliptic zero modes. The explicit homogeneous zero-spinor family in Proposition K.3 supplies a nonempty comparison class independently of any additional charged-spinor seed construction.

Theorem K.6 (Fully finite approximation branch). *Fix integers $s \geq 5$, $p \geq 1$, and put $q = s + p + 8$. Under Assumption K.5, for the constraint-compatible initial class below, each spatial cutoff N has finite initial data $U_{0,N}$ satisfying*

$$\sup_N \|U_{0,N}\|_{H^q} \leq M_q, \quad \|U_{0,N} - U_0\|_{H^{s+4}} \leq CN^{-p}. \quad (\text{K.7})$$

Let

$$G_N(U) = P_N\{F(U) - A^i(U)\partial_i U\}, \quad U^{j+1} = U^j + \tau G_N\left(\frac{U^{j+1} + U^j}{2}\right), \quad U^0 = U_{0,N}. \quad (\text{K.8})$$

If

$$\tau(N+1) \leq c_*, \quad \tau \leq \tau_*, \quad (\text{K.9})$$

then the fully finite recursion exists on a common interval $[0, T]$ and

$$\max_j \|U^j\|_{H^q} \leq R_q. \quad (\text{K.10})$$

Its cubic-Hermite physical reconstruction $z_{N,\tau}$ satisfies

$$\|z_{N,\tau} - z\|_{C_t H^{s+2}} + \|\partial_t z_{N,\tau} - \partial_t z\|_{C_t H^{s+1}} \leq C(N^{-p} + \tau^2), \quad (\text{K.11})$$

$$\|\text{Riem}(g_{N,\tau}) - \text{Riem}(g)\|_{L_t^\infty H^s} + \|\mathcal{R}_B(z_{N,\tau})\|_{L_t^\infty H^s} \leq C(N^{-p} + \tau), \quad (\text{K.12})$$

$$\|\mathcal{R}_D(z_{N,\tau})\|_{C_t H^{s+1}} \leq C(N^{-p} + \tau^2). \quad (\text{K.13})$$

Moreover the finite comparison action on the same generated record obeys

$$|DS_{N,\tau}^{\text{cmp}}[V]| \leq C(N^{-p} + \tau)\|V\|_{0,\tau}. \quad (\text{K.14})$$

The continuum solution z is used only for the a-posteriori estimate; the finite initial solver and recursion do not sample its future values.

K.2.1 A constrained initial class and its finite approximation

Work on $\mathbb{T}^3$ near a smooth expanding constant background. Fix a constant Higgs position H_* , a nonzero radial Higgs normal momentum p , and a spatially homogeneous gauge background A^* , so that the scalar energy coefficients below are constant. Put

$$A_*^0 = \kappa p^2, \quad G_*^0 = \kappa |D_{A^*} H_*|^2, \quad B_*^0 = \kappa |B_{A^*}|^2, \quad (\text{K.15})$$

and choose the constant mean curvature so that the unperturbed Hamiltonian constraint holds. In the conformal scalar equation the derivative at the unperturbed solution $w = 1$ is

$$L_* = -8\Delta + \mu_*, \quad \mu_* = 12A_*^0 + 4G_*^0 + 8B_*^0 > 0. \quad (\text{K.16})$$

Thus $L_* : H^{r+2} \rightarrow H^r$ is an isomorphism including the zero Fourier mode.

The Gauss and vector-momentum equations are solved first by the compatible elliptic inverses of [Assumption K.5](#). For a small transverse-traceless tensor seed σ and a small standing spinor seed $\chi = \varepsilon f \chi_*$ with real f , assume the complete momentum and gauge-charge compatibility conditions of [Assumption K.5](#); the resulting source coefficients in the scalar conformal equation obey

$$\|\delta A\|_{H^r} + \|\delta B\|_{H^r} + \|\Upsilon\|_{H^r} \leq C(\|\sigma\|_{H^{r+1}}^2 + \|\chi\|_{H^{r+1}}^2). \quad (\text{K.17})$$

The Hamiltonian equation can be written

$$-8\Delta w - A w^{-7} - G_*^0 w - B w^{-3} - \Upsilon w^{-1} + C_*^0 w^5 = 0, \quad (\text{K.18})$$

where C_*^0 is fixed by the background constraint.

Proposition K.7 (Constructive constrained initial data). *Under [Assumption K.5](#), for every integer $r \geq 2$, sufficiently small compatible σ, χ as above determine a unique solution w of (K.18) in a fixed H^{r+2} neighborhood of one, with*

$$\|w - 1\|_{H^{r+2}} \leq C(\|\sigma\|_{H^{r+1}}^2 + \|\chi\|_{H^{r+1}}^2), \quad \frac{1}{2} < w < \frac{3}{2}. \quad (\text{K.19})$$

Together with the Gauss and vector solutions this gives exact Einstein, gauge and matter initial constraints. Nonzero-spinor data are included only when their complete sources satisfy the stated compatibility and conformal hypotheses; their existence is not inferred from the standing-wave ansatz alone. Nonconstant admissible source coefficients may produce inhomogeneous metric response.

If the seeds have p additional derivatives, Fourier projection of the seeds followed by the same finite elliptic construction gives finite initial tuples $U_{0,N}$ satisfying (K.7); the Gauss and vector constraints have the accuracy supplied by that compatible finite solver, and the scalar Hamiltonian residual is $O(N^{-p})$ in the indicated lower Sobolev norm. No spacetime solution is used in this finite initial solver.

Proof. Write $u = w - 1$ and subtract the linear term L_*u from (K.18). The remaining pure-background nonlinearities are quadratic in u , while the perturbed source coefficients satisfy (K.17). Hence

$$u = L_*^{-1}[\delta A(1+u)^{-7} + \delta B(1+u)^{-3} + Y(1+u)^{-1} - R_0(u)].$$

On a small H^{r+2} ball, Sobolev multiplication and the convergent power series for the negative integer powers give

$$\|\mathcal{T}(0)\|_{H^{r+2}} \leq C\eta, \quad \|\mathcal{T}(u) - \mathcal{T}(v)\|_{H^{r+2}} \leq C(\eta + R)\|u - v\|_{H^{r+2}},$$

where $\eta = \|\sigma\|^2 + \|\chi\|^2$. Banach contraction proves (K.19). The Gauss and vector equations use bounded elliptic inverses of their compatible sources, so all initial constraints hold after substitution into the conformal identities. For the finite construction, project the finite seeds, solve those elliptic equations exactly, and apply the projected contraction to the scalar equation. The contraction constant remains uniform because P_N is a contraction in Sobolev space and commutes with L_* . Subtracting the exact and projected fixed-point equations and using the ordinary Fourier tail estimate gives $O(N^{-p})$. The same argument applies to the derived harmonic initial completion and actual first jets. $\square$

Lemma K.8 (Physical identification of the symmetric extension). *Let U_0 belong to the constrained Sobolev class of Proposition K.7, with sufficiently many finite derivatives for the estimates below. The independent symmetric system*

$$U_t + A^i(U)\partial_i U = F(U) \tag{K.20}$$

has a common local Sobolev solution, and on a possibly smaller interval its head fields satisfy the complete torsion-free Einstein–Standard-Model equations and the harmonic/temporal-gauge constraints. Thus the physical solution z used in Theorem K.6 is identified from the constrained initial data; it is not an external trajectory supplied to the finite recursion.

Proof. Approximate the finite-Sobolev seeds by Fourier-polynomial seeds and apply the exact, unprojected contraction of Proposition K.7 to each approximation. This gives a sequence of analytic initial tuples satisfying all physical constraints exactly and converging to U_0 in the Sobolev topology. For each tuple in the declared constraint-compatible analytic core, use its physical analytic germ in harmonic coordinates and temporal internal gauge. Its actual first jets satisfy (K.20) by Proposition J.4. Uniqueness for the symmetric system therefore identifies the independent analytic solution with those actual jets on their common local interval.

The symmetric energy estimate for (K.20) depends only on a bounded Sobolev ball and the fixed nondegenerate chart, not on the analytic radius. Together with the physical-Cauchy continuation input in Assumption K.5, this gives a common smaller interval for the physical analytic approximants. Their actual-jet states converge in one lower Sobolev order to the symmetric-system solution with initial datum U_0 . The defining-jet identities, Einstein and matter residuals, Gauss constraint and harmonic constraint vanish on every analytic approximant. The convergence of fields, first derivatives and the equation then passes these identities distributionally to the Sobolev limit; the algebraic reconstruction identifies the actual connection, curvature and complete stress. This proves physical identification. It is an analytic-core argument for the constrained class, not a claim that arbitrary auxiliary initial jets are physical. $\square$

K.2.2 Finite symmetric evolution and midpoint construction

Let the coupled actual-jet system of [Proposition J.4](#) be written on its fixed chart as

$$U_t + A^i(U)\partial_i U = F(U), \quad G(U) = F(U) - A^i(U)\partial_i U. \quad (\text{K.21})$$

For the finite initial data above define the semidiscrete system

$$\dot{U}_N = P_N G(U_N), \quad U_N(0) = U_{0,N}. \quad (\text{K.22})$$

The symmetry of the principal matrices implies that the H^q energy estimate for [\(K.22\)](#) has constants independent of N ; the orthogonal projection disappears from the energy pairing. A first-exit argument therefore gives a common existence interval and a uniform H^q bound. On that interval,

$$\max_{0 \leq a \leq 2} \|\partial_t^a (U_N - U)\|_{C_t H^{s+4-a}} \leq C N^{-p}, \quad (\text{K.23})$$

where U is the physical Sobolev solution with initial datum U_0 . This comparison identifies the limit only after the finite ODE has been constructed.

For the fully finite evolution use [\(K.8\)](#). On $\text{Ran } P_N$ the inverse inequality gives

$$\|G_N(V) - G_N(W)\|_{H^q} \leq C_R(N+1)\|V - W\|_{H^q}.$$

Thus [\(K.9\)](#) makes the midpoint fixed-point map a contraction. The exact discrete symmetric energy identity then removes the apparent N -dependence from the propagated bound and discrete Gronwall gives [\(K.10\)](#). If two midpoint trajectories have a normalized forcing difference r_j , the same energy calculation at any $3 \leq m \leq q-1$ gives

$$\max_{j, \tau \leq T} \|U^j - V^j\|_{H^m} \leq C_T \left(\|U^0 - V^0\|_{H^m} + \tau \sum_j \|r_j\|_{H^m} \right), \quad (\text{K.24})$$

with C_T independent of N, τ . Applying this estimate to the local truncation residual of the exact semidiscrete solution yields

$$\max_j \|U^j - U_N(t_j)\|_{H^{s+4}} \leq C\tau^2, \quad \max_j \|U^j - U(t_j)\|_{H^{s+4}} \leq C(\tau^2 + N^{-p}). \quad (\text{K.25})$$

K.2.3 Physical Hermite readout and equation residuals

On each interval $[t_j, t_{j+1}]$, reconstruct the physical head by the cubic Hermite polynomial having endpoint values equal to the generated physical heads and endpoint slopes equal to the corresponding components of $G_N(U^j)$ and $G_N(U^{j+1})$. The resulting $z_{N,\tau}$ is C^1 in time. The Hermite basis, [\(K.25\)](#), and the uniform higher time derivatives of the semidiscrete system give

$$\begin{aligned} \|z_{N,\tau} - z\|_{C_t H^{s+2}} + \|\partial_t z_{N,\tau} - \partial_t z\|_{C_t H^{s+1}} &\leq C(N^{-p} + \tau^2), \\ \|\partial_t^2 z_{N,\tau} - \partial_t^2 z\|_{L_t^\infty H^s} &\leq C(N^{-p} + \tau). \end{aligned} \quad (\text{K.26})$$

The Riemann tensor is linear in second metric derivatives plus smooth quadratic first-derivative terms, giving [\(K.12\)](#). The wave-type Yang–Mills and Higgs rows have the same derivative count. The Dirac row is first order and therefore inherits the sharper estimate [\(K.13\)](#). The harmonic, Gauss and normal Einstein constraints use only the controlled first jets and spatial derivatives and are $O(N^{-p} + \tau^2)$ in the corresponding lower norms.

Finally let $S^{(1)}$ be the complete first-derivative torsion-free Einstein–Standard-Model action representative with fixed-collar boundary completion, and let $\mathcal{R}$ be the Hermite reconstruction from independent nodal values and slopes. Define

$$S_{N,\tau}^{\text{cmp}} = S^{(1)} \circ \mathcal{R}, \quad \|V\|_{0,\tau}^2 = \tau \sum_j \left(\|v_j\|_2^2 + \tau^2 \|\dot{v}_j\|_2^2 \right). \quad (\text{K.27})$$

At the generated record, integration by parts expresses $DS_{N,\tau}^{\text{cmp}}$ through the physical rows just estimated. Since $\|D\mathcal{R}[V]\|_{L^2} \leq C\|V\|_{0,\tau}$ and the temporal gauge and all metric entries remain independent variations before evaluation, this gives (K.14). Keeping the first-derivative variational representative and using (K.11) gives the sharper $O(N^{-p} + \tau^2)$ continuum first-variation comparison. In particular the complete symmetric spinor stress is included and converges strongly.

L Finite source budgets and same-record closure

The finite-action theorem above is deterministic and applies to any complete record satisfying the displayed physical budgets. We now give a finite source-level entrance that produces those budgets on one *actually observed* record. The construction does not replace a record by its expectation, by a projected state, or by a trajectory of an inserted evolution equation.

At spacing h , let $X_0, \dots, X_J$ be the records of an actual finite accepted process. The state may retain the entire finite past, so no Markov assumption is required. Let $K_{h,j}^{\text{leg}}(x, y)$ denote the transition probabilities restricted by a predeclared legal-edge mask for the raw-action chart and action identification; an attempted illegal transition is counted as failure and the diagnostic stopped process retains its last legal action value. Write μ_j for the resulting unconditioned surviving mass and $p_h^{\text{exit}} = 1 - \sum_x \mu_j(x)$. Every legal state has a complete physical record $u_h(x)$ and define

$$g_h(x) = \|E_h^{\text{raw}}(u_h(x))\|_{0,h}, \quad W_h(x) = (i_{h,k}(x) + \gamma_{h,k}(x))^2, \quad (\text{L.1})$$

together with a nonnegative native energy readout $V_h(x)$. Put

$$M_{h,J} = \frac{1}{J} \sum_{j < J,x} \mu_j(x) V_h(x), \quad A_{h,J} = \frac{1}{J} \sum_{j < J,x} \mu_j(x) W_h(x). \quad (\text{L.2})$$

Let $a_h(x) = S_h^{\text{loc}}(u_h(x))$ and suppose $A_h^- \leq a_h(x) \leq A_h^+$ on the legal chart, with action span $D_h^A = A_h^+ - A_h^-$. The stopped conditional action increment is

$$\mathfrak{d}_{h,j}(x) = \sum_y K_{h,j}^{\text{leg}}(x, y) [a_h(y) - a_h(x)]. \quad (\text{L.3})$$

Theorem L.1 (Same-record selection from the actual signed action drift). *Assume there are $\delta_h > 0$, $\kappa_A > 0$ independent of h , and nonnegative finite remainders $r_{h,j}$ such that every active legal row satisfies*

$$\mathfrak{d}_{h,j}(x) \leq -\kappa_A \delta_h g_h(x)^2 + \delta_h r_{h,j}(x). \quad (\text{L.4})$$

Set

$$\mathcal{R}_{h,J} = \frac{1}{J} \sum_{j < J,x} \mu_j(x) r_{h,j}(x). \quad (\text{L.5})$$

Then

$$\frac{1}{J} \sum_{j < J,x} \mu_j(x) g_h(x)^2 \leq \frac{D_h^A}{\kappa_A \delta_h J} + \frac{\mathcal{R}_{h,J}}{\kappa_A}. \quad (\text{L.6})$$

For thresholds $s_h, R_h, \alpha_h > 0$, select on every surviving prefix an observed index minimizing

$$\frac{g_h(X_j)^2}{s_h^2} + \frac{V_h(X_j)}{R_h^2} + \frac{W_h(X_j)}{\alpha_h^2}. \quad (\text{L.7})$$

The probability of failing to output one observed record with

$$g_h \leq s_h, \quad V_h \leq R_h^2, \quad i_{h,k} + \gamma_{h,k} \leq \alpha_h \quad (\text{L.8})$$

is at most

$$\boxed{p_h^{\text{exit}} + \frac{D_h^A}{\kappa_A \delta_h J s_h^2} + \frac{\mathcal{R}_{h,J}}{\kappa_A s_h^2} + \frac{M_{h,J}}{R_h^2} + \frac{A_{h,J}}{\alpha_h^2}}. \quad (\text{L.9})$$

No invariant distribution, mixing estimate, independence between records, or uniform high-order bound on every member of the prefix is required.

Proof. The stopped-action telescope and the deterministic score argument are proved in [Section M.1](#). The key point is that killing is recorded as failure rather than being credited as a decrease of the signed action. $\square$

The native energy must control the *physical reader actually entering* (5.1). This relation has an exact finite-dimensional criterion.

Proposition L.2 (Exact native-energy to physical-tail certificate). *Let $\mathbf{H}_h^{\text{src}}$ be a finite real source-coordinate Hilbert space, let $\mathbf{Q}_h \succeq 0$ be a positive native energy form, and let $\xi_h(x) \in \mathbf{H}_h^{\text{src}}$ be the source coordinate of a legal record. Suppose the physical reader has the affine form*

$$u_h(x) = u_h^{\text{soft}} + \mathbf{R}_h \xi_h(x). \quad (\text{L.10})$$

Let $\mathbf{T}_{h,K,m}$ map a source vector to the high-frequency Fourier coefficients of $\mathbf{R}_h \xi$, with the weights of (5.1). A finite constant $c_{h,K,m}$ such that

$$\|\mathbf{T}_{h,K,m} \xi\|_{\ell^1(w)} \leq c_{h,K,m} \langle \xi, \mathbf{Q}_h \xi \rangle^{1/2} \quad (\xi \in \mathbf{H}_h^{\text{src}}) \quad (\text{L.11})$$

exists if and only if

$$\mathbf{T}_{h,K,m} \ker \mathbf{Q}_h = 0. \quad (\text{L.12})$$

When this holds, the optimal constant in (L.11) is

$$c_{h,K,m} = \|\mathbf{T}_{h,K,m} \mathbf{Q}_h^{\dagger/2}\|_{2 \rightarrow \ell^1(w)}. \quad (\text{L.13})$$

In particular every legal record obeys the affine estimate

$$\tau_{h,m}(K; u_h(x)) \leq \tau_{h,m}(K; u_h^{\text{soft}}) + c_{h,K,m} \langle \xi_h(x), \mathbf{Q}_h \xi_h(x) \rangle^{1/2}. \quad (\text{L.14})$$

The equivalence concerns the linear estimate on the entire source space; it is not inferred from testing only a finite list of legal states. Thus the leakage assumption is a finite, directly auditable incidence condition between the native energy and the declared physical reader; finite-dimensional norm equivalence alone is not used.

Proof. See [Section M.1.2](#). $\square$

Corollary L.3 (Sobolev positive-form reader criterion). *Let L_h be the diagonal multiplier $(1+|\ell|_2^2)^{1/2}$ on the all-mode Fourier reader. If, on the retained source coordinates,*

$$R_h^* L_h^{2q} R_h \preceq C_R Q_h, \quad \|u_h^{\text{soft}}\|_{H^q} \leq C_R, \quad (\text{L.15})$$

then for every $0 \leq j \leq m < q - 2$,

$$\|u_h(x)\|_{H^q} \leq C(1 + \sqrt{V_h(x)}), \quad \tau_{h,j}(K; u_h(x)) \leq CK^{2-q}(1 + \sqrt{V_h(x)}), \quad (\text{L.16})$$

whenever $\langle \xi_h, Q_h \xi_h \rangle \leq V_h$. The power K^{2-q} is sharp for this four-dimensional Sobolev-to-weighted-tail route; in particular the composition with the k th-order forcing estimate requires $q > k + 5$ if no additional cancellation is used.

Proof. The positive-form inequality gives the H^q estimate. The sharp four-dimensional weighted Fourier bound is proved in [Section M.1.2](#); inserting $m = k + 2$ into the forcing budget yields the stated threshold. $\square$

Corollary L.4 (Same-record action, reader, and initial-constraint selection). *Assume the signed action-drift hypotheses of [Theorem L.1](#). In addition, let every legal record carry a nonnegative composite initial constraint residual*

$$c_h(x) = |\Phi_h(x, 0)|, \quad (\text{L.17})$$

where [Definition 5.3](#) holds for the normalized actual initial record. Thus Φ_h is the retained exact continuum constraint map, and its complementary constraints vanish throughout the correction family. A certified upper bound such as $|\tilde{\Phi}_h(x, 0)| + \zeta_{0,h}(x)$ from [Corollary I.2](#) may replace $c_h(x)$ in both the score and its occupation bound. Put

$$C_{h,J} = \frac{1}{J} \sum_{j < J, x} \mu_j(x) c_h(x)^2. \quad (\text{L.18})$$

For thresholds $s_h, R, \beta_h > 0$, select on each surviving prefix an observed index minimizing

$$\frac{g_h(X_j)^2}{s_h^2} + \frac{V_h(X_j)}{R^2} + \frac{c_h(X_j)^2}{\beta_h^2}. \quad (\text{L.19})$$

Then the probability of failing to output one actual unfiltered record with

$$g_h \leq s_h, \quad V_h \leq R^2, \quad c_h \leq \beta_h \quad (\text{L.20})$$

is at most

$$\boxed{p_h^{\text{exit}} + \frac{D_h^A}{\kappa_A \delta_h J s_h^2} + \frac{\mathcal{R}_{h,J}}{\kappa_A s_h^2} + \frac{M_{h,J}}{R^2} + \frac{C_{h,J}}{\beta_h^2}}. \quad (\text{L.21})$$

No independence, stationarity, or conditioning on successful selection is used.

Proof. Equation (L.6) bounds the occupation of g_h^2 . On survival, the minimum of the nonnegative score (L.19) is at most its arithmetic mean along the observed prefix. If that mean is at most one, each of the three selected terms is at most one. Markov's inequality bounds failure on survival by the expectation of the mean score, and adding the original exit event gives (L.21). $\square$

Theorem L.5 (Same-source selected exact physical shadow). Assume [Assumption 5.2](#), the hypotheses of [Corollary L.4](#), and the reader criterion [Corollary L.3](#) for some indices satisfying (5.12). Require every legal record used by the selector to lie in the common buffered field-value and inverse-metric chart and to admit the regular harmonic/temporal normalization of [Theorem 5.4](#); violations are included in p_h^{exit} . Assume also the exact family and basepoint identification of [Definition 5.3](#), the uniform initial-constraint margins (I.1)–(I.2) and the reconstruction Lipschitz constant B_* . Require the corrected data to remain in $C_k^{\text{reg}} \cap \mathcal{O}_k$ with the certified initial regularity; any failure of this condition is included in the legal domain exit event.

On the success event of [Corollary L.4](#), define

$$\varepsilon_h = C_R(s_h + h), \quad \eta_{h,k} = C_R \left(\varepsilon_h^{1-k/s} + \varepsilon_h^{1-(k+1)/s} \right), \quad (\text{L.22})$$

and

$$\bar{d}_h = 2B_*\gamma_*^{-1}\beta_h + C_R\sqrt{T}\eta_{h,k}. \quad (\text{L.23})$$

If $\bar{d}_h$ is below the uniform Cauchy-tube stability radius, then the selected actually observed record admits the unique exact physical Einstein–Standard-Model solution $z_{*,h}$ with the corrected initial datum from [Proposition I.1](#), on the common slab, and

$$\begin{aligned} & \|\mathcal{U}(\hat{z}_{h,j_*}) - \mathcal{U}(z_{*,h})\|_{C_t H_x^k} + \|\text{Riem}(\hat{g}_{h,j_*}) - \text{Riem}(g_{*,h})\|_{L_t^2 H_x^{k-1}} \\ & + \|T^{\text{SM}}(\hat{z}_{h,j_*}) - T^{\text{SM}}(z_{*,h})\|_{L_t^2 H_x^k} \leq C_R \bar{d}_h. \end{aligned} \quad (\text{L.24})$$

The probability that the selection stage fails before this deterministic conclusion is bounded by (L.21). Thus the same native-law record supplies the finite Euler row, the regularity reserve, and the near-compatible initial datum from which its exact physical comparison future is constructed.

Proof. The selected record satisfies (L.20). [Corollary L.3](#) and [Proposition I.4](#) give the forcing budget, and [Proposition I.1](#) gives the exact compatible initial datum. Apply [Theorem 5.4](#); the probability bound is (L.21). $\square$

Theorem L.6 (Same-source selected finite-action Einstein–Standard-Model closure). Let $k \geq 4$ and fix the same regular reference solution as in [Theorem 5.1](#). Assume the hypotheses of [Theorem L.1](#). Every legal record used here is required to exist on the common slab, to be in temporal internal gauge, and to satisfy the buffered field-value, inverse-metric and reconstruction-chart conditions of [Theorem H.2](#). The legal-domain failure probability includes violations of these conditions. Suppose the same legal records and the same native energy satisfy, for $j = 2, m = k + 2$,

$$\tau_{h,j}(K_h; u_h(x)) \leq t_{h,j} + c_{h,j} \sqrt{V_h(x)}, \quad (\text{L.25})$$

with certified finite reader constants. Put $T_{h,j} = t_{h,j} + c_{h,j}R_h$ and define

$$\begin{aligned} \bar{\varepsilon}_{c,h} &= s_h + hK_h^3 + K_h^2 T_{h,2}, \\ \bar{L}_{h,k} &= 1 + \log \left(2 + \frac{C_k K_h^{k+5}}{\bar{\varepsilon}_{c,h}} \right), \\ \bar{F}_{h,k} &= \bar{\varepsilon}_{c,h} K_h^{k+1} \bar{L}_{h,k}^{k+1} + K_h^{k+2} T_{h,m}, \quad \bar{D}_{h,k} = \alpha_h + \bar{F}_{h,k}. \end{aligned} \quad (\text{L.26})$$

Assume $K_h \rightarrow \infty$, $hK_h \rightarrow 0$, $T_{h,m} \rightarrow 0$, $\bar{\varepsilon}_{c,h} \rightarrow 0$ and $\bar{D}_{h,k} \rightarrow 0$. Outside the event in (L.9), the selected actually observed record satisfies

$$\begin{aligned} & \|\mathcal{U}_{h,j_*} - \mathcal{U}_*\|_{C_t H_x^k} + \|\text{Riem}(g_{h,j_*}) - \text{Riem}(g_*)\|_{L_t^2 H_x^{k-1}} \\ & + \|T^{\text{SM}}(z_{h,j_*}) - T^{\text{SM}}(z_*)\|_{L_t^2 H_x^k} \leq C \bar{D}_{h,k}. \end{aligned} \quad (\text{L.27})$$

Its zero-order physical residual is at most $C(s_h + hK_h^3)$ and its literal Cartan discrepancy is at most ChK_h^3 . If the failure bounds and $\bar{D}_{h_n,k}$ are summable along a common cofinal chain, then under any common coupling the selected successful records eventually have summable adjacent state, curvature and stress differences almost surely. No independence between cutoffs is required.

Proof. Selection gives $\sigma_h \leq s_h$, initial/gauge mismatch at most α_h , and $V_h \leq R_h^2$ on the same record. The reader certificate then gives $\tau_{h,j} \leq T_{h,j}$. Apply [Theorems 5.1](#) and [H.2](#) with these upper budgets. The first Borel–Cantelli lemma and comparison through the common reference prove eventual success and summable adjacent differences. $\square$

Corollary L.7 (An explicit polynomial selected-source regime). *Under the signed-drift, common-slab and legal-chart hypotheses of [Theorem L.6](#), fix $k \geq 4$, put $m = k + 2$, and suppose [Corollary L.3](#) holds for some $q > k + 5$. Assume, for positive $a, b, c, \epsilon, d, d_A \geq 0$,*

$$\begin{aligned} \delta_h &\geq c_\delta h^d, & D_h^A &\leq Ch^{-d_A}, \\ J_h &\geq c_J h^{-(d+d_A+2b+2\epsilon)}, & \mathcal{R}_{h,J_h} &\leq Ch^{2b+2\epsilon}, \\ M_{h,J_h} &\leq C, & A_{h,J_h} &\leq Ch^{2c+2\epsilon}, \\ p_h^{\text{exit}} &\leq Ch^{2\epsilon}. \end{aligned} \tag{L.28}$$

Choose

$$s_h = h^b, \quad R_h = h^{-a}, \quad \alpha_h = h^c, \quad K_h \asymp h^{-\beta}, \tag{L.29}$$

where

$$0 < \beta < \min \left\{ \frac{b}{k+1}, \frac{1}{k+4} \right\}, \quad 0 < a < \beta(q - k - 5). \tag{L.30}$$

Then the selection failure probability is $O(h^{2a} + h^{2\epsilon})$ and, on success,

$$\text{left side of (L.27)} \leq Ch^\rho (1 + |\log h|)^{k+1}, \quad \rho = \min\{c, b - \beta(k+1), 1 - \beta(k+4), \beta(q - k - 5) - a\} > 0. \tag{L.31}$$

Along $h_n \asymp 2^{-n}$ the successful selected branch is eventually almost sure and has summable adjacent differences whenever the displayed probability bound is summable.

Proof. See [Section M.1.4](#). $\square$

Corollary L.8 (Weak-sector example of a reader certificate). *On a nonvanishing Higgs chart of the weak $SU(2)$ sector, a cutoff-uniform bound on the finite covariant-jet energy of [Section M.2](#) yields, after algebraic Higgs normalization and a causal finite temporal gauge, a bounded all-mode H^q reader and hence the tail estimate*

$$\tau_{h,m}(K) \leq CK^{2-q}, \quad q > m + 2, \tag{L.32}$$

without requiring high ordinary derivatives of the original site frame or small original links. Exact finite gauge invariance preserves the norm of the full physical action covector under this normalization, and logarithmic link coordinates are uniformly equivalent to the physical link-tangent norm on the normalized chart. This result supplies one source-auditable realization of the reader hypothesis for the weak sector; color, determinant-line, coframe and other independent reader components require their own corresponding bounds.

Proof. The finite covariant-jet, temporal-normalization and full-covector identities are proved in [Section M.2](#). $\square$

Alternative same-action prefix balance. The direct signed action-drift entrance above is not the only way to obtain a small native Euler row. A strongly convex operational potential gives the following independent sufficient criterion. It is useful when its force-balance discrepancy is directly available, but is not imposed on source laws for which action drift can be verified more economically.

Give the free physical nodal variables the mass norm

$$\|v\|_{0,h}^2 = h^4 \sum_{x \in Q'_h} |v(x)|^2. \quad (\text{L.33})$$

Let $\mathcal{A}_h = S_h^{\text{loc}}$ on a compact convex field chart Θ_h . Suppose a positive operational potential Ψ_h has

$$g_* I \preceq \nabla_h^2 \Psi_h(y) \preceq g^* I, \quad 0 < g_* \leq g^* < \infty, \quad (\text{L.34})$$

and the local action obeys

$$\|\nabla_h^2 \mathcal{A}_h(y)\|_{0,h \rightarrow 0,h} \leq K_0 h^{-2}, \quad \text{osc}_{\Theta_h} \mathcal{A}_h \leq D_0. \quad (\text{L.35})$$

Theorem L.9 (Alternative finite-prefix stationarity criterion). *Let $\theta_0, \dots, \theta_J \in \Theta_h$ be an accepted finite sequence whose connecting segments stay in the chart. Put $E_h = \nabla_h \mathcal{A}_h$, choose $v_h = v_0 h^2$ with $v_0 K_0 < g_*/2$, and define*

$$b_j = E_h(\theta_{j+1}) + v_h^{-1} \{\nabla_h \Psi_h(\theta_{j+1}) - \nabla_h \Psi_h(\theta_j)\}. \quad (\text{L.36})$$

Then

$$\frac{1}{J} \sum_{j=0}^{J-1} \|E_h(\theta_j)\|_{0,h}^2 \leq C_0 h^{-2} \frac{D_0}{J} + D_1 \frac{1}{J} \sum_{j=0}^{J-1} \|b_j\|_{0,h}^2. \quad (\text{L.37})$$

Consequently the smallest actually observed full native slope satisfies

$$\|E_h(\theta_{j_*})\|_{0,h} \leq C \left[h^{-1} \sqrt{D_0/J} + \left(J^{-1} \sum_{j < J} \|b_j\|_{0,h}^2 \right)^{1/2} \right]. \quad (\text{L.38})$$

In particular $D_0/J = O(h^4)$ and $J^{-1} \sum \|b_j\|_{0,h}^2 = O(h^2)$ give $\sigma_h = O(h)$ without an invariant probability distribution or an assumed small selected action covector.

Proof. The same-action descent calculation is given in [Section M.3](#). □

Corollary L.10 (Finite-table certification for the prefix-balance entrance). *Suppose the accepted transitions are represented by finite tables, a safe set carries the chart and balance hypotheses above, and p_h^{exit} is the probability of leaving that safe set before J accepted comparisons. Put*

$$\mathcal{Q}_{h,J}^{\text{bal}} = \frac{1}{J} \sum_{j < J} \mathbb{E} \left[\mathbf{1}_{\text{safe}} \|b_j\|_{0,h}^2 \right]. \quad (\text{L.39})$$

For any $0 < \alpha \leq 1$, the probability that the best safe observed record fails a fixed-constant $O(h^\alpha)$ full native-row bound is at most

$$p_h^{\text{exit}} + C \frac{D_0}{J h^{2+2\alpha}} + C h^{-2\alpha} \mathcal{Q}_{h,J}^{\text{bal}}. \quad (\text{L.40})$$

If this bound is summable along a dyadic cutoff family, eventual certification holds almost surely without an independence assumption between cutoffs.

Proof. See [Section M.3](#). □

Corollary L.11 (Nonemptiness of the controlled finite class). *A separate comparison-action approximation is available under the explicit constrained-initial-solver hypotheses of [Section K.2](#). For that class, the fully finite spectral-midpoint evolution there exists on a common local slab and produces records $z_{N,\tau}$ satisfying*

$$\|z_{N,\tau} - z\|_{C_t H^{s+2}} + \|\text{Riem}(g_{N,\tau}) - \text{Riem}(g)\|_{L_t^\infty H^s} + \|\mathcal{R}_B(z_{N,\tau})\|_{L_t^\infty H^s} \leq C(N^{-p} + \tau), \quad (\text{L.41})$$

with the Dirac residual $O(N^{-p} + \tau^2)$ and summable dyadic transport. The construction starts from finite constrained initial data and does not sample a prescribed future solution. Its purpose is a generated comparison-action realization; it does not by itself establish the raw local-action budgets or the source-drift conditions. Nonemptiness for the raw local action is treated in [Corollary K.4](#), and the conditional source-native theorem is [Theorem L.6](#).

Proof. This is the content of [Section K.2](#). □

Remark L.12 (What remains outside the quantitative branch). For one actual accepted finite law, [Theorem L.6](#) derives the selected Euler-row, leakage and initial/gauge budgets jointly from explicit source-level conditions. What is not proved is that an arbitrary independently specified microscopic source satisfies the required signed action-drift, reader-energy, legal-domain and full coupled gravity-gauge-chiral-matter conditions. In the Renewal Geometry realization assembled from the Lorentzian construction [\[24\]](#), the differential/spin construction [\[25\]](#), and the structural gravity-matter construction [\[26\]](#), this becomes a separate provenance and source-selection problem; the complete microscopic-source Einstein-Standard-Model implication is not assumed here.

Remark L.13 (All-sector reader hypothesis). The tails in [\(5.1\)](#) concern the full tuple $(e, A, H, \Psi, \bar{\Psi})$ on a fixed calibrated chart, including color, weak, hypercharge and spin-geometric components. The weak-sector normalization of [Section M.2](#) is only an example for the $\text{SU}(2)$ block. Neither it nor finite gauge invariance supplies an estimate for the other blocks. One must verify the full-reader certificate [Proposition L.2](#), or independent component certificates combined as in [Proposition M.1](#). No all-sector normalization theorem is inferred from the Higgs frame.

M Source-selection proofs and physical-reader certificates

M.1 Native source selection and physical reader certificates

This subsection proves the finite source-level results used in [Appendix L](#). All probabilities refer to the actual accepted table; failures are retained, and no transition law is conditioned on success.

M.1.1 Stopped signed-action drift and same-record selection

Let Z_j denote the diagnostic stopped action value: on a legal transition it equals $a_h(X_j)$, while after the first attempted illegal transition it retains the last legal value. Shifting the action by the constant A_h^- gives $0 \leq Z_j \leq D_h^A$ without changing any increment or Euler covector. Conditional on an active legal row,

$$\mathbb{E}[Z_{j+1} - Z_j \mid X_j = x] = \mathfrak{d}_{h,j}(x). \quad (\text{M.1})$$

Summing [\(L.4\)](#) against the surviving subprobability masses and telescoping gives

$$\kappa_A \delta_h \sum_{j < J, x} \mu_j(x) g_h(x)^2 \leq D_h^A + \delta_h \sum_{j < J, x} \mu_j(x) r_{h,j}(x), \quad (\text{M.2})$$

which is (L.6).

For the selection step define

$$\mathcal{S}_j = \frac{g_h(X_j)^2}{s_h^2} + \frac{V_h(X_j)}{R_h^2} + \frac{W_h(X_j)}{\alpha_h^2}.$$

On a surviving prefix, if j_* minimizes $\mathcal{S}_j$ and $J^{-1} \sum_{j < J} \mathcal{S}_j \leq 1$, then every term in (L.8) is bounded by its threshold. Hence selection failure on survival implies that the mean score exceeds one. Markov's inequality, (L.6), and the definitions (L.2) give exactly (L.9). Notice that no union bound over all candidates is used; the high-order energy and alignment conditions are required only of the selected record.

M.1.2 Exact reader constant and sharp Fourier transfer

For the affine reader (L.10), collect the high-frequency coefficient maps in

$$\mathsf{T}_{h,K,m}\xi = (T_{h,\ell}\xi)_{|\ell|_\infty > K}, \quad w_\ell = (1 + |\ell|_1/K)^m.$$

If $\xi \in \ker \mathbf{Q}_h$, the energy side of (L.11) vanishes; hence (L.12) is necessary. A bound only on a prescribed finite set of record coordinates would not imply this universal kernel condition. Conversely restrict to $(\ker \mathbf{Q}_h)^\perp$ and write $\xi = \mathbf{Q}_h^{\dagger/2}y$. Then

$$\|y\|^2 = \langle \xi, \mathbf{Q}_h \xi \rangle,$$

and weighted block- ℓ^1 duality gives

$$\sup_{\langle \xi, \mathbf{Q}_h \xi \rangle \leq 1} \sum_{|\ell|_\infty > K} w_\ell \|T_{h,\ell}\xi\| = \|\mathsf{T}_{h,K,m}\mathbf{Q}_h^{\dagger/2}\|_{2 \rightarrow \ell^1(w)}. \quad (\text{M.3})$$

This proves Proposition L.2; the affine offset is handled by the triangle inequality.

For completeness, on the normalized four-torus Parseval and Cauchy–Schwarz give

$$\tau_{h,m}(K; u) \leq A_{N,K,m,q} \|u\|_{H^q}, \quad A_{N,K,m,q}^2 = \sum_{K < |\ell|_\infty \leq N} \frac{(1 + |\ell|_1/K)^{2m}}{(1 + |\ell|_2^2)^q}. \quad (\text{M.4})$$

Dyadic shells in four dimensions contain $O(R^4)$ lattice points. On a shell of radius $R \asymp 2^a K$, the contribution to (M.4) is bounded by $CK^{-2m}R^{2m-2q+4}$. Thus

$$\tau_{h,m}(K; u) \leq C_{q,m} K^{2-q} \|u\|_{H^q} \quad (q > m + 2), \quad (\text{M.5})$$

and the exponent is sharp on shell-supported examples. At $q = m + 2$ the dyadic contributions are comparable and the constant diverges logarithmically with the ultraviolet cutoff. Combining (L.15) with (M.5) proves Corollary L.3.

M.1.3 A directly verifiable action-drift mechanism

The drift hypothesis (L.4) is itself testable against the actual transition law and the same action. Let $E = \nabla \mathcal{A}$ in the physical mass norm and $\Delta = Y - x$ for one legal transition. Suppose

$$m(x) := \mathbb{E}[\Delta \mid x] = -\delta M(x)E(x) + \delta e(x), \quad \frac{M + M^*}{2} \succeq \lambda I, \quad (\text{M.6})$$

with $\lambda > 0$. Taylor's integral identity gives

$$\mathbb{E}[\mathcal{A}(Y) - \mathcal{A}(x) \mid x] = \langle E(x), m(x) \rangle + \mathcal{W}(x), \quad \mathcal{W}(x) = \mathbb{E} \left[\int_0^1 (1-t) D^2 \mathcal{A}(x + t\Delta) [\Delta, \Delta] dt \mid x \right]. \quad (\text{M.7})$$

Therefore

$$\mathfrak{d}(x) \leq -\frac{\lambda}{2} \delta \|E\|^2 + \delta \left(\frac{\|e\|^2}{2\lambda} + \frac{[\mathcal{W}(x)]_+}{\delta} \right). \quad (\text{M.8})$$

Only the signed action work of the innovation is charged. In particular, if the actual transition factors as $K(x, dy) = \int D(x, dz) N_x(z, dy)$ and the second kernel is action-neutral,

$$\int \mathcal{A}(y) N_x(z, dy) = \mathcal{A}(z), \quad (\text{M.9})$$

then its stochastic variance contributes nothing to the action-drift remainder. The selected record itself, including that retained noise, is still tested by V_h and W_h . These identities do not authorize inserting a gradient step or an action-neutral component into a source that does not actually possess them.

M.1.4 Polynomial regime

Under (L.28)–(L.30), the five terms in (L.9) are respectively $O(h^{2\epsilon})$, $O(h^{2\epsilon})$, $O(h^{2a})$, $O(h^{2\epsilon})$, and the declared exit term. Hence the failure probability is $O(h^{2a} + h^{2\epsilon})$. The Sobolev reader gives

$$T_{h,j} \leq Ch^{-a} K_h^{2-q}, \quad j = 2, k+2. \quad (\text{M.10})$$

Consequently

$$\bar{\varepsilon}_{c,h} \leq C(h^b + hK_h^3 + h^{-a}K_h^{4-q}), \quad \bar{L}_{h,k} \leq C(1 + |\log h|),$$

and the terms in $\bar{F}_{h,k}$ have orders

$$h^b K_h^{k+1}, \quad hK_h^{k+4}, \quad h^{-a} K_h^{k+5-q}, \quad h^{-a} K_h^{k+4-q},$$

up to the fixed logarithmic factor. The fourth exponent is larger than the third by β ; adding $\alpha_h = h^c$ gives precisely (L.31). Positive powers of a dyadic spacing times a fixed logarithmic power are summable.

Proposition M.1 (Assembly of full-reader bounds from component certificates). *In a fixed compatible atlas write the physical reader as the finite tuple $u = (u^{(1)}, \dots, u^{(B)})$, with blocks for the independent geometric, gauge and matter components. Suppose on the same source record*

$$\tau_{h,m}(K; u^{(b)}) \leq t_b + c_b \sqrt{V_b}, \quad V = \sum_{b=1}^B V_b.$$

Then, up to the fixed equivalence constants of the component norms,

$$\tau_{h,m}(K; u) \leq \sum_{b=1}^B t_b + \left(\sum_{b=1}^B c_b^2 \right)^{1/2} \sqrt{V}. \quad (\text{M.11})$$

The conclusion applies to local Lie-algebra components of $S(\text{U}(3) \times \text{U}(2))$ provided their bundle transitions and gauges have already been identified consistently. It neither chooses those gauges nor establishes any missing component estimate.

Proof. The Euclidean norm of a Fourier coefficient of the tuple is at most the sum of its component norms. Sum with the nonnegative tail weights, apply the component estimates, and use Cauchy–Schwarz in the finite block index. The finite central quotient changes no local Lie-algebra estimate; global compatibility remains the declared bundle hypothesis. $\square$

M.2 Gauge-covariant realization of the weak-sector reader

The Fourier reader used in the quantitative theorem is a coordinate object, but its control need not originate from ordinary derivatives of an arbitrary gauge frame. We record a finite gauge-covariant entrance on a nonvanishing Higgs chart. This appendix concerns the weak $SU(2)$ sector only; it is not a substitute for independent color, determinant-line or gravitational reader estimates.

On a periodic grid let $U_i(x) \in SU(2)$ and define covariant shifts and differences

$$T_i^U F(x) = \rho(U_i(x))F(x + e_i), \quad D_i^U F = h^{-1}(T_i^U F - F), \quad D_I^U = D_{i_1}^U \cdots D_{i_\ell}^U. \quad (\text{M.12})$$

For a fixed integer L put

$$\mathcal{J}_{h,L}(F; U)^2 = \sum_{\ell=0}^L \sum_{|I|=\ell} \|D_I^U F\|_{2,h}^2. \quad (\text{M.13})$$

Expanding the ordered product $h^{-\ell}(T_{i_1}^U - I) \cdots (T_{i_\ell}^U - I)$ gives, for each word I ,

$$D_I^U F(x) = h^{-\ell} \sum_{S \subset \{1, \dots, \ell\}} (-1)^{\ell-|S|} F_S(x), \quad (\text{M.14})$$

where all transported endpoints $F_S(x)$ transform by the same unitary at x . Hence $|D_I^U F(x)|^2$ is a fixed contraction of the positive mixed path Gram $G_I(x)_{ST} = \langle F_S(x), F_T(x) \rangle$. The energy (M.13) is therefore gauge invariant and can be audited from same-record mixed path data.

The discrete Kato inequality

$$|\delta_i|F|(x)| \leq |D_i^U F(x)| \quad (\text{M.15})$$

and ordinary Sobolev inequalities on the fixed grid imply, for $d \leq 4$,

$$\|F\|_{\infty,h} \leq C_d \mathcal{J}_{h,3}(F; U), \quad \max_{|I| \leq L-3} \|D_I^U F\|_{\infty,h} \leq C_{d,L} \mathcal{J}_{h,L}(F; U), \quad (\text{M.16})$$

with constants independent of the links.

Assume now $|H_x| \geq \rho_* > 0$. Let $Q_x \in SU(2)$ be the algebraic frame sending e_1 to $H_x/|H_x|$ and put

$$W_i(x) = Q_x^* U_i(x) Q_{x+e_i}, \quad a_i = h^{-1}(W_i - I), \quad J_I = Q_x^* D_I^U H. \quad (\text{M.17})$$

The exact identities

$$\delta_i J_I = J_{iI} - a_i T_i J_I, \quad |H_{x+e_i}| = |H_x| e_1 + h J_i(x) \quad (\text{M.18})$$

close the ordinary difference hierarchy in terms of covariant jets. On a fixed lower-radius chart, repeated finite-difference chain rules therefore give

$$\mathcal{J}_{h,s+4}(H; U) + \sum_b \mathcal{J}_{h,s+3}(F_b; U) \leq R \implies \|H\|_{C_h^{s+1}} + \sum_i \|h^{-1} \log W_i\|_{C_h^s} + \sum_b \|Q^* F_b\|_{C_h^s} \leq C_{s,\rho_*,R}. \quad (\text{M.19})$$

No derivative of the original Q_x and no smallness of the original links is assumed.

The algebraic Higgs frame need not be temporal gauge. Starting with $g_{0,x} = I$ and defining causally

$$g_{j+1,x} = g_{j,x} W_0(j, x) \quad (\text{M.20})$$

sets the transformed temporal links to the identity. Iterated product rules and discrete Gronwall show that one additional covariant derivative controls the transformed spatial logarithmic links through order s on a fixed physical-time interval. All transformed matter fields retain the same order of control. Parseval and the discrete-difference symbol estimate then yield an all-mode H^q bound and, by (M.5), the tail estimate (L.32).

Finally, exact finite gauge invariance prevents this normalization from hiding the action row. If E_h is the Riesz gradient of the *full* physical action in the mass norm, then for every fixed site gauge transformation g ,

$$E_h(g \cdot z) = T_g E_h(z), \quad \|E_h(g \cdot z)\|_{0,h} = \|E_h(z)\|_{0,h}. \quad (\text{M.21})$$

When $g = g(z)$ is the algebraic Higgs normalization, differentiating $g(z) \cdot z$ adds a vertical gauge tangent; the exact finite Ward identity annihilates that tangent, so the complete first variation is unchanged. On the normalized logarithm chart $W = e^{hA}$ with $h\|A\| \leq c$, the physical left link variation and $\dot{A}$ are related by

$$J_h(A) \dot{A} = \int_0^1 \text{Ad}_{e^{thA}} \dot{A} dt, \quad \|J_h - I\| \leq Cc, \quad (\text{M.22})$$

so the logarithmic and physical link-source norms are uniformly equivalent for fixed sufficiently small c . A derivative restricted to the radial Higgs slice alone is not used as a substitute for this full covector.

M.3 Finite-prefix stationarity from the same local action

This subsection proves [Theorem L.9](#) and [Corollary L.10](#). The argument does not define a new field evolution and does not replace the indefinite Lorentzian action by a least-squares cost. It studies any accepted finite prefix using the same local action (D.6) and a positive operational potential supplied by the source geometry.

Lemma M.2 (Shifted-first-jet normal form of the unchanged local action). *On a bounded nondegenerate coframe/logarithm chart, the action (D.6) has the exact interior representation*

$$S_h^{\text{loc}}[y] = h^4 \sum_{x \in Q'_h} F_h^{(1)}(\Xi_h^{(1)} y(x)) + \mathcal{B}_h[y], \quad (\text{M.23})$$

where $\Xi_h^{(1)} y$ consists of a fixed finite set of shifted field values and normalized first differences, $F_h^{(1)}$ is smooth with fixed-order derivatives bounded uniformly for small h , and $\mathcal{B}_h$ depends only on the fixed collars for the interior variation problem. Therefore the interior Euler covector and all of its mixed physical derivatives are exactly those of the shifted-first-jet bulk representative; no continuum approximation is used in this identity.

Proof. Only the gravitational plaquette needs a reduction. Write its curvature logarithm as the exact discrete curl of the coframe connection plus a remainder that is a regular function of shifted connections. If $C^{\mu\nu}(e)$ is the Palatini coframe coefficient, the discrete product identity

$$\langle C, \delta_\mu^+ W \rangle = -\langle \delta_\mu^- C, W \rangle + \delta_\mu^- \langle C, T_\mu W \rangle \quad (\text{M.24})$$

converts both curvature-curl terms into first-jet bulk expressions plus an exact lattice divergence. Moreover, with $e_- = T_{-\mu}e$ and $p_- = T_{-\mu}\delta_\mu^+e$,

$$\delta_\mu^- C(e) = \int_0^1 DC(e_- + thp_-)[p_-] dt, \quad (\text{M.25})$$

which is a smooth shifted-first-jet function with no inverse power of h . Summing the divergence leaves only the fixed collar functional $\mathcal{B}_h$.

The Yang–Mills plaquette is already a uniformly regular function of shifted gauge values and first differences by [Lemma H.3](#); the Higgs and Dirac link terms in [\(D.3\)–\(D.4\)](#) are shifted first-jet functions directly. Thus all sectors share the representation [\(M.23\)](#). Since the transformation is an exact discrete summation-by-parts identity, interior first and higher physical variations are unchanged. $\square$

Lemma M.3 (Uniform upper Hessian scale of the local action). *On a compact convex subchart of the finite field coordinates, with all shifted first-jet arguments of [\(D.5\)](#) remaining in a common compact coefficient set, there are constants C_1, K_0, D_0 independent of h and of the number of nodes such that*

$$|D^2 \mathcal{A}_h(y)[u, v]| \leq C_1 \|u\|_{1,h} \|v\|_{1,h}, \quad (\text{M.26})$$

$$\|\nabla_h^2 \mathcal{A}_h(y)\|_{0,h \rightarrow 0,h} \leq K_0 h^{-2}, \quad (\text{M.27})$$

$$\text{osc}_{\Theta_h} \mathcal{A}_h \leq D_0, \quad (\text{M.28})$$

where

$$\|u\|_{1,h}^2 = \|u\|_{0,h}^2 + \sum_{\mu=0}^3 \|\delta_\mu^+ u\|_{0,h}^2.$$

These are upper bounds only; no positivity or inverse for the Lorentzian action Hessian is asserted.

Proof. By [Lemma M.2](#), the interior local density is exactly a smooth function of finitely many shifted first jets. Differentiating the finite sum twice gives h^4 times a sum of uniformly bounded coefficient bilinear forms applied to shifted values and first differences of u, v . Each shift occurs only a fixed number of times. Cauchy–Schwarz gives [\(M.26\)](#). The elementary inverse estimate $\|\delta_\mu^+ u\|_{0,h} \leq 2h^{-1} \|u\|_{0,h}$ gives [\(M.27\)](#). Finally the density is uniformly bounded on the compact chart and h^4 times the number of lattice cells in the fixed cylinder is uniformly bounded, giving [\(M.28\)](#). The coframe/spin-connection chain is contained in the smooth density and is differentiated throughout. $\square$

Proof of Theorem L.9. Let $x = \theta_j$, $y = \theta_{j+1}$, $d = y - x$, and write $p_h = \nabla_h \Psi_h$. By [\(M.27\)](#), Taylor’s integral formula gives

$$\mathcal{A}_h(x) - \mathcal{A}_h(y) \geq -(E_h(y), d)_{0,h} - \frac{1}{2} K_0 h^{-2} \|d\|_{0,h}^2. \quad (\text{M.29})$$

The lower bound in [\(L.34\)](#) yields

$$(p_h(y) - p_h(x), d)_{0,h} \geq g_* \|d\|_{0,h}^2.$$

Substitute [\(L.36\)](#) and put

$$a_h = g_* v_h^{-1} - \frac{1}{2} K_0 h^{-2} = a_0 h^{-2} > 0, \quad \ell_h = g^* v_h^{-1} + K_0 h^{-2} = \ell_0 h^{-2}.$$

Young's inequality gives

$$\mathcal{A}_h(x) - \mathcal{A}_h(y) \geq \frac{1}{2}a_h\|d\|_{0,h}^2 - \frac{1}{2a_h}\|b_j\|_{0,h}^2. \quad (\text{M.30})$$

Moreover

$$E_h(x) = b_j - v_h^{-1}(p_h(y) - p_h(x)) + (E_h(x) - E_h(y)),$$

so

$$\|E_h(x)\|_{0,h} \leq \|b_j\|_{0,h} + \ell_h\|d\|_{0,h}.$$

Combining with (M.30), squaring, and summing over j gives

$$\sum_{j < J} \|E_h(\theta_j)\|_{0,h}^2 \leq \frac{4\ell_h^2}{a_h} \{\mathcal{A}_h(\theta_0) - \mathcal{A}_h(\theta_J)\} + \left(2 + \frac{2\ell_h^2}{a_h^2}\right) \sum_{j < J} \|b_j\|_{0,h}^2.$$

Since $4\ell_h^2/a_h = C_0h^{-2}$, the second coefficient is a cutoff-independent constant, and the telescoped action decrement is at most the uniform oscillation D_0 , this is (L.37). Taking the minimum of the finite list gives (L.38). $\square$

Proof of Corollary L.10. Let G be the event that the accepted prefix stays inside the declared safe set. On G , (L.37) holds pathwise. Multiplying by $\mathbf{1}_G$, taking expectations, and enlarging the nonnegative balance sum to all safe occupied edges gives

$$\mathbb{E} \left[\mathbf{1}_G \frac{1}{J} \sum_{j < J} \|E_h(\theta_j)\|_{0,h}^2 \right] \leq C_0h^{-2} \frac{D_0}{J} + D_1 \mathcal{Q}_{h,J}^{\text{bal}}.$$

If the best safe slope exceeds h^α , the mean square exceeds $h^{2\alpha}$. Markov's inequality therefore gives the last two terms of (L.40); adding $\mathbb{P}(G^c) = p_h^{\text{exit}}$ proves the bound. Summability along dyadic cutoffs and the first Borel–Cantelli lemma yield eventual certification without independence. $\square$

M.4 Complete physical variations on constrained source charts

The source estimates must concern the same complete action covector as the continuum variation. A restriction to a constraint surface, a selected principal block, or a partial reader does not by itself provide this identification. The following finite statements separate these requirements.

Proposition M.4 (Tangential stationarity and the missing normal row). *Let S_h be differentiable on a finite positive Hilbert space X_h , and let $C_h : X_h \rightarrow Z_h$ be differentiable with surjective derivative at x_h . Assume $B_h : Z_h \rightarrow X_h$ is a right inverse of $DC_h(x_h)$ and put $P_h = I - B_h DC_h(x_h)$. Suppose every lifted unit physical test satisfies*

$$\|P_h I_h v\| \leq L_h^{\text{tan}}, \quad \|DC_h(x_h) I_h v\| \leq L_h^{\text{nor}}.$$

If

$$|DS_h(x_h)[w]| \leq a_h \|w\| \quad (w \in \ker DC_h(x_h)), \quad |DS_h(x_h)[B_h q]| \leq b_h \|q\|,$$

then

$$\sup_{\|v\| \leq 1} |DS_h(x_h)[I_h v]| \leq a_h L_h^{\text{tan}} + b_h L_h^{\text{nor}}. \quad (\text{M.31})$$

In particular exact stationarity of S_h restricted to $C_h = 0$ does not imply complete physical stationarity unless the normal contribution also vanishes or all physical lifts are tangent.

Proof. Since $DC_h B_h = I$, $P_h I_h v$ lies in the tangent kernel and $I_h v = P_h I_h v + B_h DC_h I_h v$. Apply the two covector bounds. For the failure of the unrestricted implication, take $S(x, y) = x^2 + y$ and $C(x, y) = y$. The restriction is stationary at $(0, 0)$, whereas $DS(0, 0)[(0, 1)] = 1$. $\square$

Proposition M.5 (Residual-preserving extension of a constrained identity). *Let a finite-dimensional regular constraint chart have coordinates (s, c) , with the constraint surface $c = 0$. Let $J(s, c)$ be a C^1 matrix-valued map. Then*

$$J(s, c) = J(s, 0) + \int_0^1 D_c J(s, tc)[c] dt. \quad (\text{M.32})$$

If $c_a(t)$ satisfies the exact tracking equation $\dot{c}_a = A_a(t)c_a + r_a(t)$, with $\int_0^T \|A_a(t)\| dt \leq M$ and $\|c_a(0)\| + \|r_a\|_{L^1(0, T)} \leq Ca^p$, then

$$\sup_{0 \leq t \leq T} \|c_a(t)\| \leq Ce^M a^p. \quad (\text{M.33})$$

A uniformly bounded $D_c J$ consequently gives an $O(a^p)$ difference between the full and constrained principal matrices on these paths. This conclusion gives no estimate for an independently occurring state-only source $f(s, c)$ without a bound or an identity for that source.

Proof. The first identity is the fundamental theorem of calculus in the normal coordinate. The variation-of-constants inequality followed by Gronwall gives (M.33). Substitution into (M.32) gives the matrix estimate. For the last assertion take $J(s, c)$ constant while $f(s, c)$ is any fixed nonzero covector, or any cutoff-dependent multiple of it. The principal identity then holds exactly and imposes no restriction on the source. $\square$

These observations apply before invoking either variational closure or the coupled stability estimate. A constraint-compatible principal operator must be accompanied by control of the full source, including its normal and state-only rows. Likewise, an exact algebraic decision procedure for stationarity does not establish the existence of a stationary record until its actual coefficients, physical-domain conditions and complete covector have been supplied and evaluated. No such source-existence conclusion is used in the continuum theorems.

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